FINANCIAL FRAUD DETECTION USING MULTI-CLASSIFIER SYSTEMS

A thesis submitted in fulfilment of the requirements for the degree of Doctor of Philosophy in Information and Communication Technology

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Abstract

With the advent of digital technologies, electronic commerce, or e-commerce, has transformed the limited traffics in physical stores into an open online traffics in e-stores. Leveraging the Internet connectivity, e-commerce allows consumers to choose their desirable items in terms of price, quantity, type among various online stores and services across the world. In line with the recent Covid-19 pandemic in which people avoid visiting physical stores due to safety concerns, the popularity and usage of e-commerce has risen exponentially.

In e-commerce, online payment cards constitute one of the common transaction methods. Although this business method is convenient for consumers, it opens up opportunities for fraudsters to engage in illegal activities. Indeed, as online financial activities increase, the value of fraudulent transactions has been on the rise, leading to billions of dollars in loss for merchants every year. This financial burden is passed on to consumers, which in turn increases the product prices. Using a dataset comprising over 60,000 financial records from transactions across 23 countries, it was discerned that real-world data often requires pre-processing due to its inherent inconsistencies. Steps such as data cleaning and feature selection were essential. Given these challenges, it is both critical and timely for financial institutions to detect and prevent fraudulent transactions.

To minimize financial loss, there is a growing need to develop a reliable, effective, and efficient system for financial fraud detection. This is increasingly important with the reliance on using payment cards for online transactions. Payment cards, which consist generally of credit and debit cards, have attracted fraudsters to innovate and deploy different methods to beat the current fraud prevention mechanisms by financial institutions. The general fraud rules established by financial institutions become outdated rapidly as fraudsters constantly find and exploit the vulnerability of the existing systems.

One advanced methodology currently being widely investigated for mitigating financial fraud cases is the use of machine learning models. Researchers continue to create effective (in terms of accuracy) and efficient (in terms of time) intelligent solutions for fraud detection in the financial sector as machine learning advances. In this thesis, a multi-classifier system is designed and developed based on machine learning models to undertake payment card fraud detection. Specifically, the Behavior-Knowledge Space (BKS) method is exploited to aggregate the predictions from multiple machine learning classifiers in an ensemble manner. The original BKS model is enhanced in two variants. The first involves the use of confidence levels for the predicted outputs, instead of mere outputs from the constituent classifiers, while the second exploits a threshold of the confidence level to improve performance.

For performance evaluation, publicly accessible data sets and actual financial records are employed in empirical studies. With a total of 30 benchmark data sets, the results indicate that the devised ensemble models are more effective than the frequently employed majority voting method for combining predictions from various classifiers in addressing noisy data and fraud detection problems. In addition, a real-world case study with noise-induced financial data set is conducted. The results with up to 40% noise in the data samples demonstrate the efficacy of the BKS-based ensemble models in addressing payment card fraud detection in real-world environments, outperforming majority voting by up to 19 times at certain noise levels. Specifically, at a threshold of 0.90, BKS accuracy rates were observed to be 0.02 higher than without the threshold, even maintaining a rate of 0.9444 at 40% noise level, compared to 0.9238 without the threshold.

Given the large volume of transactions performed daily, computational speed is paramount for fraud detection in credit card transactions utilizing machine learning. Traditional fraud detection methods rely on manual review, which is both time-consuming and costly. In contrast, machine learning-based fraud detection systems can swiftly analyse vast volumes of transaction data, delivering faster and more precise results. However, the time required for training and optimizing these models can be substantial. For example, when comparing computational speeds at two different thresholds, it was observed that BKS required a marginally longer duration than voting. Yet, the standout accuracy rates of BKS more than compensated for this slight delay, underscoring its efficacy.

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Candidate Declaration

I, Kuldeep Kaur, declare that this thesis is my own original work and does not contain material that has been accepted for the award of any other degree or diploma.

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Kuldeep

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List of Abbreviations

ANFIS	Adaptive Neuro-Fuzzy Inference System
ANN	Artificial Neural Networks
API	Application Programming Interface
AUC	Area Under the ROC Curve
AVS	Address Verification System
BKS	Behaviour Knowledge Space
BNC	Bayesian Network Classifier
BNM	Bank Negara Malaysia
CNN	Convolution Neural Network
CNP	Card Not Present
CV	Cross Validation
CVV	Card Verification Code
DL	Deep Learning
DT	Decision Tree
EEG	Electroencephalogram
ELM	Extreme Learning Machine
GB	Gradient Boosting
GB	Gradient Boosting
HMM	Hidden Markov Model
kNN	*k*-Nearest Neighbours
LDA	Linear Discriminant Analysis
LR	Logistic Regression
LR	Logistic Regression
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
MCS	Multi-Classifier System
ML	Machine Learning
MLP	Multi-Layer Perceptron
MSE	Mean Squared Error

NASDAQ	National Association of Securities Dealers Automated Quotations
NB	Naïve Bayes
NN	Neural Networks
PCA	Principal Component Analysis
RBF	Radial Basis Function
RF	Random Forest
RMSE	Root Mean Squared Error
RNN	Recurrent Neural Network
RNN	Recurrent Neural Network
ROC	Receiver Operating Characteristic
SGD	Stochastic Gradient Descent
SOM	Self-Organizing Maps
SVM	Support Vector Machine
UCI	University of California, Irvine
XGB	Extreme Gradient Boosting

1 Introduction

This chapter provides an overview of the background and motivations for the study on financial fraud detection using multi-classifier systems. Problem statements, research objectives, and scope are all presented. The thesis structure is provided at the end of the chapter.

1.1 Overview

The term Machine Learning (ML) is commonly heard today. Over six decades ago in 1959, Arthur Samuel described ML as the:

> *"field of study that gives computers the ability to learn without being explicitly programmed."*

Programming computers in order to learn from experience will eventually remove the need for detailed efforts in programming (Awad and Khanna, 2015).

Basing on an inductive analytical paradigm, ML methods derive conclusions from patterns observed in data, without the need to define assumptions such as that in linear distribution (Severino and Peng, 2021). It can be used as an advanced data-driven approach that provides a framework in mapping complicated and non-linear relationships from the input to output parameters in a process. The ML models are capable in achieving higher prediction accuracies as it is able to generalize relations between variables, without the need for explicit system knowledge (Abbasi, Moghaddam and Kowsari, 2022).

In general, ML models can be divided into supervised and unsupervised learning. A key difference between the both is labelled data is used in supervised learning while unsupervised learning uses unlabelled datasets. Using labelled data in supervised learning, the process is divided into model training and prediction. An overview is shown in Fig. 1.1.

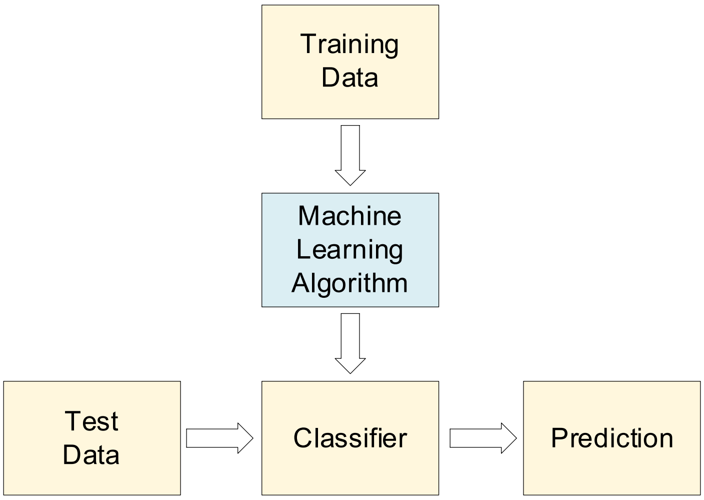

Fig. 1.1 Overview of pattern recognition process

Training data is typically larger than the test data. There have been various algorithms proposed for the ML models, such as Multi-Layer Perceptron (MLP), Support Vector Machine (SVM), and Decision Tree (DT) (Ding *et al.*, 2023). The ML models can be categorized into individual, hybrid and ensemble models. Individual models have been used in various predictions in different fields. However, issues such as overfitting, local minima may be present in the models. In reducing these issues, hybrid and ensemble approaches have been developed (Pourpanah *et al.*, 2019).

These models are developed by integrating multiple algorithms, which directly improves predictions by consolidating advantages of individual models. Some of the ensemble examples include Random Forest (RF), Gradient Boosting (GB) and Extreme Gradient Boosting (XGB). A hybrid model combines multiple individual models, which may include sampling and feature selection methods. For instance, PayPal, an online payment platform uses models which include DT, Neural Networks (NN) and Logistic Regression (LR) that they believe has been the most effective approach (Kim *et al.*, 2019).

ML algorithms have been widely used in intrusion detection, pattern recognition and fraud detection. With the growth of e-banking and online payments, the increase in financial fraud has been on the rise (Forough and Momtazi, 2021). Financial fraud covers a wide range of area, and is further discussed in Chapter 1.4. One of the areas of financial fraud is in the area of payment cards. A variety of payment cards, which include credit, charge, debit, and prepaid cards, are widely available nowadays. In some nations, they are the most widely used form of payment. Moreover, improvements in digital technology have paved the road for money transfer, particularly in payment methods that have shifted from being a physical activity to digital transactions via electronic means. This has transformed the landscape of monetary policy, as well as the corporate plans and operations of both major and small businesses.

E-commerce has greatly increased in popularity in recent years, largely because cross-border purchases and online credit card transactions are so simple (Zhang *et al*., 2021). Customers are no longer limited by the offers and terms of localised merchants, but instead have access to a diverse variety of global merchants and the ability to compare their items, offering quality, price, services, and so on in only a few clicks.

The population of Malaysia has been consistently growing year after year. Figs. 1.2 and 1.3 display the volume of payment card transactions and the value of such transactions per capita, respectively, adjusted for population. To calculate the per capita value, one would take the total value or volume of the transactions and divide it by the total population.

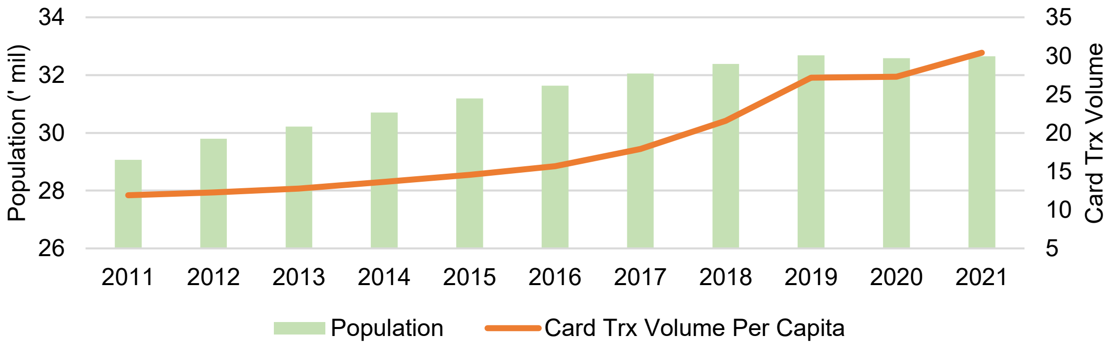

Fig. 1.2 Payment card transaction (Trx) volume per capita by population (BNM, 2021)

Based on Fig. 1.2, it can be seen that a yearly increase of 1.85 mil transactions volume is recorded (BNM, 2020). There was a stagnant period in 2020, which is most likely caused due to the Covid impact, as the volume picked up in the year after. The card transaction value however, has been recording a steady increase, year on year. Card transaction value and volume have been steadily rising, providing opportunity to analysis transaction records. Consumers are now spending more using payment cards as compared with the situation in the past.

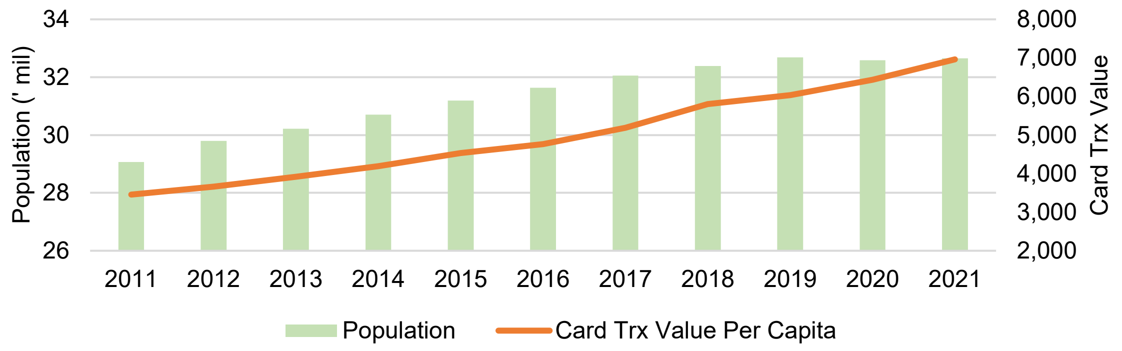

Fig. 1.3 Payment card transaction (Trx) value per capita by population (BNM, 2021)

While e-commerce is currently a mature market with many participants, online payment security lags behind (Van Vlasselaer *et al*., 2015). Credit card fraud is an important issue and has considerable cost for banks and card issuer companies (Halvaiee and Akbari, 2014). The problem has been getting worst, and it is expected to be worst in the coming years.

Using a variety of security measures, financial institutions work to eliminate fraudulent activity on customer accounts. The more intricate the security measures are, the more adept the fraudsters become; to put it another way, con artists are constantly adapting their strategies to stay one step ahead.

The term of “credit card fraud” can be coined as follows (Carneiro, Figueira and Costa, 2017),

> *“The action of an individual who uses a credit card for personal reasons without the consent of the owner of the credit-card and with no intention of repaying the purchase made.”*

Contrary to popular belief, merchants must pay the bill when a fraudster steals products or services in a consumer-not-present transaction, such as an online payment. The widespread use of credit cards, as well as the different transaction scenarios, would certainly result in billions of dollars in credit card fraud losses.

The main cause is the rapid expansion of online commerce, which generates a large number of Card Not Present (CNP) transactions, a type of payment that attracts the attention of criminals who attempt to trick the system by posing as someone else. Along with this, credit card issuers require an automated system that stops the pursuit of an incoming transaction if it is particularly susceptible to fraud, i.e., if it deviates from customary consumer behaviour.

Zooming into Malaysia specifically, as shown in Fig. 1.4, the number of fraud cases and losses are also increasing. These increases are despite a number of stringent security measures which had been implemented over the years. Across the globe, ‘Chip and Pin’ verification, card verification code, and address verifications have been implemented by card issuers (Olowookere and Adewale, 2020). Back to Malaysia, financial institutions started introducing real-time fraud detection system in 2011. This helped for a short period as in 2012, the authentication for online credit card transactions was made available.

The authentication is based on a 6-digits PIN that is sent to the cardholder to authenticate the online transactions. The 'Chip and PIN' verification for POS (Point of Sales) payment was introduced in 2017. All magnetic stripes of cards were replaced with chips, which were deemed to be more secure. However, it can be seen till today that the fraud cases have not reduced.

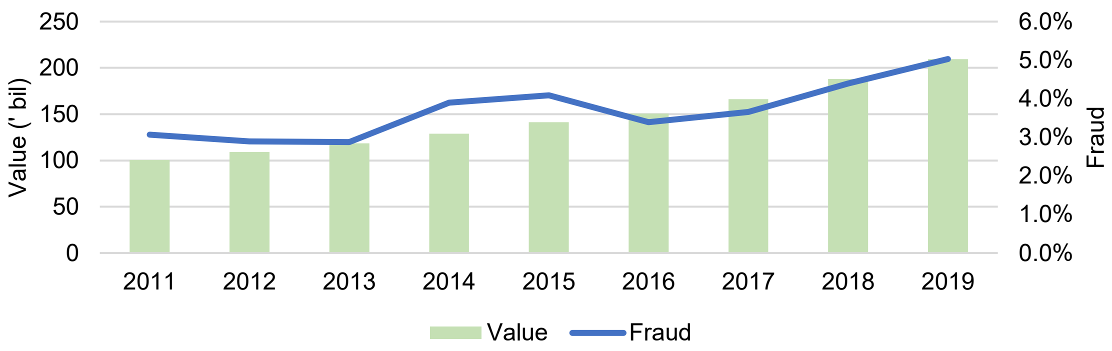

Fig. 1.4 Payment card fraud as a percentage of transactions value[1] (BNM, 2021)

Fraud detection has become an essential task in order to reduce the impact of fraudulent transactions on service delivery, costs, and the company's reputation (Carneiro, Figueira and Costa, 2017). Fraud is cost and detecting it before the transaction is registered will reduce this cost significantly, which needs a very accurate system with quite few false alarms.

A chargeback occurs when a customer claims that he or she did not get the desired products or services, or that the order was placed by a fraudster (Carneiro, Figueira and Costa, 2017). If the merchant is unable to counter this claim, the money must be returned to the customer's account, and the product must be forfeited (if it has been shipped). Should the chargeback rate exceed the standards set by the card organisations, merchants may be assessed chargeback costs and fines.

[1] There has been no fraud data published by Bank Negara Malaysia after 2019.

Fraud detection is a difficult topic because fraudsters go to great lengths to make their actions appear legitimate (Carneiro, Figueira and Costa, 2017). Another issue is that there are significantly more valid records than fraudulent instances. The data analyst must take extra precautions when dealing with such uneven collections. Creating dynamic systems that can adjust to new fraud patterns is the key to precise fraud detection. As fraudsters are always one-step ahead, efficient and productive methods that do the work must be designed (Roseline *et al*., 2022).

As a result, deploying robust fraud detection solutions is critical for any company that issues credit cards or handles online transactions in order to avoid losses and boost consumer confidence (Fiore *et al*., 2019). When illicit transactions are mixed with valid ones, modern fraud detection systems use advanced analytics and data mining approaches to uncover suspect usage patterns in transaction logs. This entails mining very large datasets for insights and performing binary classification to distinguish between legal and fraudulent transactions.

To provide a clearer understanding of the research's progression and main areas of focus, a simplified flowchart summarizing the primary processes and topics is given in Fig. 1.5. This flowchart offers a visual summary of the research's trajectory, guiding readers through the subsequent detailed chapters.

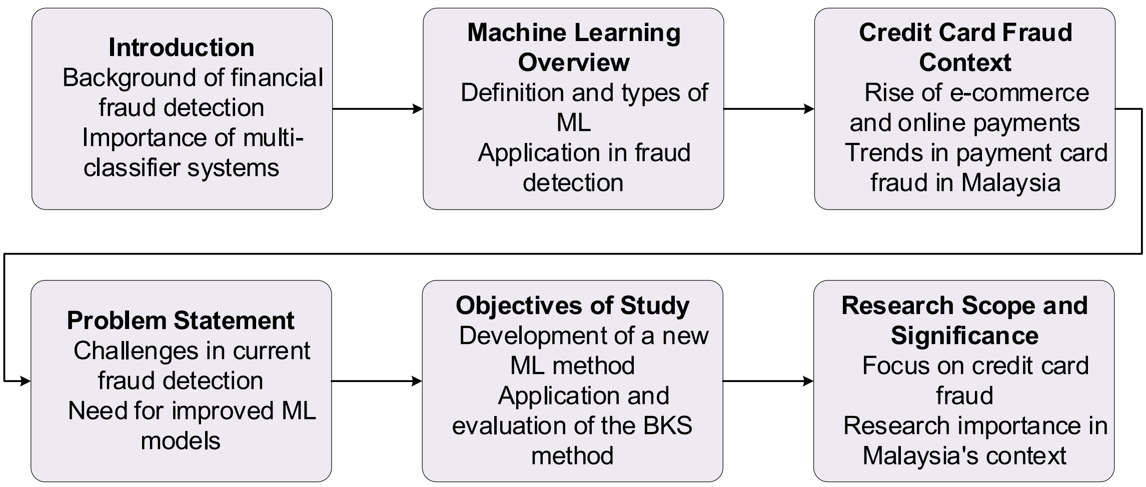

Fig. 1.5 Simplified overview of the research process in fraud detection

1.2 Problem Statement

One striking example of the increasing prevalence of digitization in everyday life and society over the course of the past few decades is the widespread adoption of credit cards as the primary method of payment (Hilal, Gadsden and Yawney, 2022). The widespread usage of credit cards brings up the issue of credit card fraud, which refers to the complex schemes and processes that are utilised to steal money and assets.

Although it is difficult to gain a precise assessment of the loss since card issuers are typically unwilling to share figures, there is some publicly available data that shows the severity of credit card fraud. Card transaction fraud losses totalled $28.65 billion in 2020, with the United States responsible for one-third of the total with $9.62 billion (Hilal, Gadsden and Yawney, 2022).

CNP fraud has surged considerably in recent years as ecommerce has grown in popularity and fraudsters have changed their attacks from Card Present (CP) to CNP transactions (Wang, Chen and Chen, 2019). CNP fraud now accounts for more than 70% of all card fraud worldwide. It should be noted that banks normally bear the losses for CP fraud, whereas merchants typically face the losses for CNP fraud.

There is a rising concern that there will be a rise in the number of fraudulent attempts as the use of credit cards becomes more widespread and popular. Credit card fraud that goes undetected can have severe repercussions because it has been linked to the funding of organised crime, the distribution of illegal drugs, and even acts of terrorism.

Credit card issuers can't manually verify each transaction to detect fraud due to the enormous volume of transactions (Zhang *et al*., 2021). Various technologies, such as the Address Verification System (AVS), Chip and Pin verification, and Card Verification Code (CVV), have been employed to prevent fraud (Carneiro, Figueira and Costa, 2017). Even the most advanced systems, however, are prone to failure.

Fraud detection and prevention have long been high priorities for card issuers and academic researchers, as discovering and halting even a small portion of fraudulent activity would save millions of dollars. Credit card fraud is divided into two types: application fraud and behavioural fraud, commonly known as offline and online fraud (Hilal, Gadsden and Yawney, 2022).

Application fraud is commonly associated with identity fraud since it typically involves fraudsters attempting to obtain new credit cards from credit organisations using the personal information of others. Behavioural fraud includes mail theft, stolen or lost cards, counterfeit cards, CNP fraud, and bankruptcy fraud.

Credit card fraud is detected using machine learning (ML) models by card issuers and network providers. A fraud detection model can be thought of as a transactional prediction model in which prior transactional behaviour is utilised to predict the authenticity of current transactional behaviour (Zhang *et al.*, 2021). Feature engineering in fraud detection is expected to generate feature variables that effectively summarise and describe transaction behaviour information from raw transaction records. A good feature engineering model serves as the foundation for an effective fraud detection model, and hence is important to the success of a fraud detection system.

A typical fraud detection system incorporates both an automatic and a manual method (Jurgovsky *et al.*, 2018). The automated algorithm is based on fraud detection techniques. It evaluates all new incoming transactions and assigns them a fictitious score. The manual method was developed by fraud detectives. They concentrate on transactions with a high fraudulent score and give binary feedback (fraud or real) on all transactions investigated. To fuel fraud detection systems, expert-driven rules, data-driven rules, or a combination of both types of rules can be employed. Expert-driven attempt try to recognise specific fraud scenarios discovered by fraud investigators.

Several outstanding issues exist in automatically recognising these transactions (de Sá, Pereira, and Pappa, 2018). Among these include the large number of transactions that must be handled in near real time, as well as the fact that fraud does not occur frequently, resulting in severely skewed datasets. Moreover, the number of normal transactions in the normal class is significantly greater than the number of fraudulent transactions in the abnormal class.

For every legitimate decrease, there are up to 40 false declines, implying that 97% of transactions marked as high-risk could be real (Wang, Chen and Chen, 2019). Consumers are inconvenienced, dissatisfied, and have bad purchasing experiences as a result of the large number of false declines. 61% of cardholders who receive incorrect rejects reduce or terminate card use. False declines cost retailers a lot of money. Likewise, fraudulently refused card transactions cost US ecommerce retailers $8.6 billion in 2016, considerably outweighing the $6.5 billion saved by fraud detection.

Merchants are put in a "merchant's dilemma" as a result of the unreliability of the machine learning (ML) fraud detection models that are currently in use. The merchant must decide whether to accept the false positives generated by the ML models or whether to accept all transactions and take the losses that result from fraudulent transactions (Wang, Chen and Chen, 2019).

1.3 Objectives of Study

The aim of this research is to design and develop a machine learning method to detect fraudulent payment card transactions, contributing towards preventing huge loss by financial institutions. The specific research objectives are:

1) To develop a multi-classifier system model using machine learning algorithms for data classification;
2) To apply the Behaviour Knowledge Space (BKS) method for combination of predictions from multiple classifiers and analyse the performance with benchmark data sets;
3) To enhance the BKS method with confidence measures for effective combination of predictions from multiple classifiers;
4) To ascertain the efficacy of the devised multi-classifier system with enhanced BKS for payment card fraud detection with noise-induced transactions records.

The multi-classifier system model must work well, especially on the financial data, as that is the main focus of the thesis. As the financial data typically has various features, and some might not be ideal, it is paramount to select the right features and derive new features from it, such as the use of feature aggregation.

The ultimate goal is to make sure that the model that was created can identify fraudulent behaviour patterns, particularly with regard to financial data. As fraudulent activity is so expensive for the financial sector, preventing it should be a top priority.

1.4 *Research Scope and Significance*

The financial fraud scope is large, covering securities, bank, insurance and others, as shown in Fig. 1.5. In securities fraud, it relates to securities and commodities fraud, financial statement fraud, among others. Healthcare and automobile fall under insurance fraud. In the bank fraud, loan, credit card and money laundering looked at. In this thesis, the credit card fraud is of interest.

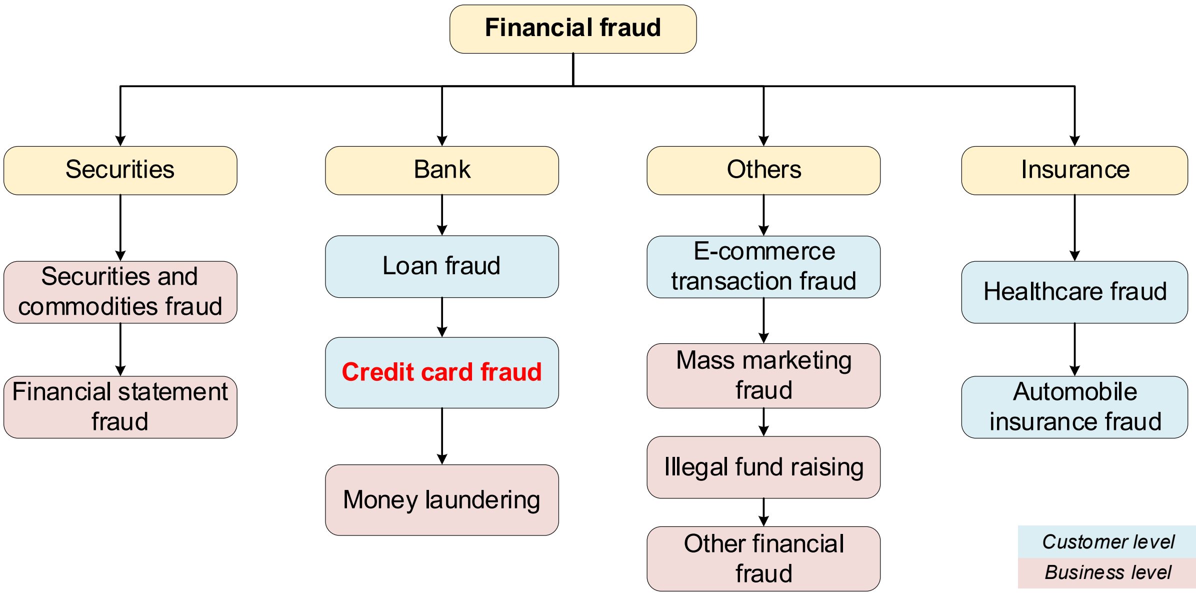

Fig. 1.6 The classification of financial fraud types (Zhu *et al*., 2021)

The significance of this research lies in the design and development of a hybrid neural network for the purpose of detecting credit card fraud. This network will have the capability of handling challenges connected with class imbalance, missing data, and real-time detection. The system that was built provides a low-cost local technology that can be adapted to detect fraudulent spending patterns of cardholders in Malaysia.

1.5 Thesis Organization

The following is an outline of this dissertation's organisation. In Chapter 2, a review of the previous research is carried out. The articles that were examined provide research on the identification of fraudulent activity on credit and payment cards, using both benchmark and real-world data sets.

The research design is elaborated upon in Chapter 3, detailing the standard algorithms used. The design and development of the BKS approach are also formulated within this chapter. Chapter 4 presents a series of systematic experiments using publicly available financial and credit card data sets. Various experimental setups were conducted, and their results were subsequently compared with existing literature.

The real-world financial data is evaluated in Chapter 5. Further modifications are made to the BKS, and more experiments are conducted. Results are then discussed. Conclusions are drawn in Chapter 6. The contributions that this research has made are discussed, and a number of potential directions for further research are recommended.

2 Literature Review

This chapter presents an overview of the literature on credit and payment card fraud detection. The data set has an extreme class imbalance because the number of fraud incidents is extremely modest in comparison to the number of genuine transactions. Most algorithms perform well if the sample numbers in each class is roughly equal, because the algorithms are designed to optimise accuracy while minimising error. Data imbalance, a prevalent challenge in fraud detection, can be overcome utilising sampling approaches.

Oversampling works by including more minority classes in the data. It is useful when there isn't a lot of data to work with. Undersampling eliminates part of the observations in the majority class. When there is too much data, this can be a useful option; however, valuable data may be deleted. This may result in the data set being underfitted.

A variety of measures are available to assess classifier performance. The confusion matrix is a popular example. True Negative (TN) reflects the number of regular transactions that were correctly identified as normal, whereas False Negative (FN) represents the number of fraudulent transactions that were incorrectly flagged as normal, resulting in missed fraud cases. True Positive (TP) are fraudulent transactions that have been correctly marked as fraudulent, i.e., detected fraud incidents, whereas False Positive (FP) are the number of normal transactions that have been incorrectly flagged as fraudulent. The Area Under the Curve (AUC) has been applied in a variety of fields. There are two forms of AUC in the literature. TP is plotted versus FP using the Receiver Operating Characteristic (ROC)-AUC. Precision is plotted against recall in the Precision Recall (PR)-AUC. Furthermore, the f-measure (or F1-score) is the harmonic average of precision and recall, with a maximum score of 1 (perfect precision and recall) and a minimum score of 0.

In the following sub-chapters, the literature review is divided as follows. The detection of fraudulent card transactions is first done. This is followed by ensembles and multi-classifier system (MCS).

2.1 Metrics

Metrics are quantitative assessment measures that are often used to evaluate or compare performance. There are various metrics used throughout the review, and a summary of the metrics are here given.

The most commonly used metric, is the accuracy. It is the measure of all correctly identified cases, and it is given as:

$$\mathrm{Accuracy} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{FP} + \mathrm{TN} + \mathrm{FN}} \tag{2.1}$$

True Positive (TP) and True Negative (TN) is a test result that correctly indicates the presence or the absence of a condition, respectively. False Positive (FP) and False Negative (FN) is a test results which wrongly indicates a particular condition is present or absent, respectively.

Precision is defined as the proportion of accurately detected positive cases among all anticipated positive cases, given as

$$\mathrm{Precision} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{2.2}$$

Recall is a measure of the proportion of correctly detected positive cases among real positive cases, given as

$$\mathrm{Recall} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{2.3}$$

The F1-score, or known also as F-score or F-measure is the harmonic mean of precision and recall. It can be calculated using

$$\mathrm{F1} = 2 \times \frac{\mathrm{Precision} \times \mathrm{Recall}}{\mathrm{Precision} + \mathrm{Recall}} \tag{2.4}$$

A model will receive a high F1-score is both precision and recall are high, and vice versa.

The use of programming language such as Python simplifies the calculation of the commonly used metrics. For example, the calculation of accuracy, precision, recall and F1-scores is listed in Algorithm 2.1. It shows that just a single line of code is sufficient for the calculation, as compared to using formulas when done manually.

Algorithm 2.1 Codes for metrics

```
from sklearn.metrics import accuracy_score, precision_score, recall_score,
f1_score

# assume y_true = truth labels and y_pred = predicted labels

# accuracy calculation
accuracy = accuracy_score(y_true, y_pred)

# precision calculation
precision = precision_score(y_true, y_pred)

# recall calculation
recall = recall_score(y_true, y_pred)

# F1-score calculation
f1 = f1_score(y_true, y_pred)
```

In addition to the commonly used metrics, calculation of various error rates is also made. Mean Absolute Error (MAE) gives the average of the absolute difference between the actual and predicted values in the dataset, and is calculated using

$$\mathrm{MAE} = \frac{1}{N}\sum_{i=1}^{N}|y_i - \hat{y}| \tag{2.5}$$

where $\hat{y}$ is the predicted value of y.

Mean Squared Error (MSE) gives the average of the squared difference between the original and predicted values in the data set, and is calculated using

$$\mathrm{MSE} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2 \qquad (2.6)$$

Root Mean Squared Error (RMSE) is the square root of MSE, as given in equation (2.6). It is given as

$$\mathrm{RMSE} = \sqrt{\mathrm{MSE}} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2} \qquad (2.7)$$

Coefficient of determination (R^2) provides the coefficient of how well the values fit compared to the original values, and given by

$$R^2 = 1 - \frac{\sum(y_i - \hat{y})^2}{\sum(y_i - \bar{y})^2} \qquad (2.8)$$

where $\bar{y}$ is mean value of y. Values of R^2 can be between 0 to 1, and closer the value is to 1, the better the model is. On the other hand, the lower the value of MAE, MSE, and RMSE, the better the model is.

2.2 *Ensemble Methods*

Ensemble indicates a group of models working together, which could be classifiers for function approximators. In general, the ensemble will leverage the capabilities of the underlying classifier models to generate a high-quality pattern recognition system that outperforms individual classifiers. Bagging, boosting, and stacking-based ensemble methods are examples of traditional ensemble methods. In this chapter, literature on various areas of ensemble methods, which include finance, engineering, medical, and other sectors is reviewed.

An example of ensemble methods is boosting, which is given in Fig. 2.1. Boosting is an iterative method to adjust an observation's weight based on the previous classification.

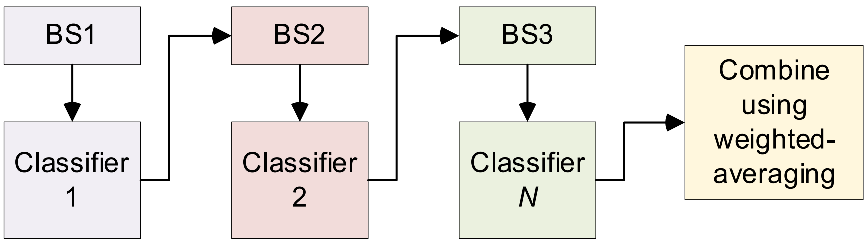

Fig. 2.1 The boosting algorithm (Bramer, 2007)

The boosting algorithm tries to improve prediction power using a training a sequence of weak models, where each model will compensate the weakness of the previous model. It is important to note that boosting is a general algorithm rather than a specific model, as it will require the use of an ML algorithm.

AdaBoost or Adaptive Boosting is a type of boosting algorithm. It combines multiple “weak classifiers” into a single “strong classifier”. An example of the implementation in Python is given in Algorithm 2.2. The sklearn library is used to generate the synthetic dataset and to build the classifier. Evaluation of the AdaBoost classifier is done using the accuracy score.

Algorithm 2.2 AdaBoost in Python

```
from sklearn.datasets import make_classification
from sklearn.ensemble import AdaBoostClassifier
from sklearn.model_selection import train_test_split

# generate a synthetic dataset
X, y = make_classification(n_samples=1000, n_features=10, n_informative=5,
n_classes=2, random_state=42)

# split the dataset into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,
random_state=42)

# create an AdaBoost classifier with a decision tree base estimator
clf = AdaBoostClassifier(n_estimators=50, random_state=42)

# fit the classifier on the training set
clf.fit(X_train, y_train)

# make predictions on the testing set
y_pred = clf.predict(X_test)

# evaluate the performance of the classifier
score = clf.score(X_test, y_test)
print("Accuracy:", score)
```

2.2.1 Finance Applications

A hybrid data mining model for credit scoring was presented in Koutanaei, Sajedi and Khanbabaei (2015). The hybrid model consisted of four feature selection methods and an ensemble model, with SVM as the base classifier. In the second stage, the Principal Component Analysis (PCA) algorithm achieved the best results. In the third stage, it showed that AdaBoost with SVM achieved the best accuracy rates, at 91.1% as compared to the other classifiers. Credit scoring using feature selection algorithm and ensemble classifiers is detailed in Nalić, Martinović and Žagar (2020). A total of 5 feature selection algorithms were used and the ROC with accuracy were combined based on voting. A total of 8 ensemble models were created using a combination of three classifiers, where a combination of GLM, SVM, Naïve Bayes (NB) and DT were taken. The best combination was that of GLM and DT, which outperformed all models with an accuracy rate of 82.01%.

A non-ferrous metal price prediction using an ensemble was devised in Guo *et al.* (2022). The ensemble incorporated data decomposition, phase space reconstruction, multi-objective optimization, point prediction, and interval prediction. The ensemble consisted of a kernel Extreme Learning Machine (ELM) based on the improved multi-objective lion swarm optimization algorithm. Using data from the London Metal Exchange, the aluminium data was experimented using various algorithms, and the proposed ensemble achieved the lowest Root Mean Squared Error (RMSE) at 13.9566 and 16.4382, for both experiments, respectively.

2.2.2 Engineering Applications

In fault diagnosis of rolling bearing with multi-source domains, a reinforcement ensemble is proposed in Li *et al*. (2022). New kernel maximum mean discrepancies (kMMDs) were used in constructing deep transfer learning ensemble. Using unified metric as a reward, the reinforcement learning method generates effective combination rules. The ensemble resulted in an accuracy of 92.4%, was compared to 75.83% to 87.38% in other methods. For fault detection of gearbox, a RF hybrid classifier is proposed in Tang *et al*. (2021). An improved Dempster–Shafer information fusion method is used in fusing results from the hybrid classifier. In various evaluations that were conducted, it showed that single faults could be detected from the compound faults effectively. In a number of experiments, perfect accuracy rates were achieved by the hybrid classifier.

In detecting crashes in real-time, a deep ensemble modelling approach is proposed in Taghipour *et al*. (2021). The first ensemble consisted of MLP and Deep Learning (DL) while the second ensemble consisted of RF and DL. The data source includes loop detector, crash information and weather information from 241 crash and 6038 non-crash cases. Experimental results show that the first and second ensemble model acquired AUC of 97.2% and 95.2% respectively, as compared to 93.7% using standalone DL. In wind speed forecasting, a novel ensemble model was introduced in Yang, Tian and Hao (2022). The ensemble forecasting model is designed by kernel-based ELM and multi-objective optimization. It included mixed-frequency data, which are used in predicting future wind speed changes in addition to the machine learning models. Experimental results indicate error rates of 4.4286%, 6.6154%, 5.2740% and 3.8682% for wind speed forecasting during the four seasons.

In obtaining safety factors and evaluating slope stability more accurately, a stacking ensemble learning model with Bayesian optimization is proposed in Sun *et al*. (2022). The ensemble model is built using DT, SVM and nearest neighbour. Optimal parameters are developed using a multi-objective mayfly optimization algorithm. Slope data from 210 groups which included a number of features were used. Results indicate a small MSE rate of 0.0916 from the ensemble model, as compared to the single models. For prediction of monthly electricity consumption in the economic sector, an ensemble learning approach is developed in Hadjout *et al*. (2021). A total of 3 models are combined, consisting of Long Short-Term Memory (LSTM), gated recurrent unit and temporal convolutional networks. The dataset contained 14 years of monthly electricity consumption from 2000 clients in Algeria. Experimental results show the proposed ensemble achieved better performance as compared to other classifiers, with the lowest RMSE at 67.42.

Prediction of the remaining fatigue life (RFL) of structures with stochastic parameters using an ensemble was proposed in Feng *et al*. (2022). Datasets associated with the structural responses were generated using the extended finite element method. The ensemble consisted of multiple algorithms, where details of the algorithms were not provided. Analysis results show that the ensemble was able to predict the RFL of structures accurately. Weather forecasts are discussed in Yagli, Yang and Srinivasan (2022) using an ensemble forecasting model. The ensemble model is developed using a dropout NN model with Monte Carlo sampling. To further improve the ensemble, a generalized additive model was applied. Four years of half-hourly data in the United states were used. Experimental results indicate up to 66% accuracy scores were achieved, and it could be useful to solar power plants for their forecasting.

Approach in classifying soil using machine learning models is approached in Eyo and Abbey (2021). The standalone models consisted of LR, NN, DT and ensembles (stacking and voting). Experiments show that the soil clay fraction mostly influenced the prediction while the carbonate content had the least influence in the dataset. Standalone models had the lowest performance of 45% accuracy as compared to the ensembles with 75% accuracy. The work of Qu *et al*. (2022) proposes a multi-parameter approach to fire detection that makes use of feature depth extraction and a stacking ensemble learning model. Fire feature parameters, such as temperature, smoke concentration and CO concentration, are collected as the original data. The stacking ensemble model is made up of LSTM, DT, RF, SVM, *k*-Nearest Neighbours (*k*NN) and XGB. Experimental results show an accuracy rate of 93% from the ensemble model.

Prediction of the sulphate resistance for recycled aggregate concrete using ensemble methods is approached in Liu *et al*. (2022a). Various ensemble methods, such as RF, AdaBoost, GB and XGB were used. 10 variables related to the material properties from 143 samples were used, with a 5-fold CV. Experimental results indicate the ensemble methods, especially the XGB predicts sulphate resistance with high accuracy and obtain superior performance over individual models. Icing faults in wind turbine is predicted using a combination of models in Xiao *et al*. (2022a). A total of three models were constructed with the Convolution Neural Network (CNN), such as the Recurrent Neural Network (RNN), LSTM, and gated recurrent unit. The group method of data handling was applied in cost-sensitive ensemble. Two wind turbine datasets collected using a data acquisition system were used. The best accuracy rate from the experiments were at 90.93%.

A slope prediction method using margin distance minimization selective ensemble, MDMSE, is proposed in Zhang *et al*. (2022a). A large number of differentiated individual learners are built by the method of data sample and algorithm parameter perturbation, followed by searching the optimal subset of individual learners. Using 422 groups of slope samples, the performance of MDMSE is compared with individual models. Experimental results show that MDMSE acquires an accuracy rate of 88.10% and F1-score of 90.38%, higher than other models. The prediction of driver injury outcomes in rear-end crashes is done using a hybrid classifier in Chen *et al*. (2016). The hybrid classifier was made up of a decision table and NB classifier, which can select the deterministic attributes. Attributes including weather, lighting conditions, driver behaviour were found to be significant. Estimations done were used in developing counter-measures to reduce rear-end crash injuries and improve traffic system safety. Experimental results showed an accuracy rate of 74.06%.

A novel intrusion detection method using a hybrid model is proposed in Song *et al*. (2021) for smart grids. The hybrid model consisted of LSTM and XGB, where results are fused based on accuracies of both models. Optimization of parameters is done using the Bayesian method. The experimental show best accuracy rate of 89.21%. In classifying bulking states in biological wastewater treatment plants (WWTPs) effluent, a hybrid DL-based method that uses a sparse constraint stacked autoencoder (SAE) model is proposed in Safder *et al*. (2022). The weights in SAE were adjusted using an ELM technique. Deep-belief network was applied in extracting feature space of ELM to classify the current state of secondary classifier. Experimental results show accuracy rate of 90.7% with precision of 91%.

Correlation of structural performance and mechanical aspects of carbon nano-tubes composites using ensemble models were proposed in Bagherzadeh and Shafighfard (2022). The flexural and compressive strength of the tubes were predicted using RF and GBM in Python. The best accuracy was achieved by GBM at 86% while RF received slightly lower score at 82%. In predicting the resilient modulus of pavement subgrade soil, ensemble techniques were used in Kardani *et al*. (2022). Datapoints from 12 compacted soils were collected, and a number of ensemble methods such as meta-model, stacking and bagging were developed. The ensemble model with the best predictive performance was the bagging ensemble, which had a RMSE of 0.0190.

The prediction of energy consumption in smart homes was performed using four standard ML models and ensemble in Priyadarshini *et al*. (2022). The ensemble was made up of DT, RF, and XGB, which was then compared with the standard ML models. Using multiple datasets, the results show that the proposed ensemble outperformed all other models with a R^2 score of 0.99 and RMSE of 0.00072. A framework using ensemble climate and ML models to assess the impact of hydropower generation at the Nangbeto plant was developed in Obahoundje *et al*., 2022. The dataset was based on climate models output data, which produced a future generation from 2020 to 2099. The ensemble was developed using 8 different types of ML algorithms, from bagging to linear regression. Results indicate the model performs well with R^2 score of 0.87, while for inflow, the R^2 score was 0.82.

An ensemble method based on MLP, Radial Basis Function (RBF), Self-Organizing Maps (SOM) and Cascade Forward Back Propagation (CFBP) with crest regression was developed in Carneiro *et al*. (2022). The model was then applied to wind and solar forecasting at two locations with different climate profiles. Using data from Brazil and Spain, the ensemble achieves the Mean Absolute Percentage Error (MAPE) values of 14.191% and 11.261%. It is shown that the ensemble reduces forecast errors and it is important in optimizing intermittent solar and wind resources. In mitigating the negative impacts of cracks in rotating machines, a study by Zhong and Ban (2022) proposed an ensemble model. The ensemble model consisted of RF and an AdaBoost tree, and was compared with a regular decision tree. The dataset consisted of bearing and gear system faults. Experimental results show that as compared with the single model, the ensemble model accuracy was up to 4% higher. The same is seen when noise is added into the data.

To predict heating value of biomass, 3 different ensemble algorithms were used in Dodo *et al*. (2022). There are 2 stages in the algorithm, where in the first stage, 4 individual model were used in predicting the heating value. In the second stage, the outputs from the models were aggregated for the ensemble learning. Inputs to the model consisted of 7 combinations of biomass proximate analysis components. Results from the ensemble show a low MSE rate of 0.0032, which is up to 15% lower as compared to individual models. Warpage prediction using ensemble models were developed in Ahmed *et al*. (2020) for injection moulded Polyvinyl chloride components. RF and Gradient Boosted (GB) trees were used, in which 40 experiments were carried out. To identify the most significant inputs, ANOVA was used. Results shows that MAPE of RF is much lower at 3.25% as compared to GB tree at 9.37%. The model allows production managers to regulate the warpage with minimum waste.

The tuning of parameters in a power system stabilizer using an ensemble method was discussed in Shahriar *et al*. (2022). The ensemble combines ELM, neurogenetic system and a multi-gene genetic programming. Two systems were used, which were infinite bus power and unified power flow controller. Experimental results of the gain parameters (K) and time-constant parameters show a RMSE of 0.1021 and 0.0003, respectively. In improving tool condition monitoring systems for new machine conditions, an ensemble technique was devised in Schueller and Saldaña (2022). Milling tool life experiments were done with numerous machine conditions and signals were collected. Machine tool wear levels were collected and evaluated using 4 individual ML and 5 ensemble models. The experimental results show that the extremely randomized trees ensemble model achieved the best accuracy score of 98.9%, using 10-fold CV and 92.4% using leave-one-group-out CV.

Fourier transform infrared spectroscopy is a useful method in detecting microplastics, according to Yan *et al*. (2022). In their work, they utilized individual and ensemble ML models for automated detection of microplastics. The available datasets are imbalanced; hence the quality of the classification may reduce. The ensemble model used took advantage of the individual models in improving the overall score. Using the Kedzierski and Jung datasets, the accuracy rates achieved were 94.19% and 94.5%, respectively, which was much higher as compared to individual ML models. In detecting faulty waters during semiconductor manufacturing process, an approach for fault detection and classification method was proposed in Mellah *et al*. (2022). The approach was based on a number of ensemble techniques, which includes RF, GB and XGB. Most important features from each time-series were extracted to build concatenated features. Results show that the XGB model classifies the best, with an accuracy of 98.79%.

2.2.3 Medical Applications

In predicting overall cognitive function of schizophrenia, a bagging ensemble framework was established in Lin, Lin and Lane (2022). An analysis of cognitive domain and test scores of 302 schizophrenia patients in Taiwan was done. The ensemble framework was compared with LR, SVM, and RF. Among the cognitive domains and tests, the overall cognitive function in schizophrenia was best predicted using the proposed bagging ensemble framework. An ensemble technique was used in screening tuberculosis using Chest X-ray images in Dey *et al*. (2022). Base learners are built using fully connected and softmax layers. Decisions from 3 base learners are combined using type-1 Sugeno fuzzy integral-based ensemble. Experimental results indicate the highest classification accuracy at 99.75%.

The work in Nanglia *et al*. (2022) presents a heterogeneous ensemble machine learning approach, to detect breast cancer in the early stages. A stacking ensemble model was built using *k*NN, SVM, and DT. The performance was then compared with single classifiers of LR, Artificial Neural Networks (ANN), NB, Stochastic Gradient Descent (SGD), and RF. A total of five features were derived using Chi-Square. Proposed ensemble model provides best accuracy rate of 78% with p-value of 0.014. An ensemble method based on multiple objective particle swarm optimization for electroencephalogram (EEG)-based emotion recognition is developed in Li *et al*. (2022). Two public datasets were used, where 13 features were extracted from the EEG signals to create a feature vector. The average accuracies for arousal and valence are 65.70% and 64.22%, respectively for the DEAP database while it was at 84.44% on the SEED database.

Process of detecting COVID-19 infection from Lung Computerized Tomography (CT) scan images was done in Shaik and Cherukuri (2022). A total of 2 datasets from patients infected and not infected were used. The ensemble model works on aggregating multiple DL models. Experimental results show that accuracy rate of 98.99% was achieved on the SARS-CoV-2 dataset while accuracy of 93.33% on the COVID-CT dataset. In classifying Trichotillomania (TTM) and obsessive-compulsive disorder (OCD), a hybrid approach was employed in Erguzel *et al*. (2015). A quantitative electroencephalography (QEEG) ordnance values with

19 electrodes from 10 brain regions in 4 frequency bands was used. The hybrid method used ANN, SVM, *k*NN and NB in classifying 39 TTM and 40 OCD patients. Experimental results show the hybrid approach achieves an overall accuracy of 81.04%, up from 67.12% from single classifiers.

A hybrid classifier, BSMAIRS, for estimating brain metastasis from lung cancer is developed in Wang *et al*. (2015). BSMAIRS combines borderline synthetic minority oversampling technique (BSM) and artificial immune recognition system (AIRS) with the nearest neighbour algorithm used as a local classifier. The dataset of lung cancer metastasis to the brain collected from Taiwan was used in the experiments. Accuracy rates of 92.66% was achieved by BSMAIRS as compared to 83.64% to 89.08% using other classifiers. In characterizing coronary plaque regions in intravascular ultrasound (IVUS) images, a hybrid ensemble classifier was proposed in Hwang *et al*. (2018). GA was used in selecting the optimal feature set. The hybrid ensemble consisted of feed-forward NN, RF, and AdaBoost. Using a 10-fold CV, the hybrid ensemble model achieved best accuracy of 81% and 75% for the two datasets used.

Improvement diagnosis of melanoma and dysplastic lesions using a hybrid classifier is proposed in Zakeri and Hokmabadi (2018). Features are extracted using intensity variations within the lesion of pigment distribution and texture features. The hybrid classifier consisted of log-linearized Gaussian mixture neural network, *k*NN, Linear Discriminant Analysis (LDA), and SVM. Using 792 dermoscopy image, the recorded diagnostic accuracies were 96.8%, 97.3%, and 98.8% for melanoma, dysplastic, and benign lesions, respectively.

A hybrid classifier for discriminating Fourier transform infrared (FTIR) spectrum data was proposed in Kanwal, Mehmood and Butt (2021). The hybrid classifier consisted of partial least squares and SVM, with a number of SVM kernel-based variants used. CV was used to avoid overfitting. Using the various datasets, experimental results show the hybrid classifier achieved accuracy rates of 100% for coffee, 92% for ILS, and 98% for olive oil data sets.

In assisting the disabled, brain-computer interfaces (BCI) frequently uses electroencephalogram-based motor imagery. Pawar and Dhage (2023) proposed an ensemble learning method based on extreme learning machine, which offers fast learning speed for classification of motor imagery data. Accuracy rates of pruning and without pruning are at 63.67% and 57.13%, respectively, which are higher as compared to other techniques. An early prediction of diabetes mellitus using different ensemble models was developed in Ganie and Malik (2022). Synthetic minority oversampling was used for balancing the class while k-fold CV was used in the experiments. A feature engineering process was done for the lifestyle parameters. Among the various ensemble methods, the bagged decision tree received the best accuracy rate at 99.41% and AUC of 99.07%.

In exploiting structural features of neuroimaging data for the Alzheimer's disease, an ensemble framework was proposed by Nguyen *et al*. (2022). The framework combined a deep learning model based on 3D-ResNet and an XGB algorithm. Predictions from the ensemble was combined with patient demographics and cognitive test scores, which results in a final diagnosis prediction. MRI brain images of the Alzheimer's disease were utilized. Using a 5-fold CV, the AUC score during the testing phase was at 96%. To determine the risk of sepsis-associated acute knee injury, an ensemble model was developed in Zhang *et al*. (2022). Data from patients with sepsis in the US was used for the modelling. The ensembled was designed using SVM, RF, NN and XGB using a stacking algorithm. Experimental results show the AUC values were between 0.774 to 0.788, which demonstrated good performances as compared to other methods.

A hybrid ensemble model to identify lung and colon cancer was introduced in Talukder *et al*. (2022). The extraction includes deep feature extraction with filtering for cancer image datasets. The ensemble consisted of SVM, LR and MLP models. Experimental results indicate the model can detect lung, colon and both together with accuracy rates of 99.05%, 100%, and 99.30%, respectively. Results demonstrate the proposed model outperforms existing model significantly. To enhance CT images, Contrast Limited Histogram Equalization was applied by Islam and Nahiduzzaman (2022). Based on the CT images, a Convolutional Neural Network (CNN) was used in extracting 100 features from each image. The ensemble consisted of Gaussian NB, SVM, DT, LR and RF, where the outputs were combined using a voting method. The proposed ensemble achieved an accuracy rate of 99.73%, which outperformed other models.

The analysis of categorical and numerical features for heart disease prediction is done in Pan *et al*. (2022). Ensemble algorithms are used in the analysis, where it is composed of the base classifier of SVM, DT, LR, RF, NN together with GB, XGB, AdaBoost and CatBoost. A soft voting ensemble is also used in the prediction. Using the Cleveland heart disease dataset, SVM and AdaBoost acquired the best accuracy rates of 85.24% and 83.60%, respectively. The Ensemble of SVM and AdaBoost then provided the optimum performance. A number of steps was taken for the diagnosis of coronary artery disease in Kolukisa and Bakir-Gungor (2022). The first, 7 computational and a domain knowledge-based feature selection method was first used. This was followed by an ensemble method, which was made up from a selection of 6 classification algorithms and a choice of 4 voting algorithms. The base classifier of MLP received the best accuracy rates of 91.78%, 85.55%, and 85.47% accuracy for the Z-Alizadeh Sani, Statlog, and Cleveland data sets, respectively.

In the prediction of compressive strength of construction demolition waste geopolymers, various ensemble methods were employed in Shen *et al*. (2022). The methods used include RF, GB and XGB. A total of 8 input features based on the compressive strength were analysed. Performances of XGB were the best, with RMSE of 4.176, followed by GB at 4.442. RF came in last with the highest RMSE at 5.032.

2.2.4 Other Applications

Zhao *et al*. (2020) offer WHMBoost, a weighted hybrid ensemble approach for categorising imbalanced data in binary classification applications. Two data sampling methods and base classifier are combined, where each is given corresponding weights. WHMBoost performance was tested on a total of 40 benchmark imbalanced datasets. Results from the experiments show that WHMBoost is effective when dealing with imbalance data. In benchmark sentiment analysis, heterogeneous ensembles are constructed and comparatively evaluated in Kazmaier and van Vuuren (2022). A greedy approach where *k* base learners with largest mean test accuracy were selected, and *k* is arbitrarily set to 5. Performance improvements up to 5.53% were observed in the experiments.

An ensemble method for multi-label drifting streams using self-adjusting nearest neighbour subspaces is proposed in Alberghini, Junior and Cano (2022). It uses self-adjusting *k*NN with advantage of ensembles for concept drift in multi-label environment. Each base classifier is provided with unique subset of features for training. Experimental study shows the proposed method achieves better performances as compared with 30 other classifiers. A prediction method for carrot mass based on machine vision and machine learning algorithms was discussed in Xie *et al*. (2022). A stacked ensemble model is built using a number of machine learning models. Six features of each carrot were extracted using image processing techniques, from a total of 1254 carrots. Using a 5-fold CV, experimental results showed the proposed stacked ensemble model to be accurate and an efficient method for grading and packaging carrots.

Ensemble of deep learning framework for text classification was proposed in Mohammed and Kora (2021). A meta-learning ensemble method that fuses baseline deep learning models using 2-tiers of meta-classifiers. Public benchmark text classification datasets were used. Experimental results show the ensemble approach enhances performance of the baseline classifiers, and even outperformed stacking and weighted vote ensemble methods. For construction of an unsupervised anomaly detection ensemble, an Item Response Theory (IRT) was detailed in Kandanaarachchi (2022). Using IRT mapping, an ensemble that can downplay noisy, non-discriminatory methods is constructed. Benchmark datasets were used in the experiments. Results from the experiments show the IRT ensemble performs well even when the detection methods have low correlation values, with AUC rate of up to 0.1 higher achieved as compared to regular models.

For multi-class classification tasks, hybrid mining models are introduced in Farid *et al*. (2014). The hybrid models consisted of DT and NB classifiers. DT induction is used in selecting important subset of attributes. Experiments were conducted using the hybrid model and individual models, with a 10-fold CV on benchmark datasets from UCI. The best average accuracy rate of 88.3% was achieved from the hybrid model, as compared to 83.5% using DT alone. To detect hate speech, Khanday *et al*. (2022) used ML and ensemble techniques during Covid-19. Data from Twitter was extracted using trending hashtags, via API. The extracted tweets were then manually annotated into 2 categories from different factors. In the standard ML techniques, the DT classifier was found to be effective. The best classifier was the Stochastic GB which outperformed all others with a 98.04% rate.

An electronic nose approach to beef quality classification was proposed in Wijaya *et al*. (2022). Ensemble methods are used in two parts, the first on the sensor array optimization. A total of 3 filter-based feature selection methods are used to build the first ensemble. Then several ensemble algorithms are used. Classification results indicate that AdaBoost received the highest accuracy at 99.90% and 99.82% when using all the sensors and optimized sensors, respectively. For the detection of falsification attack in cognitive radio network, an ensemble model was proposed by Kumar and Reddy (2022). The ensemble is based on a hierarchical extreme learning model that is used in analysing security threats in the network. Different attacks from random yes, random no and random false were considered. MATLAB was used for the simulation and compared with existing models such as deep NN, RNN and SVM. The proposed model achieved the best accuracy rate of 99.7%.

Strength prediction of iron ore tailings when used as supplementary cementitious materials was researched by Cheng, Yang and Zhang (2022). A total of 4 ensemble models were used in the prediction. The impact of material strength influence was done using Shapley additive explanations. All the R^2 values of the models were above 0.83 during the test phase. Based on the various ensemble models, the GB regression tree had the lowest

RMSE value of 4.86. Factors that affect the intention of using a third-party logistic provider during the Covid-19 pandemic was studied in German *et al*. (2022). An ensemble consisting of ANN and RF was constructed. Findings show that attitude is a critical factor affecting the consumers' behaviour intention. The best results achieved was an accuracy of 98.56%.

Ecological aspects which include land and water are of interest using ensemble models. In explaining social ecological aspects of the land use and cover at the Okavango basin, the study by Kavhu, Mashimbye and Luvuno (2022) utilized ensemble models. Climate-base data from the year 2002, 2013 and 2020 were used in the experiments. A total of 5 ensemble models were used, which included RF and GBM. Experimental results indicate ensembles performs better with an AUC of 0.964 as compared to individual models. The study also indicated that wetlands to woodlands will gradually decrease in the future. For spatial mapping of gully erosion susceptibility in Hinglo river basin, Roy and Saha (2022) presented the use of an ensemble. The ensemble consisted of an RBF with random sub-space and rotation forest. A total of 120 gullies were grouped into a 4-fold, which includes 23 different properties. The proposed ensemble accuracy increased up to 3% with the ensemble, as compared to individual classifiers.

Forecast of future drought and simulated impact over crop yield in South Asia was studied by Prodhan *et al*. (2022). The study looked at a number of climate models. Structure of the ensemble was made up of RF and GBM. Experiments showed that higher and longer drought is projected in some of the regions. The proposed ensemble had a minimum RMSE values of 0.358 to 0.390, which is much lower if compared to RF and GBM used independently. In modelling and mapping soil nutrient reduction in the highlands of 10 Rwandan agro-ecological zones, the authors in Uwiragiye *et al*. (2022) utilized ensemble ML methods. The calculated soil nutrient balances and environmental covariates in the Rwanda were used in calibrating the nutrient depletion model, using ensemble models. A 10-fold CV was used in the experiments. Experimental results indicate the R^2 value of the models were 62%, which shows an improvement up to 5% as compared to other models.

Soil organic carbon sequestration is a method for capturing CO2 from the atmosphere while providing benefits in improving soil functions. The work in Xiao *et al*. (2022) looked at improving pedotransfer functions to predict soil mineral using ensemble ML methods. Using 352 mineral topsoil samples across Europe, prediction is done using an ensemble consisting of RF, GBM with predictor selection such as recursive feature elimination. The feature selection reduced the number of predictors to 12, from 21. Results of the experiments were R^2 of 0.877 to 0.9 with a RMSE value of 2.994 to 3.269. Classification of soil using ensemble using individual and ensemble models was done in Eyo and Abbey (2022). Among ML models used include LR, ANN, DT, stacking and voting ensembles. Experiments show that soils' clay fraction was the biggest influencer on the prediction while specific surface and carbonate content was the least. The LR and ANN models achieved accuracy rates at 45%, tree-based at 68% while ensembles achieve the highest at 75%.

In preparing spatial groundwater potential maps of Malda district in India, Das and Saha (2022) proposed a hybrid ensemble. A number of models were used, which include ANN with bagging, RF and SVM. Data from 93 well and 17 groundwater locations were selected. The spatial maps show that only 23% of the area has high potential for groundwater. Experimental results show that the ANN with bagging ensemble had the highest AUC rate, at 0.96 followed by the other individual models.

2.2.5 Summary of Ensemble Methods

A summary of the ensemble methods is given in Table 2.1. It lists the type of applications being looked at, including the metrices used for the results. A number of literatures without any reported results were removed. It can be seen that accuracy is the most commonly used metric.

Table 2.1 Summary of ensemble methods

Authors	Application	Results
Koutanaei, Sajedi and Khanbabaei (2015)	Credit scoring	Accuracy 91.1%
Nalić, Martinović and Žagar (2020)	Credit scoring	Accuracy 82.01%
Guo *et al.* (2022)	Non-ferrous metal price prediction	RMSE 13.9566
Li *et al*. (2022)	Fault diagnosis of rolling bearing	Accuracy 92.4%
Tang *et al*. (2021)	Fault detection of gearbox	Accuracy 100%
Taghipour *et al*. (2021)	Detecting crashes in real-time	AUC of 95.2-97.2%
Yang, Tian and Hao (2022)	Wind speed forecasting	Error rates 3.86-6.61%
Sun *et al*. (2022)	Safety factors and evaluating slope stability more accurately	MSE 0.0916
Hadjout *et al*. (2021)	Prediction of monthly electricity consumption in the economic sector	RMSE 67.42
Yagli, Yang and Srinivasan (2022)	Weather forecasts using an ensemble forecasting model	Accuracy 66%
Eyo and Abbey (2021)	Classifying soil	Accuracy 75%
Qu *et al*. (2022)	Multi-parameter approach to fire detection	Accuracy 93%
Xiao *et al*. (2022a)	Icing faults in wind turbine	Accuracy 90.93%
Zhang *et al*. (2022a)	Slope prediction method	Accuracy 88.10%, F1-score 90.38%
Chen *et al*. (2016)	Prediction of driver injury outcomes in rear-end crashes	Accuracy 74.06%
Song *et al*. (2021)	Novel intrusion detection for smart grids	Accuracy 89.21%

Safder *et al*. (2022)	Classifying bulking states in biological wastewater treatment plants	Accuracy 90.7%, Precision 91%
Bagherzadeh and Shafighfard (2022)	Correlation of structural performance and mechanical aspects of carbon nano-tubes composites	Accuracy 86%
Kardani *et al*. (2022)	Predicting the resilient modulus of pavement subgrade soil	RMSE 0.0190
Priyadarshini *et al*. (2022)	Prediction of energy consumption in smart homes	RMSE 0.00072, R^2 0.99
Obahoundje *et al*. (2022)	Impact of hydropower generation at the Nangbeto plant	R^2 0.82-0.87
Carneiro *et al*. (2022)	Wind and solar forecasting	MAPE 11.261-14.191%
Zhong and Ban (2022)	Impacts of cracks in rotating machines	Accuracy 4% higher
Dodo *et al*. (2022)	Predict heating value of biomass	MSE 0.0032
Ahmed *et al*. (2020)	Warpage prediction for injection moulded Polyvinyl chloride components	MAPE 3.25%
Shahriar *et al*. (2022)	Tuning of parameters in a power system stabilizer	RMSE 0.0003-0.1021
Schueller and Saldaña (2022)	Improving tool condition for new machine conditions	Accuracy 98.9%
Yan *et al*. (2022)	Automated detection of microplastics	Accuracy 94.19-94.5%
Mellah *et al*. (2022)	Detecting faulty waters during semiconductor manufacturing process	Accuracy 98.79%
Dey *et al*. (2022)	Screening tuberculosis using Chest X-ray images	Accuracy 99.75%
Nanglia *et al*. (2022)	Detect breast cancer in the early stages	Accuracy 78%
Li *et al*. (2022)	EEG-based emotion recognition	Accuracy 84.44%
Shaik and Cherukuri (2022)	Detecting COVID-19 infection from Lung CT scan images	Accuracy 93.33%
Erguzel *et al*. (2015)	Classifying TTM and OCD	Accuracy 81.04%
Wang *et al*. (2015)	Estimating brain metastasis from lung cancer	Accuracy 92.66%
Hwang *et al*. (2018)	Coronary plaque regions in IVUS images	Accuracy 75-81%

Zakeri and Hokmabadi (2018)	Improvement diagnosis of melanoma and dysplastic lesions	Accuracy 96.8-98.8%
Kanwal, Mehmood and Butt (2021)	Discriminating FTIR spectrum	Accuracy 92-100%
Pawar and Dhage (2023)	Classification of motor imagery data in assisting the disabled	Accuracy 63.67%
Ganie and Malik (2022)	Prediction of diabetes mellitus	Accuracy 99.41%, AUC 99.07%
Nguyen *et al*. (2022)	Exploiting structural features of neuroimaging data for the Alzheimer's disease	AUC 96%
Zhang *et al*. (2022)	Risk of sepsis-associated acute knee injury	AUC 0.774-0.788
Talukder *et al*. (2022)	Identifying lung and colon cancer	Accuracy 99.05-100%
Islam and Nahiduzzaman (2022)	Enhance CT images	Accuracy 99.73%
Pan *et al*. (2022)	Analysis of categorical and numerical features for heart disease prediction	Accuracy 83.60-85.24%
Kolukisa and Bakir-Gungor (2022)	Diagnosis of coronary artery disease	Accuracy 85.47-91.78%
Shen *et al*. (2022)	Prediction of compressive strength of construction demolition waste geopolymers	RMSE 4.176
Farid *et al*. (2014)	Multi-class classification tasks	Accuracy 88.3%
Khanday *et al*. (2022)	Detect hate speech during Covid-19	Accuracy 98.04%
Wijaya *et al*. (2022)	Electronic nose approach to beef quality classification	Accuracy 99.82-99.90%
Kumar and Reddy (2022)	Detection of falsification attack in cognitive radio network, an ensemble model	Accuracy 99.7%
Cheng, Yang and Zhang (2022)	Strength prediction of iron ore tailings	RMSE 4.86
German *et al*. (2022)	Intention of using a third-party logistic provider during the Covid-19	Accuracy 98.56%
Kavhu, Mashimbye and Luvuno (2022)	Social ecological aspects of the land use and cover at the Okavango basin	AUC 0.964
Roy and Saha (2022)	Spatial mapping of gully erosion susceptibility in Hinglo river basin	Accuracy 3% higher

Prodhan *et al*. (2022)	Forecast of future drought and impact over crop yield	RMSE 0.358-0.390
Uwiragiye *et al*. (2022)	Mapping soil nutrient reduction in the highlands of 10 Rwandan agro-ecological zones	R^2 0.62
Xiao *et al*. (2022)	Improving pedotransfer functions to predict soil mineral	R^2 0.877-0.9, RMSE 2.994-3.269
Eyo and Abbey (2022)	Classification of soil	Accuracy 75%
Das and Saha (2022)	Preparing spatial groundwater in India	AUC 0.96

2.3 Multi-Classifier System

MCS indicates a group of classifiers, which is more restrictive as compared with ensemble. It frequently incorporates a decision combination approach for combining predictions from a classifier ensemble. Many applications based on MCS models have been created throughout the years. In this chapter, literature on various areas of MCS applications, which include finance, engineering, medical, and other sectors is reviewed.

A standard architecture of an MCS is given in Fig. 2.2. It is made up of *N* classifiers and a function for parallel combination of the classifier outputs.

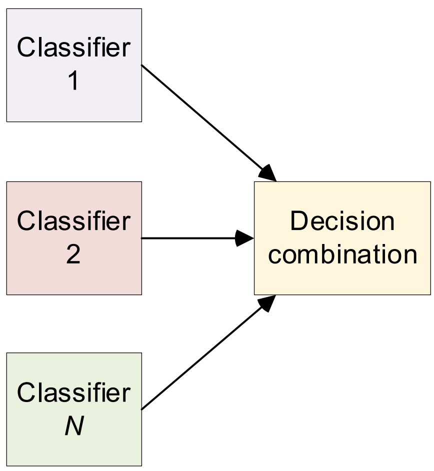

Fig. 2.2 Multi-classifier system (Roli, 2009)

2.3.1 Finance Applications

Wang, Tong and Wang (2020) developed an advanced warning system for outward foreign direct investment based on 42 Chinese enterprises utilising an MCS approach. The MCS model, which was applied to resource-based firms in China, used logit regression, SVM, NN, and decision trees. The experimental results indicated the MCS model was able to yield an accuracy score of 85.088%, as compared to 80.526% to 84.561% from other classifiers (Wang, Tong and Wang, 2020). A decision support system for trading in the financial market was presented in Thakur and Kumar (2018). An MCS model with a weighted multicategory generalized eigenvalue SVM and RF was used to generate the buy or sell signals. When tested against five index returns, including those from NASDAQ and Dow Jones, the MCS model outperformed the buy/hold strategy and other algorithms by a wide margin. The model made positive rate of return in majority of the data sets.

Ensemble models are useful for bankruptcy prediction. In the work of Du Jardin (2018), using a data set managed by Bureau van Dijk, Diane in France for balance sheets on firms, ensemble-based models were created, with 2 to 200 models on bagging and boosting. The results yielded accuracy rates of 81.13% to 81.81% for the prediction on health status of the companies. An anomaly detection in credit card data with unbalanced and overlapped classes was studied in Kalid *et al*. (2020). Two datasets of credit card frauds and default payments were used. The MCS consisted of C4.5 tree with Naïve Bayes as both works together in classifying the various samples. Best accuracy rate of 99.9% was achieved using the proposed MCS.

The work of improving effect of credit risk assessment was done by Wang, Liu and Qi (2022). A new multi-classifier assessment model for personal credit risk using information fusion from 6 ML algorithms was proposed. The model can simultaneously take the advantage of multiple classifiers while reducing interferences of uncertain data. Data from a commercial bank from China was used, with the best accuracy rate of 84.15%, which was 4.79% to 15.28% better as compared to individual classifiers.

2.3.2 Engineering Applications

Yijing *et al*. (2016) established an adaptive MCS model for recognising oil-bearing reservoirs. A total of 5 classifiers were used, namely C4.5, SVM, RBF, data gravitation-based, and *k*NN algorithms. The adaptive MCS model also contained a number of rules. The proposed approach recognised the attributes of different layers in oil logging data with perfect accuracy. In improving classification accuracy for detection of steel plate surface defects, an MCS was proposed in Gong *et al*. (2019), which included multi-class support vector hyper-spheres with insensitivity to noise. The test results showed that susceptibility of the proposed method to noise, leading to improved classification accuracy and efficiency.

The severity of abnormal aviation events was predicted based on risk levels in Zhang and Mahadevan (2019) using an MCS framework consisting of SVM and DL models. Deep learning was employed in training and the SVM was used to uncover correlations between event synopsis and outcomes. The suggested MCS model achieved 81% accuracy using cross-validation, which is up to 6% higher than standalone SVM and DL models. An MCS model was used to estimate precipitation from satellite photos, which incorporated RF, NN, SVM, NB, weighted kNN, and *k*-means in Lazri *et al*. (2020). There were six types of precipitation intensities obtained, ranging from no rain to very heavy rain. The proposed method produced a coefficient of correlation score of 0.93, whilst other methods produced just 0.46.

Deep Learning (DL) has been used to devise a variety of MCS models. In Hüthwohl, Lu and Brilakis (2019), an automated inspection and objective bridge defect classification system was proposed. The MCS consisted of three DL pre-trained networks were fine-tuned based on images from the databases extracted from US department of transportation. Experimental results show that the MCS can accurately detect concrete surface patches with a F1-score of 83.5%.

An object-oriented satellite images extraction using MCS is proposed in Tan *et al*. (2021). The MCS is build using DL algorithms, with a splitting mechanism using tree structure. In the experiments, the MCS works well when the tracking standards are low and provides better classification accuracy and recall as compared to other methods. Rotating machinery fault diagnosis was performed using an MCS optimization Li *et al*. (2020). Similarly, the MCS was built using multiple DL algorithms, which were optimized to maximize the classification discrepancy of the unlabelled target data. Based on three rotating machinery data sets, the proposed model had a testing accuracy of 99.31% to 99.75% in undertaking cross-domain fault diagnostic issues.

In the fault diagnosis of gearbox and locomotive roller bearings, a complex wavelet packet transformed is first introduced in Qu, Zhang and Gong (2016). An MCS fusion is developed with neighbourhood rough set. The final diagnosis is done using Bayesian belief method. The results from the MCS is an accuracy of 100% with sensitive features and slightly lower at 93.59% using randomly selected features. In the imbalanced samples for bearing fault diagnosis, an MCS with weighted balanced distribution adaption was proposed (Chen *et al*., 2021a). The MCS consisted of SVM with random sampling. Accuracy rates of over 90% was achieved using the proposed method.

An MCS based on an extended classifier system is used in simulation a better manufacturing process organization (Hercog, 2013). MAXCS adapts to the various configurations of bars in the experiments. When MAXCS is used with GA, it determines if there is only one or several agents throughout the experiment. Experimental results concluded that the multibar problem provides manufacturing processes with good coordination solutions. The prediction of road classes on the OpenStreetMap platform using MCS is done in Pazoky and Pahlavani (2021). Fourteen parameters having the highest impact on classes of features were calculated. Various single classifiers were used with a 5-fold CV method. The best classification accuracy of the single classifier was at 93.55% by RF. The best MCS combination was using SVM and RF with Behaviour Knowledge Space (BKS), resulting in an accuracy of 97.19%.

For remote sensing image classification, a combination of DL and MCS was used in Dou *et al*. (2021). A framework of DL-based feature extractor and MCS models was used in the classification. The MCS was based on MLP classifiers. Using Landsat images in California for Sutter and Kings, the accuracy rates were 83.51% and 79.97%, much higher as compared with other methods. In change detection for high quality remote sensing images, a strategy using ensemble of MCS is proposed in Tan *et al*. (2016). Spatial information such as texture and morphological features are used in addition to spectral information. The MCS consists of extreme learning machine (ELM), multinomial logistic regression (MLR), and *k*NN. Results show the proposed method provided good accuracies as compared to other widely used methods.

Hyperspectral image for land cover classification using MCS methods was assessed in Damodaran and Nidamanuri (2014). Airborne hyperspectral images from five different sites covering range of land cover classified by MCS. As compared to individual classifier, SVM, the accuracy rate of MCS shows an overall 5% increase across the hyperspectral images and sites. Diagnosis of electrical motor drive system faults was presented in Ding *et al*. (2022). An MCS method based on SVM and hybrid Particle Swarm Optimization (PSO) and Gravity Search Algorithm (GSA) was proposed. A number of simulations were conducted. Results of the experiments indicated the proposed method of SVM with PSO and GSA acquired an accuracy of 95.10%, which is up to 2.55% higher with other methods.

In providing more insights using application service class, the work in Zaki *et al*. (2022) utilized an MCS technique. The service class looked into inter-application, which is similar service between different parent applications and intra-applications, such as services within same parent application, mainly referring to social media. The proposed MCS called GRAIN ties 2 RF together in producing a multi-label classification using 7 statistical features. Results from GRAIN show an average F-measure of 99%, 93% and 88% at the application, inter-application and intra-application levels, respectively. The work in Su *et al*. (2022) worked on a fault-Line selection method in grounded systems using MCS. Architecture of the MCS consisted of DL with denoising autoencoder, which reduced the dimensions of the dispatching data and extracting fault features. Fault segments were marked and labelled.

The proposed MCS attained an accuracy of 97.8%, which is greater than traditional approaches.

In motor imagery recognition, an MCS using a fully connected network with multi-classifier fusion, named FCN-MCF is proposed in Cheng and Gao (2022). The fully connected network samples transfer method is used in moving suitable samples from similar types with similar conditions. The classifiers were trained based on the target domain, in addition to the transferred domains. Based on a BCI competition IV dataset 2A, FCN-MCF acquired an accuracy of 72.52%, about 4.76% higher than the closest method. For the fault diagnosis in moving industrial equipment, an MCS was proposed by Rajabi *et al*. (2022). The MCS consisted of a multi-output Adaptive Neuro-Fuzzy Inference System (ANFIS). Vibration signals are utilized for performance of bearings, and the fault condition of the bearing is done using a permutation entropy. Once the fault is detected, the exact fault type is determined using further signal processing. Best accuracy rate of 98.6% was achieved by the proposed MCS.

2.3.3 Medical Applications

An ensemble of SVMs in a hybrid serial and parallel configuration was used in classification of neuromuscular disorders (Kamali, Boostani and Parsaei, 2013). An MCS system utilizing both time and time-frequency domain features from electromyographic signals was developed. The best average accuracy of 97% was achieved by the proposed system, as compared to 88% from single SVM classifier. An ensemble classification model using *k*NN and SVM was presented in Sun *et al*. (2020) to classify electrocardiogram (ECG) signals. The suggested model achieved an accuracy score of 0.752, compared to other classifiers' scores ranging from 0.561 to 0.737.

A deep belief network-based MCS model for detecting breast cancer was used in Beevi, Nair and Bindu (2017). Specifically, the method for detection of mitotic cells in Hematoxyline and Eosin stained images was conducted for a series of segmentation and classification stages. The proposed method was evaluated using the MITOS-ATYPIA data and clinical images from a regional cancer centre. F-score rates of 84.29% and 75% were achieved from the MITOS-ATYPIA and clinic data, respectively. An MCS model to maximise the diagnostic accuracy of thyroid detection was developed by Jothi and Rajam (2017). To get the highest diagnostic accuracy, the model used SVM, NB, kNN, and nearest matching rule classifiers. In automatically distinguishing thyroid histopathology images as normal thyroid or papillary thyroid cancer, the suggested system attained an accuracy of 99.54%, compared to 99.09% for the best individual classifier.

Lin (2012) investigated an adaptive MCS model for gene expression. The ensemble model uses particle swarm optimization, a bat-inspired method, and SVM to show significant gains in classification accuracy with respect to breast cancer and embryonal tumours, with the training error decreased by up to 50%. In Fredrickson *et al*. (2019), mortality prediction of trauma patients was conducted using the National Trauma Data Bank. An ensemble classifier consisting of a four-component Optimal Temporal Common Subsequence (OTCS) model and its modifications yielded the best result, i.e. an AUC of 0.8649.

Fatigue recognition identification based on electrocardiography (ECG) and electromyography (EMG) was studied in Chen *et al*. (2021b). PCA and Grey Relational Analysis were used as indicators. SVM-RBF is compared with XGB models. The results showed that XG-Boost performed the best, with a recognition accuracy of 89.47%, sensitivity and specificity of 100%, and an AUC of 0.90. In Gao, Turek and Vitaterna (2016), EEG and EMG recordings are used to study sleep patterns in rodents. An MCS consisting of decision tree, *k*NN, Naïve Bayes, SVM, ANN and linear discriminant analysis was created. Decision tree and *k*NN was further improved using bagging and random subspace. Using single classifier, SVM had the lowest error rate of 0.054 while using MCS, best result of 0.049 was achieved.

In Sanz *et al*. (2017), survival status prediction for trauma patients in hospitals in Spain was proposed. LR and C4.5 tree is combined to form an MCS. A 10-fold CV is then applied. A 10-fold CV model was applied on LR model and C4.5 tree for an MCS. The MCS results were sensitivity 89.08%, specificity 67.03%, geometric mean 76.61%, and p-values less than 0.01. Detection of Type-2 Diabetes Mellitus (T2DM) was done using a dynamic weighted voting scheme called multiple factors weighted combination (Zhu, Xie and Zheng, 2015). The MCS consisted of multiple *k*NN classifiers with the weighted voting scheme. The MFWC achieved the best accuracy, which is 2% better as compared to regular *k* sizes and outperforms individual classifiers.

To determine biological activity of molecules for chemical and physicochemical properties, a D2-MCS is proposed (Ruano-Ordás *et al*., 2019). The D2-MCS consists of feature clustering, optimizing hyper-parameters, and molecule classification. Various classifiers were used with the MCS, and model with RF had the best accuracy result of 91.34%. Results indicate the D2-MCS performed better as compared to single classifier. In classification of thyroid disease, the work in Zhang *et al*. (2022) utilized a multi-channel CNN. Alternative methods in enhancing the diagnostic accuracy of CNN using concatenation of different scales was done. Experimental results show the multi-channel CNN achieved an accuracy of 90.9%, which was 0.7% higher as compared to a single-channel. The proposed multi-channel CNN has potential to be used in a clinical setting.

To classify neuropsychiatric disorders, a modified multi-modal brain graph network was proposed in Liu *et al*. (2022b). Neuropsychiatric disorders can be linked with functions and structures of the brain. The multi-model graph convolutional network was developed for classification of healthy and neuropsychiatric disorders patients. The proposed method achieved accuracy rate of 93.71% using an open data set by the Neuropsychiatric Phenomics. Enhancement of DL classifiers with anatomical side markers for multi-centre trials was done in Kim *et al*. (2022). The markers were first detected using an EfficientDet detection network and imprinted using a generative adversarial network. A 5-fold CV showed an improvements trend, in the pre-processing stage. Network performance were compared using DeLong's test on AUC, with accuracies of 90% to 95% achieved.

To enhance classification in correlative microscopy, Bitrus *et al*. (2022) used MCS with dynamic selection. Various spectroscopy and microscopy samples were used for the test samples. As compared to individual classifiers, the proposed MCS achieved an AUC score of 0.997 and F1-score of 0.98 on the volcanic rock samples. Hybrid CNN classifiers were utilized in breast cancer detection in Sahu, Das and Meher (2023). The probability-based weight factor and threshold value is vital in making a good hybrid model. Datasets of breast cancer modalities, where the mini-DDSM for mammogram and BUS1 and BUS2 for ultrasound were used. The accuracy rates for the mini-DDSM datasets were 99.17% for abnormality and 98.00% for malignancy detection, while for the BUS1 and BUS2 datasets, it was 96.52% and 98.13% for malignancy detection.

2.3.4 Other Applications

The studies in Mendialdua *et al*. (2015), Kang, Cho and Kang (2015) and He *et al*. (2020) used data sets from the UCI machine learning repository for evaluation of MCS models. Mendialdua *et al*. (2015) created an MCS employing stacked generalisation based on DT, kNN, and NB. The studies used a total of 20 different UCI data sets. The MCS model attained an accuracy rate of 74.83% on a breast cancer data set, compared to 71.15% for other classifiers.

Kang, Cho and Kang (2015) investigated a one-against-one technique with MCS that included NN, DT, kNN, SVM, linear discriminant analysis, and logistic regression. The MCS model produced an error rate of 0.99%, compared to 14.85% for other approaches on the zoo data set. A multi-criterion fusion classifier framework was designed in He *et al*. (2020). A total of 10 UCI data sets and two additional clinical data sets were used in the study. Given the heart data, an accuracy rate of 0.83 was achieved by the proposed MCS model, as compared with 0.81 from stacking and 0.80 from standalone RF.

Jiang, Zeng and Zhang (2013) described an MCS framework for utilising unlabelled data. The MCS model was created with NB, SVM, and kNN. The experiments made use of five different text categorization data sets. When compared to other algorithms, the MCS model achieved the greatest accuracy rate of 83.3%. A multi-domain sentiment classification algorithm using a combination of SVMs was proposed in Li, Huang and Zong (2011). The SVM classifiers were combined to produce a final decision, leading to 27.6% reduction in the average error against the performance from standalone classifiers Li, Huang and Zong (2011).

Ferrara, Franco and Maio (2012) proposed an MCS model for facial picture segmentation. In the MCS model, three Bayes and one SVM were utilised. For hair across eyes requirements, an error rate of 13.87% was attained, compared to 50% for standard classifiers. An aggregate ensemble learning model which tolerates data errors is proposed in Zhang *et al*. (2011). Studies on both synthetic and real data streams were used in the experiments, that contained noise. Using random selection with noise, the proposed method achieved average accuracy of 82.3%.

Using an ensemble, Da Silva, Hruschka and Hruschka Jr (2014) automatically classify tweet sentiment as positive or negative. The experiment makes advantage of publicly available tweet sentiment databases. Multinomial NB, SVM, RF, and logistic regression are used to create the ensemble. On a dataset trained with only 0.03% of the collected data, an accuracy rate of 81.06% was obtained. An MCS fusion integrating probabilistic SVM and improved Dempster-Shafer (DS) evidence theory in supporting risk analysis under uncertainty is developed in Pan *et al*. (2020). A Monte Carlo simulation approach is used to model the randomness and uncertainty underlying limited observations. A tunnel case in China was used in assessing the structural health risk. Results from the MCS (SVM-DS) gives an accuracy of 87.99% as compared to only 63.84% from standalone SVM.

A technique based on MCS for Turkish sentiment classification is investigated in Catal and Nangir (2017). The MCS consisted of NB, SVM, and bagging. Using MCS, the performance of individual classifiers contributes to the overall increased accuracy of MCS. Experimental results show an accuracy rate of 96.13% from the proposed MCS. In matching heterogeneous faces such as in near-infrared images, a novel method using a geometric edge-texture feature is proposed in Roy and Bhattacharjee (2016). For classification, an MCS built using fuzzy partial least square and fuzzy discriminant analysis was used. Experimental results statistically prove that MCS performs better as compared to individual classifiers, with an accuracy rates of 99.66% and 99.50% in the two experiments.

An MCS framework based on random projection, WMCRP is proposed in Nguyen *et al*. (2019). Benchmark datasets from UCI are utilized, and least squares method is used in weighing the outputs of the base classifiers. Decision trees made up the WMCRP, and it was compared with bagging, boosting and random subspace. Experimental results show the proposed WMCRP got better results as compared with other ensemble methods. An MCS consisting of self-adaptive parameter and strategy-based particle swarm optimization (SPS-PSO) is proposed in Xue *et al*. (2020). Benchmark data from UCI is used in the experiments. Individual classifiers such as *k*NN, LDA, SVM is compared with the proposed

MCS. Experimental results show that the SPS-PSO MCS achieves the best accuracy in all test sets.

Using various classifiers of SVM, C4.5 and *k*NN and different parameters settings, a number of MCS are built in Sáez *et al*. (2013). The benchmark datasets from UCI is used in the experiments. To test the MCS, noisy data is used in the experiments. Based on the multiple MCS models built, the MCS3-*k* provided the best accuracy rates. The work in Poreddy and Gopi (2023) looked at improving classification accuracy of an MCS. The improvement is sought in an under-performing classifier using discrete memoryless channel model and PSO. Both Monte-Carlo simulated synthetic data set and a real data set was used in the experiments. The classification accuracy increased by up to 5% in most of the cases.

A refinement of the feedforward network using a hybrid supervised clustering and fuzzy classifier (CRSCF) was done by Fontes (2022). The refinement of accuracy was based on the single feedforward NN (SFNN) to recognize multiple static data patterns. Accuracy rates of the proposed CRSCF was compared with SFNN, where for the vehicle datasets, it was 84.0% and 81.9%, respectively, breast tissue 77.4% and 74.7%, respectively and the Ecoli dataset at 88.3% and 88.9%, respectively. Fault diagnosis of proton exchange membrane fuel cell systems was conducted in Lu *et al*. (2022). A simultaneous algorithm called multi-label classification network is proposed. A number of experiments using a high-power system, having the same boundary conditions in the simulation model was used. The subset accuracy of the proposed network was 97.3%.

In detection of ransomworm, a multi-feature and MCS-based system was proposed by Almashhadani *et al*. (2022). A full behavioural analysis of the ransomworm network traffic and the NotPetya case study was used. Experimental results indicate a high accuracy, where it achieved 99.88% for session-based and 99.66% for time-based MCS. An MCS was developed Liu *et al*. (2022c) to effectively detect spammers. Data from popular social media platform in China was used. Base classifiers are constructed using profile, content and behaviour. MCS fusion is combined with evidential reasoning, and results of base classifiers are aggregated. The proposed method achieved an accuracy rate of 88.73%.

An enhanced threat detection method based on users' behaviour was proposed by Singh, Mehtre and Sangeetha (2022). A multi fuzzy classifier with 4 fuzzy inference engines works in parallel to allow faster processing was used. The CMU-CERT v4.2 dataset was used in the evaluation. Experimental results show an accuracy rate of 89.7% from the proposed multi fuzzy classifier, as compared to 75.3% from the next best classifier. A hybrid fuzzy multiple SVM is proposed in Yang *et al*. (2022) for high-dimensional data. The proposed hybrid method uses feature fusion based on CNN. Extracted features is captured by a clustering method and integrated into the composite kernel functions, which could improve the classification performance. Using Corona discharge images, the accuracy rate captured was 98.75%, which was up to 18% higher as compared to other methods.

2.3.5 Summary of Multi-Classifier System

A summary of the multi-classifier systems is given in Table 2.2. It lists the type of applications being looked at, including the metrices used for the results. A number of literatures without any reported results were removed. Again, accuracy is the most commonly used metric of performance measurement.

Table 2.2 Summary of multi-classifier system

Authors	Details	Results
Wang, Tong, and Wang (2020)	Warning system for outward foreign direct investment	Accuracy 85.088%
Du Jardin (2018)	Bankruptcy prediction using balance sheets on firms	Accuracy 81.13-81.81%
Kalid *et al*. (2020)	Anomaly detection in credit card data	Accuracy 99.9%
Wang, Liu and Qi (2022)	Improving effect of credit risk assessment	Accuracy 84.15%
Yijing *et al*. (2016)	Recognising oil-bearing reservoirs	Accuracy 100%
Zhang and Mahadevan (2019)	Prediction of severity of abnormal aviation events	Accuracy 81%
Lazri *et al*. (2020)	Estimate precipitation from satellite photos	Coefficient correlation 0.93
Hüthwohl, Lu and Brilakis (2019)	Automated inspection and objective bridge defect classification	F1-score 83.5%
Li *et al*. (2020)	Rotating machinery fault diagnosis	Accuracy 99.31-99.75%
Qu, Zhang and Gong (2016)	Fault diagnosis of gearbox and locomotive roller bearings	Accuracy 100%
Chen *et al*. (2021a)	Imbalanced samples for bearing fault diagnosis	Accuracy 90%
Pazoky and Pahlavani (2021)	Prediction of road classes on the OpenStreetMap platform	Accuracy 97.19%
Dou *et al*. (2021)	Remote sensing image classification	Accuracy 79.97-83.51%
Damodaran and Nidamanuri (2014)	Hyperspectral image for land cover classification	Accuracy 5% higher
Ding *et al*. (2022)	Diagnosis of electrical motor drive system faults	Accuracy 95.10%
Zaki *et al*. (2022)	Social media services within same parent application	F-score 88-99%
Su *et al*. (2022)	Fault-line selection method in grounded systems	Accuracy 97.8%
Cheng and Gao (2022)	Motor imagery recognition	Accuracy 72.52%
Rajabi *et al*. (2022)	Fault diagnosis in moving industrial equipment	Accuracy 98.6%
Kamali, Boostani and Parsaei (2013)	Classification of neuromuscular disorders	Accuracy 97%

Sun *et al*. (2020)	Classify ECG signals	Accuracy 75.2%
Beevi, Nair and Bindu (2017)	Detection of breast cancer using clinical images	F-score 84.29%
Jothi and Rajam (2017)	Maximise the diagnostic accuracy of thyroid detection	Accuracy 99.54%
Lin (2012)	Multivariate gene expression	Error reduced 50%
Fredrickson *et al*. (2019)	Mortality prediction of trauma patients was conducted using the National Trauma Data Bank	AUC 0.8649
Chen *et al*. (2021b)	Fatigue recognition identification based on ECG and EMG	Accuracy 89.47%, AUC 0.90
Gao, Turek and Vitaterna (2016)	EEG and EMG recordings are used to study sleep patterns in rodents	Error rate 0.049
Sanz *et al*. (2017)	Survival status prediction for trauma patients in hospitals in Spain	Sensitivity 89.08%, specificity 67.03%
Zhu, Xie and Zheng (2015)	Detection of Type-2 Diabetes Mellitus (T2DM)	Accuracy 2% higher
Ruano-Ordás *et al*. (2019)	Determine biological activity of molecules for chemical and physicochemical properties	Accuracy 91.34%
Zhang *et al*. (2022)	Classification of thyroid disease	Accuracy 90.9%
Liu *et al*. (2022b)	Classify neuropsychiatric disorders	Accuracy 93.71%
Kim *et al*. (2022)	Enhancement of anatomical side markers for multi-centre trials	Accuracy 90-95%
Bitrus *et al*. (2022)	Enhance classification in correlative microscopy	AUC 0.997, F1-score 0.98
Sahu, Das and Meher (2023)	Breast cancer detection using mammograms	Accuracy 98.00-99.17%
Mendialdua *et al*. (2015)	Classification of benchmark breast cancer data	Accuracy 74.83%
Kang, Cho, and Kang (2015)	Classification of benchmark zoo data	Error rate 0.99%
He *et al*. (2020)	Classification of benchmark hear data	Accuracy 83%
Jiang, Zeng, and Zhang (2013)	Diversity among base classifiers using pseudo-labelled data	Accuracy 83.3%

Li, Huang and Zong (2011)	Multi-domain sentiment classification	Error rate reduced 27.6%
Ferrara, Franco, and Maio (2012)	Facial picture segmentation	Error rate 13.87%
Zhang *et al*. (2011)	Aggregate ensemble learning model which tolerates data errors	Accuracy 82.3%
Da Silva, Hruschka, and Hruschka Jr (2014)	Classify tweet sentiment as positive or negative	Accuracy 81.06%
Pan *et al*. (2020)	Supporting risk analysis of tunnel under uncertainty	Accuracy 87.99%
Catal and Nangir (2017)	Turkish sentiment classification	Accuracy 96.13%
Roy and Bhattacharjee (2016)	Matching heterogeneous faces using geometric edge-texture feature	Accuracy 99.50-99.66%
Poreddy and Gopi (2023)	Improving classification accuracy of an MCS	Accuracy 5% higher
Fontes (2022)	Refinement of the feedforward network	Accuracy 74.7-88.9%
Lu *et al*. (2022)	Fault diagnosis of proton exchange membrane fuel cell systems	Accuracy 97.3%
Almashhadani *et al*. (2022)	Detection of ransomworm network traffic	Accuracy 99.66-99.88%
Liu *et al*. (2022c)	Effectively detect spammers on social media	Accuracy 88.73%
Singh, Mehtre and Sangeetha (2022)	Threat detection method based on users' behaviour	Accuracy 89.7%
Yang *et al*. (2022)	Classification of high-dimensional data	Accuracy 98.75%

2.4 Fraudulent Card Transactions

In this sub-chapter, various methods for detecting fraudulent card transactions are explored and analysed. A comprehensive review of relevant literature has been undertaken.

A sequence of credit card transactions was modelled from different perspectives in Lucas *et al*. (2020). Using the different perspectives, a total of 8 sequences were produced, which was then modelled using a Hidden Markov Model (HMM). The outputs were then fed into RF for classification. Experimental results show that the PR-AUC for e-commerce transactions were at 0.320 while in-person transactions at 0.145. A fraud detection system using deep learning and homogeneity-oriented behaviour analysis (HOBA) was developed in Zhang *et al*. (2021). A study was done using dataset from commercial bank in China. Experimental results indicate that using HOBA features, the best accuracy rate achieved was 98.11% as compared to 97.44% using different methods.

A publicly available credit card dataset for European cardholders was used in Fiore *et al*. (2019), Forough and Momtazi (2021) and Li *et al*. (2021). The dataset contained 0.17% fraudulent transactions. In Fiore *et al*. (2019), the Generative Adversarial Networks was used for classification, where its output mimicked minority class. Experimental results show effective fraud detection rates, where it achieved an accuracy rate of 99.962%. In Forough and Momtazi (2021), an ensemble model based on deep recurrent neural networks. The ensemble used LSTM networks as base classifiers, on top of a voting mechanism based on ANN. Experimental results demonstrate a precision rate of 0.9569, with it being efficient in its performance.

For handling the problem of class imbalance, a hybrid method with dynamic weighted entropy in handling class imbalance is proposed in Li *et al*. (2021). The model is first trained on minority samples, and then remaining samples form an overlapping subset with a low imbalance ratio. The dynamic weight entropy is used in evaluating the quality, in which good hyperparameters are selected. The proposed method achieved best F1-score of 73% in the experiments. An approach to detect fraudulent credit card transactions conducted in online stores known as APATE is proposed in Van Vlasselaer *et al*. (2015). APATE combines features from incoming transactions and network-based features using a time-suspiciousness score. The best accuracy rate from the experiments were at 99.46% with an AUC rate of 0.986.

The development fraud detection system in a large e-tail merchant is given in Carneiro, Figueira and Costa (2017). A data-mining method is implemented in the development. From the experiments, assuming that 80% of the orders in testing with lowest scores would be automatically approved, the false alarm rate was a mere 2%. In detection fraudulent cashout accurately, four feature sets based on expert knowledge was constructed in Wu, Xu and Li (2019). A real dataset with 25000 credit cards was used. The classifications were done using XGB, RF, and SVM, which considered dynamic patterns of cardholders. Precisions in the top 5% to 20% levels were improved by magnitudes of 0.049 to 0.081.

Rtayli and Enneya (2020) proposed a hybrid strategy for detecting credit card fraud. To solve the unbalanced data problem, the hybrid model used a feature removal strategy, hyper-parameter optimization, and Synthetic Minority Oversampling (SMOTE). Based on real-world data, experimental results reveal an accuracy rate of 99% and an AUPR of 0.81. For credit card fraud detection problem, a customized Bayesian Network Classifier (BNC) was proposed in de Sá, Pereira and Pappa (2018). An evolutionary algorithm organized the knowledge on BNC algorithms into the best combinations. A dataset from online payments service in Brazil was used in the experiments. Results show that the economic efficiency increased by 72.64% using the proposed BNC method.

A cost-sensitive decision tree method that minimizes the sum of misclassification cost was proposed in Sahin, Bulkan and Duman (2013). Real-world credit card data set was used in the experiments, in which the misclassification cost is taken as varying. Financial losses can be reduced using this approach, which can be seen in the results. Experimental results show that TPR for the proposed method is 92.8%, as compared to the others at 74.6% to 92.6%. Detection of credit card fraud using a modified Fisher discriminant analysis was proposed in Mahmoudi and Duman (2015). The modified fisher discriminant function tweaks the function to be more sensitive to important instances, where weighted average is used in calculating the variances. The proposed method performed better in the experiments, as compared to classical methods.

A fraud detection model for prepaid cards using a dynamic transaction model is studied in Robinson and Aria (2018). The model creates, updates, and compares hidden Markov models (HMM) of merchant terminals. It allows fraud to be detected in real-time for merchants. Experimental results show that good F-score is achieved in known fraud cases. In credit card fraud detection, a sequence classification task and LSTM network were employed in Jurgovsky *et al*. (2018). Feature aggregation strategies was integrated into the task. Using RF as the classifier, it showed that LSTM improved detection accuracy on offline transactions. Experimental results showed combination of sequential and non-sequential methods tend to detect different frauds, hence the use of combination.

A machine learning approach dealing with imbalanced and non-stationarity data, named SCARFF is presented in Damodaran and Nidamanuri (2014). A card dataset containing 0.4% fraud transactions is used. The first model is a RF trained on investigator feedback, and second model is an ensemble of balanced random trees. Transaction risk score is based on both models. Experimental results on real credit card transactions show that 24 of the 100 average alerts are correct. An Artificial Immune Systems (AIS)-based credit card detection model was proposed in Halvaiee and Akbari (2014). AIRS, an immune system inspired algorithm was used in the fraud detection model. Using this combination, the experiments showed that the accuracy rates increased up to 25% with an improved detection rate of 23%. In addition, the training time was reduced by up to 40%.

RIBIB, a Bayesian inference bagging model is developed for credit card fraud detection in Akila and Reddy (2018). The model incorporates a constrained bag creation method, Bayesian inference method and a cost-sensitive weighted voting combiner. Experiments were conducted a Brazilian bank and UCSD-FICO data. Experimental results show that RIBIB has good performances in all the proposed metrics except for the Mathew’s Correlation Coefficient. A framework combining meta-learning ensemble techniques and cost-sensitive learning paradigm for fraud detection is proposed in Olowookere and Adewale (2020). It allows base-classifiers to fit while cost-sensitive learning incorporated in ensemble learning process. Experimental results were evaluated using the Area Under the Receiver Operating Characteristic curve (AUC). The cost-sensitive ensemble classifier achieves the highest AUC at 0.99, while MLP at 0.95, DT at 0.90, and the lowest *k*NN at 0.87.

Zhang *et al*. (2021) investigate financial company fraud using text analysis of annual and quarterly reports from China's listed corporations. First, the management discussion and analysis are vectorized. As compared to alternative vector indexes, the bag-of-words (BoW) model and machine learning technique perform well. The accuracy rates for SVM were 71.39% versus 55.72% for NB.

2.4.1 Summary of Fraudulent Card Transaction

A summary of the fraudulent card transactions is presented in Table 2.3. It enumerates the types of applications being examined, including the metrics used for the results. Several pieces of literature without any reported results were excluded. In comparison to previous literature, a combination of accuracy, AUC, and F1-scores were employed as performance metrics.

Table 2.3 Summary of fraudulent card transactions

Authors	Details	Results
Lucas *et al*. (2020)	Credit card fraud detection using multi-perspective model	PR-AUC 0.145-0.320
Zhang *et al*. (2021)	Fraud detection system for commercial bank	Accuracy 98.11%
Fiore *et al*. (2019)	Credit card dataset for European cardholders with mimicking minority class	Accuracy 99.962%
Forough and Momtazi (2021)	Credit card dataset for European cardholders using voting	Precision 0.9569
Li *et al*. (2021)	Credit card dataset for European cardholders with dynamic weighted entropy	F1-score 73%
Van Vlasselaer *et al*. (2015)	Detection fraudulent credit card transactions conducted in online stores	Accuracy 99.46%, AUC 0.986
Carneiro, Figueira and Costa (2017)	Fraud detection system in a large e-tail merchant	False alarm rate 2%
Wu, Xu and Li (2019)	Detection fraudulent cashout accurately using dynamic patterns	Precision 0.081
Rtayli and Enneya (2020)	Hybrid strategy for detecting credit card fraud	Accuracy 99%, AUPR 0.81
de Sá, Pereira and Pappa (2018)	Credit card fraud detection problem	Economic efficiency 72.64% higher
Sahin, Bulkan and Duman (2013)	Cost-sensitive decision tree for credit card fraud data	TPR 92.8%
Damodaran and Nidamanuri (2014)	Dealing with imbalanced and non-stationarity data for card dataset	24/100 correct alerts correct
Halvaiee and Akbari (2014)	Artificial Immune Systems (AIS)-based credit card detection model	Accuracy 25% higher
Olowookere and Adewale (2020)	Cost-sensitive learning paradigm for fraud detection	AUC 0.99
Zhang *et al*. (2021)	Financial company fraud using text analysis of reports	Accuracy 71.39%

2.5 Literature Summary

Ensemble approaches have grown in popularity in machine learning for its capacity to improve forecast accuracy by mixing the output of numerous base models. The method has been used in a variety of applications, including credit scoring, machine fault detection, disease diagnosis, and wind speed predictions. Several metrics, including accuracy, root mean squared error (RMSE), and area under the curve (AUC), have been used to assess the performance of these models. Accuracy ranges vary by application, with credit scoring models achieving accuracies ranging from 82.01% to 91.1% and disease diagnostic models reaching accuracies ranging from 78% to 99.75%.

In multi-classifier systems, accuracy, coefficient correlation, and F1-score are used to evaluate model performance. Credit scoring, disease diagnosis, malfunction identification in equipment, sentiment classification, and threat detection based on user behaviour are all applications of multi-classifier systems. These models' accuracy spans from 72.52% to 100%, with some models surpassing benchmark data. Ensemble learning, modification of feedforward networks, and diversity across base classifiers utilising pseudo-labelled data are all techniques used to increase classification accuracy.

In recent years, there has also been increased interest in the application of machine learning to detect fraud in credit card transactions. To solve the issue, many models and methods with accuracy ranging from 71.39% to 99.962% have been created. Multi-perspective models, dynamic weighted entropy, and cost-sensitive learning paradigms are among the strategies used in these models. Some methods identify fraud using imbalanced or non-stationary data, while others use artificial immune systems or text analysis. False alarm ranges from 2%, with economic efficiency increasing by up to 72.64%.

The creation of these models reflects the continual attempts to improve fraud detection and financial loss prevention. The findings demonstrate that machine learning approaches can be useful in detecting fraudulent credit card transactions. The application of ensemble approaches and multi-classifier systems can increase the accuracy of these models even further. However, identifying fraud in real-time remains a big difficulty, and additional research is needed to develop more efficient and reliable fraud detection systems. Overall, advances in machine learning techniques for fraud detection show great potential for increasing financial security and safeguarding consumers from fraudulent activities.

2.6 Card Features and Flow

There is numerous information collected in the card transaction. A review is conducted in Bahnsen *et al*. (2016) to summarize the possible features that have been used. These features comply with international financial reporting standards. Table 2.1 shows the most commonly used raw features based on the literature review.

Table 2.4 Summary of typical raw features in credit card datasets

No	Features	Description
1	Transaction ID	Transaction reference number
2	Time	The transaction's date and time
3	Account number	Customer identification number
4	Card number	Card identification number
5	Transaction type	Internet, ATM, or POS card identification
6	Entry mode	Magnetic stripe or chip and pin
7	Amount	Transaction amount
8	Merchant code	Identifying the merchant type
9	Merchant group	Identification of a merchant group
10	Country 1	Transaction country
11	Country 2	Country of origin
12	Type of card	Visa, MasterCard, and American Express
13	Gender	The cardholder's gender
14	Age	The cardholder's age
15	Bank	The card's issuer bank

The credit card transaction authorization system, comprising the transaction path and decision-making process, is depicted in Fig. 2.3. There are five parties involved: the consumer, the merchant, the (merchant) acquirer, the network (provider), and the (card) issuer.

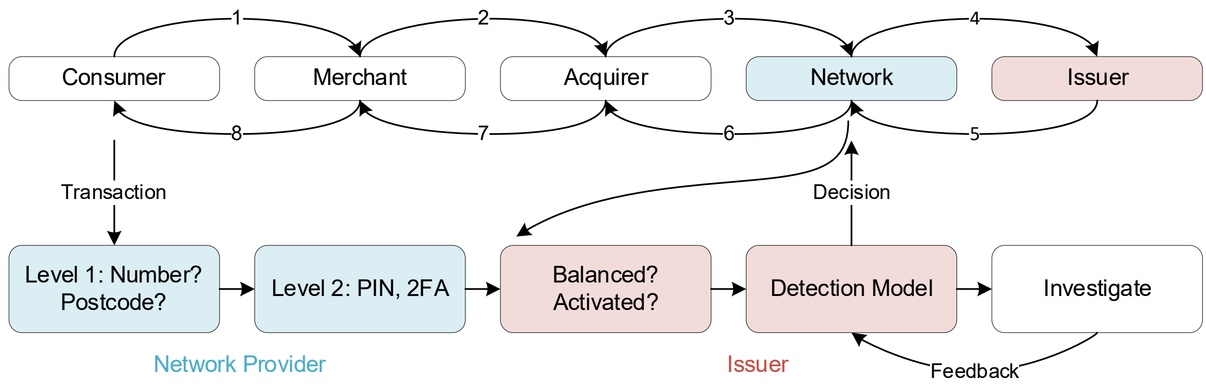

Fig. 2.3 Credit card transaction authorization network (Wang, Chen and Chen, 2019)

Customers make purchases at stores like Aeon with credit cards provided by issuers like Maybank, which are part of a larger network of financial institutions (i.e., Visa, Mastercard). Typically, the issuer is a bank or other financial institution that issues and sells payment cards with the logo of a network service provider. A bank or other financial institution handles the merchant acquiring function in the context of accepting credit card payments. It paves the way for stores to take credit card payments from customers using the issuer of a given network.

A branded platform for the execution of payment transactions is provided by the network provider, also known as the card association. A transaction is authorised by the system after eight stages: (1) customer makes purchase request to merchant; (2) merchant sends request to acquirer; (3) acquirer sends request to issuer for authorization via network provider; (5) issuer returns authorization code (approve or decline) to acquirer; (6) acquirer authorises transaction; (7) customer receives product or service; (8) customer pays for transaction.

2.7 Summary

A number of actions must be completed progressively in order to build a prediction model. The needs for developing the model should be reviewed first. Then, data pre-processing or cleaning of data should be undertaken (removing inconsistent data, data transformation, feature selection, or sample selection). Finally, to maximise pattern extraction, a suitable prediction model that takes feature qualities into account should be used (Mohammed, Onieva and Woniak, 2020).

It has been experimentally demonstrated that building a flawless classifier using a single paradigm is frequently difficult (Mendialdua *et al.*, 2015). As a result, designers have attempted to integrate numerous classifiers, resulting in the development of Multiple-Classifier Systems (MCS), which try to combine the benefits of different individual classifiers, frequently generated using different training paradigms, to achieve better results. Classifier combination is a potential alternative to utilising a single classifier and has evolved into a well-established research subject based primarily on heuristic answers.

Intuitively, a collection of classifiers should produce better outcomes than a single decision maker. Such systems mitigate the consequences of errors induced by its components, particularly for complex samples (Mohammed, Onieva and Woniak, 2020). MCSs should be designed to ensure a balance of diversity and compatibility among individual members in order to minimise redundancy and achieve robustness.

3 Research Design

In this study, three primary algorithms are employed. An overview of each algorithm is provided as follows.

3.1 Algorithms

A total of three main algorithms consisting of Random Forest (Chapter 3.1.1), Generalized Linear Models (Chapter 3.1.2) and Gradient Boosting Machines (Chapter 3.1.3) are detailed. Each sub-chapter elucidates the algorithm's details, structure, and associated equations.

3.1.1 Random Forest

Random Forest (RF) is a supervised learning algorithm that performs classification and regression using the ensemble learning method. Random forest is a bagging method, not a boosting method. RF have trees that grow in parallel. There is no interaction between the trees during the formation process.

The RF model generates a set of random trees. The number of trees is determined by the user, and the resulting RF model employs voting to determine the final classification decision based on the predictions from all trees formed. Fig. 3.1 depicts the RF structure. The number of classes is given as k, while the number of trees is given as T. The bagging approach is used in the building of RF, with random attribute selection.

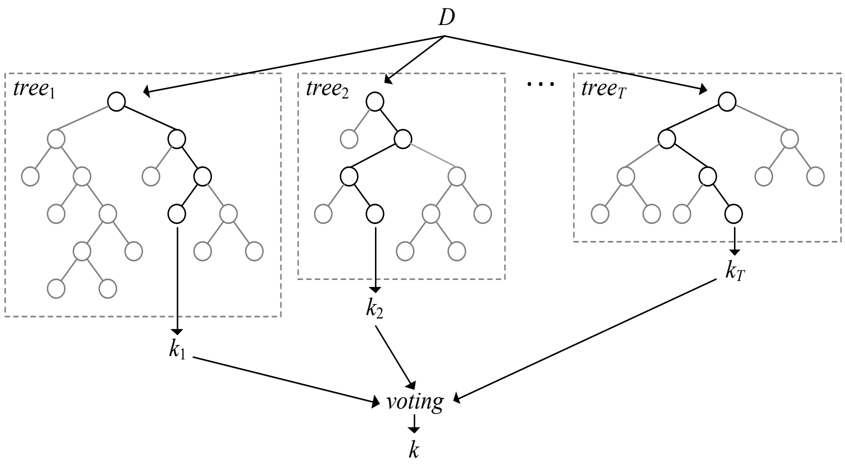

Fig. 3.1 Structure of Random Forest

The default settings were a gain ratio criterion and a maximum depth of 20. Other default settings included a tree count of 10, pruning confidence of 0.25, a gain of 0.1, and a leaf size of 2.

An example of the implementation in Python is given in Algorithm 3.1. All the algorithms used are based on the H20.ai platform, hence the import h2o code is needed to load the packages.

Algorithm 3.1 Random Forest in Python

```
import h2o
from h2o.estimators.random_forest import H2ORandomForestEstimator

rfe1 = H2ORandomForestEstimator(seed=seed1)
rfe1.train(x=x, y=y, training_frame=train1)
result_ml1v1 = rfe1.predict(validate1)
```

3.1.2 Generalized Linear Models

Generalized linear models (GLMs) are models that expand the usual linear regression model to incorporate variables that are not necessarily normally distributed and where the explanatory variables are linear as a function of the response mean. The classical linear regression model takes the following form

$$\boldsymbol{y} = \boldsymbol{X\beta} + \boldsymbol{\epsilon} \tag{3.1}$$

where $\boldsymbol{y}$ are results of the random vector $\boldsymbol{Y}$ and is an $\boldsymbol{n} \times \boldsymbol{1}$ vector of observations $\boldsymbol{y_i}$ where $\boldsymbol{i} = \boldsymbol{1}, \ldots, \boldsymbol{n}$; $\boldsymbol{X}$ is an $\boldsymbol{n} \times \boldsymbol{p}$ matrix of the observations of the explanatory variables; $\boldsymbol{\beta}$ is an $\boldsymbol{p} \times \boldsymbol{1}$ vector of coefficients of the explanatory variables; and $\boldsymbol{\epsilon}$ is an $\boldsymbol{n} \times \boldsymbol{1}$ vector of the error terms.

GLMs are a broad class of statistical models that includes linear models as a subset. As a result, the linear model's constraints in terms of normality, constant variance, and effect additivity are lifted. The response variable Y, on the other hand, is considered to be a member of the exponential family of distributions. GLMs allow the variance to fluctuate with the distribution mean, and the influence of explanatory variables on the response variable is considered to be additive on a transformed scale. The GLM-analog to the preceding linear model assumptions can be summarised as follows:

Random component: The components of $\boldsymbol{Y}$ are independent and belong to the exponential distribution family.

Systematic component: The $\boldsymbol{p}$ explanatory variables of the model are merged to get the linear predictor $\boldsymbol{\eta}$, such that $\boldsymbol{\eta} = \boldsymbol{X\beta}$

Link function: The link function, *g*, specifies the relationship between the random and systematic components, that is differentiable and monotonic so that $\boldsymbol{E}[\boldsymbol{Y}] = \boldsymbol{\mu} = \boldsymbol{g}^{-\boldsymbol{1}}(\boldsymbol{\eta})$

The example implementation in Python is listed in Algorithm 3.2. Similarly, the H20.ai packages are utilized.

Algorithm 3.2 Generalized Linear Models in Python

```
import h2o
from h2o.estimators.glm import H2OGeneralizedLinearEstimator

glm1 = H2OGeneralizedLinearEstimator(seed=seed1)
glm1.train(x=x, y=y, training_frame=train1)
result_ml5v1 = glm1.predict(validate1)
```

In each of the numerous groups, a different seeds number is used. In light of this, the seed numbers are shuffled about for each of the groups.

3.1.3 Gradient Boosting Machines

Gradient boosting is a sort of boosting in machine learning. It assumes that the best next model, when merged with past models, minimises the overall prediction error. To reduce error, the fundamental idea is to describe the desired outcomes for this next model.

Decision tree models frequently exhibit high variance and are quite sensitive to changes in the training data. To address this issue, ensemble methods can be utilised, which integrate numerous weaker models to build a more powerful model. There are other ensemble approaches, but GBMs are among the most widely used. GBMs are well-known in the ML community for producing good results on virtually any given task and frequently winning ML competitions.

In the predictive learning issue, a response variable *y* and a set of explanatory variables $\boldsymbol{x} = (\boldsymbol{x_1} \dots, \boldsymbol{x_p})$ is used. Using a training sample of observations $(\boldsymbol{x_1}, \boldsymbol{y_1}), (\boldsymbol{x_2}, \boldsymbol{y_2}), \dots, (\boldsymbol{x_n}, \boldsymbol{y_n})$ the goal is to get an estimate $\hat{\boldsymbol{f}}(\boldsymbol{x})$ of $\boldsymbol{f}^*(\boldsymbol{x})$ which maps *x* to *y* and that minimizes the expected value of some specified loss function $\boldsymbol{L}(\boldsymbol{y}, \boldsymbol{f}(\boldsymbol{x}))$ over the joint distribution $(\boldsymbol{x}, \boldsymbol{y})$, according to

$$\boldsymbol{f}^* = \arg\min_{\boldsymbol{f}} \boldsymbol{E}[\boldsymbol{L}(\boldsymbol{y}, \boldsymbol{f}(\boldsymbol{x}))] \tag{3.2}$$

As to the limited number of training examples available, an approximation to equation (3.2) can be produced as

$$\hat{\boldsymbol{f}} = \arg\min_{\boldsymbol{f}} \frac{\boldsymbol{1}}{\boldsymbol{n}} \sum_{\boldsymbol{i=1}}^{\boldsymbol{n}} \boldsymbol{L}(\boldsymbol{y_i}, \boldsymbol{f}(\boldsymbol{x_i})) \tag{3.3}$$

GBMs forecast non-linear and interacting correlations more accurately, and they also deliver accurate answers for each variable's incremental influence.

As with the previous two examples, the implementation for GLM is given in Algorithm 3.3. A The codes are similar to each other as the same main package it used.

Algorithm 3.3 Gradient Boosting Machines in Python

```
import h2o
from h2o.estimators.gbm import H2OGradientBoostingEstimator

gbm1 = H2OGradientBoostingEstimator(seed=seed1)
gbm1.train(x=x, y=y, training_frame=train1)
result_ml4v1 = gbm1.predict(validate1)
```

3.2 Classification Methods

Many common machine learning models from H2O.ai were used in this thesis to build an MCS model. Python software was utilised in the Google Colab environment. The majority voting and the BKS model of Huang and Suen (1995) for decision combination are explained in the following sub-chapters.

3.2.1 Majority Voting

Given $\boldsymbol{M}$ target classes, where every class is represented by $\boldsymbol{C_i}, \forall_i \in \boldsymbol{\Lambda} = \{\mathbf{1}, \mathbf{2}, \ldots, \boldsymbol{M}\}$. The classifier task is to categorize an input sample, $\boldsymbol{x}$, to one of the $(\boldsymbol{M}+\mathbf{1})$ classes, with the $(\boldsymbol{M}+\mathbf{1})$th class denoting that the classifier rejects $\boldsymbol{x}$.

Majority voting is an approach for integrating several classifier outputs. With $\boldsymbol{K}$ classifiers, denoted by $\boldsymbol{e_1}, \ldots, \boldsymbol{e_K}$, the goal is to have a combined result, $\boldsymbol{E}(\boldsymbol{x}) = \boldsymbol{j}, \boldsymbol{j} \in \{\mathbf{1}, \mathbf{2}, \ldots, \boldsymbol{M}, \boldsymbol{M}+\mathbf{1}\}$ from all $\boldsymbol{K}$ predictions, $\boldsymbol{e_k}(\boldsymbol{x}) = \boldsymbol{j_k}, \boldsymbol{k} = \mathbf{1}, \ldots, \boldsymbol{K}$. The number of votes can be counted using a binary function (Jedrzejowicz and Jedrzejowicz, 2020), i.e.,

$$\boldsymbol{V_k}(\boldsymbol{x} \in \boldsymbol{C_i}) = \begin{cases} \mathbf{1}, & \textbf{if } \boldsymbol{e_k}(\boldsymbol{x}) = \boldsymbol{i}, \boldsymbol{i} \in \boldsymbol{\Lambda} \\ \mathbf{0}, & \textbf{otherwise.} \end{cases} \tag{3.4}$$

Then, sum the votes from all $\boldsymbol{K}$ classifiers for each $\boldsymbol{C_i}$

$$\boldsymbol{V_K(x \in C_i) = \sum_{k=1}^{K} V_k(x \in C_i)}, \qquad \boldsymbol{i = 1, \ldots, M} \tag{3.5}$$

and the combined result, $\boldsymbol{E(x)}$, can be determined by

$$\boldsymbol{E(x)} = \begin{cases} \boldsymbol{j}, & \boldsymbol{if\ V_E(x \in C_j) = \mathbf{max}_{i \in \Lambda}(x \in C_i)\ \mathbf{and}\ \frac{V_E(x \in C_j)}{K} \geq \lambda} \\ \boldsymbol{M+1}, & \mathbf{otherwise} \end{cases} \tag{3.6}$$

where $\boldsymbol{0 \leq \lambda \leq 1}$ is a user-defined criterion that governs trust in the final decision (Xu, Krzyzak, and Suen, 1992).

A framework with *N* agent groups using majority voting, where each group has multiple individual agents providing the inputs is shown in Fig. 3.2.

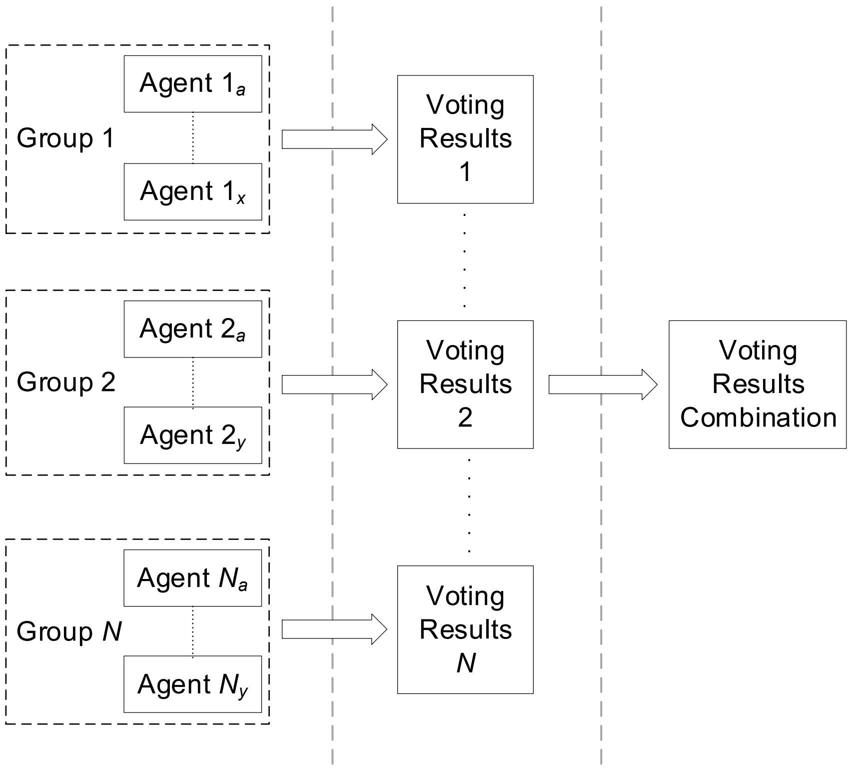

Fig. 3.2 A hierarchical majority voting agent-based framework

The codes in Python for the creation of the voting table is given in Algorithm 3.4.

Algorithm 3.4 Creation of majority voting in Python

```
def majority_voting(validate_mlrfe1, validate_mlgbm1, validate_mlglm1):

    vote = validate_mlrfe1 + validate_mlgbm1 + validate_mlglm1
    if vote >= 2:
        return 1
    else:
        return 0
```

This function takes in the predictions from classifiers as arguments, and returns the majority voted prediction (either 0 or 1). In this implementation, the majority vote is determined by summing the three predictions, and if the sum is 2 or greater, the function returns 1, otherwise it returns 0.

3.2.2 Behavior Knowledge Space

A Behavior Knowledge Space (BKS) is a $\boldsymbol{K}$-dimensional space, which each dimension represents the prediction (i.e., predicted class) from a single classifier. The interchapter of the decisions from $\boldsymbol{K}$ different classifiers occupy a single unit in the BKS, e.g., $\mathbf{BKS}(\boldsymbol{e_1}(\boldsymbol{x}) = \boldsymbol{j_1}, \dots, \boldsymbol{e_K}(\boldsymbol{x}) = \boldsymbol{j_K})$ represents one unit in which each $\boldsymbol{e_k}$ produces a prediction $\boldsymbol{j_k}, \boldsymbol{k} = \mathbf{1}, \dots, \boldsymbol{K}$. Foe every BKS unit, there are $\boldsymbol{M}$ partitions (cells), that aggregate the number of data samples truly belonging to $\boldsymbol{C_i}$.

Consider the following example which has two classifiers. As shown in Table 3.1, a two-dimensional (2-D) BKS can be built.

Table 3.1 Two-dimensional BKS

e_2 \ e_1	1	2	...	$M+1$
1	U_{11}	U_{12}	...	$U_{1(M+1)}$
2	U_{21}	U_{22}	...	$U_{2(M+1)}$
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
$M+1$	$U_{(M+1)1}$	$U_{(M+1)2}$	...	$U_{(M+1)(M+1)}$

Each BKS unit, $\boldsymbol{U_{ij}}$, contains $\boldsymbol{M}$ cells, i.e., $\boldsymbol{n_1^H}, \dots, \boldsymbol{n_M^H}$, where $\boldsymbol{H}$ shows the total prediction $\boldsymbol{e_1}(\boldsymbol{x}) = \boldsymbol{j_1}, \dots, \boldsymbol{e_K}(\boldsymbol{x}) = \boldsymbol{j_K}$. The complete number of data samples that belong in each class is captured in each $\boldsymbol{n_1^H}, \boldsymbol{i} = \mathbf{1}, \dots, \boldsymbol{M}$. When an input sample, $\boldsymbol{x}$, is displayed, one of the BKS units is activated (referred to as the focal unit) after the collection of decisions from all $\boldsymbol{K}$ classifiers. For instance, $\boldsymbol{U_{34}}$ becomes active as the focal unit if $\boldsymbol{e_1}(\boldsymbol{x}) = \mathbf{3}$ and $\boldsymbol{e_2}(\boldsymbol{x}) = \mathbf{4}$. It is possible to acquire the total number of samples included within the focal unit by utilising

$$\boldsymbol{T}(\boldsymbol{H}) = \sum_{\boldsymbol{i}=\mathbf{1}}^{\boldsymbol{M}} \boldsymbol{n_i^H} \tag{3.7}$$

and the one with the highest number of samples is identified

$$\boldsymbol{R}(\boldsymbol{H}) = \boldsymbol{j}, \;\; \textbf{where} \;\; \boldsymbol{n_j^H} = \mathbf{max}_{\boldsymbol{i}\in\boldsymbol{\Lambda}}\left(\boldsymbol{n_i^H}\right) \tag{3.8}$$

The decision rule for determining the final outcome is

$$\boldsymbol{E}(\boldsymbol{x}) = \begin{cases} \boldsymbol{R}(\boldsymbol{H}), & \textbf{if } \boldsymbol{T}(\boldsymbol{H}) > \mathbf{0} \textbf{ and } \dfrac{\boldsymbol{n}^{\boldsymbol{H}}_{\boldsymbol{R}(\boldsymbol{H})}}{\boldsymbol{T}(\boldsymbol{H})} \geq \boldsymbol{\lambda} \\ \boldsymbol{M} + \mathbf{1}, & \textbf{otherwise} \end{cases} \tag{3.9}$$

where $\mathbf{0} \leq \boldsymbol{\lambda} \leq \mathbf{1}$ is a user-defined confidence threshold.

The BKS is comparable to the confusion matrix. In order to calculate the joint probability of *K* events, the Bayesian methodology calls for the multiplication of evidence derived from confusion matrices whenever the predictions are combined. This phase is skipped in the BKS approach, which makes a final selection by directing the input sample to the class with the greatest number of samples. As demonstrated by Huang and Suen (1995) for classification of unconstrained handwritten numerals, this simple BKS approach provides a rapid and efficient method for integrating diverse judgements.

As shown in Fig. 3.3, a hierarchical agent-based architecture using the BKS for decision combination is proposed.

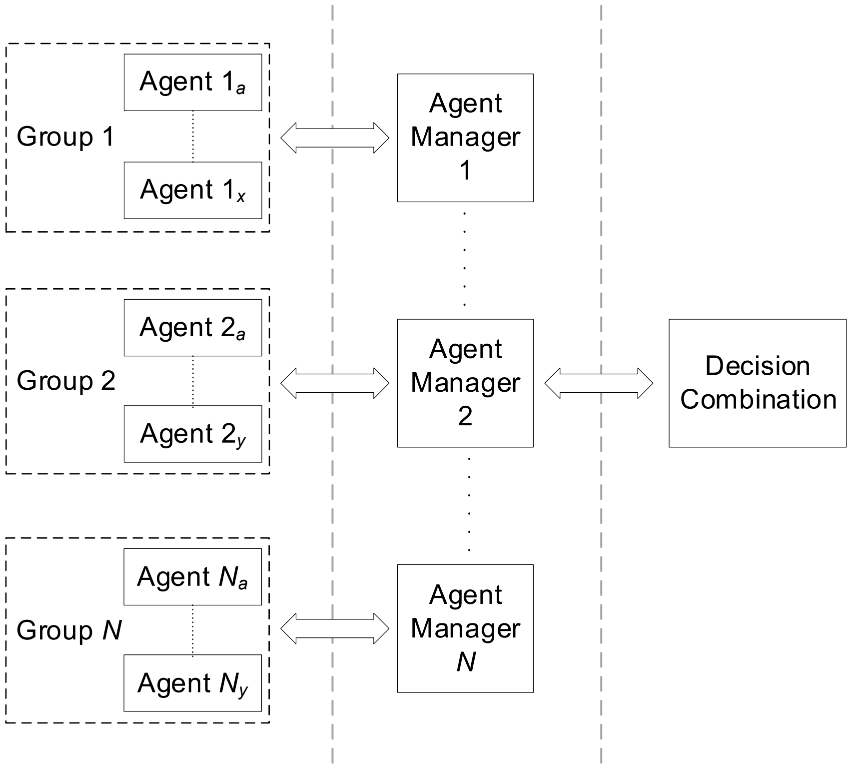

Fig. 3.3 A hierarchical agent-based framework with the BKS

The framework's basic layer contains *N* agent groups, each of which contains numerous individual agents. Machine learning models, statistical approaches, and other categorization algorithms can all be used as agents. Each group of agents is assigned a manager agent, who is tasked with using a BKS to aggregate the combined predictions from all of the groups. When many managers make predictions, they all get sent to a decision combination module, which is often another BKS in the top layer.

In first creating the BKS table, an example of the Python code is provided in Algorithm 3.5. The data being used is the validation of the training set. Data from each agent (three in total) was concatenated. It was then converted to the pandas' format. Any exact duplicates of the patterns were then removed.

Algorithm 3.5 Creation of BKS table

```
# BKS table #1: combining prediction into single dataframe
combine_results = pd.concat([validate_mlrfe1, validate_mlgbm1,
validate_mlglm1, validate_actual1], axis=1, sort=False)

combine_results["combined"] =
combine_results[combine_results.columns[0:]].apply(lambda x: '
'.join(x.dropna().astype(str)),axis=1)

combine_summary1 = combine_results.groupby(["combined"])
[["combined"]].describe()

combine_summary1 = combine_summary1.values.tolist()
combine_summary1 = pd.DataFrame(combine_summary1)
combine_summary1 = combine_summary1.drop([0, 1], axis=1)
combine_summary1 = combine_summary1.sort_values(by=3, ascending=False)

left = combine_summary1[2].str[:5]
right = combine_summary1[2].str[-1:]

combine_summary2 = pd.concat([left, right, combine_summary1[3]], axis=1,
sort=False)
combine_summary2.columns = ["pattern", "bks_class", "count"]

combine_summary3 = combine_summary2.drop_duplicates(subset="pattern",
keep="first")
combine_summary3 = combine_summary3.sort_values(by="pattern")
bks_table1 = combine_summary3.reset_index(drop=True)
```

In validating the results, the Python code is listed in Algorithm 3.6. As compared with the previous algorithm, the data being used is the validation of the test set, hence the first few lines look similar. Merging of the initial BKS table (Table 1 as created in Algorithm 3.5) was done. A final data frame containing the BKS results is produced at the end.

Algorithm 3.6 Validating results and combining BKS table

```
# Validate shuffle #1: combining prediction into single dataframe
combine_results_t = pd.concat([validate_mlrfe1, validate_mlgbm1,
validate_mlglm1, validate_actual1], axis=1, sort=False)
combine_results_t["pattern"] =
combine_results_t[combine_results_t.columns[0:3]].apply(lambda x: '
'.join(x.dropna().astype(str)),axis=1)
combine_results_t["mlx/3"] = (combine_results_t[['y_rfe1', 'y_gbm1',
'y_glm1']].sum(axis=1))/3

combine_results_tjoin = pd.merge(combine_results_t, bks_table1, on =
"pattern", how = "outer") # apply using BKS table 1
combine_results_tjoin = combine_results_tjoin.drop(["count"], axis=1)
combine_results_tjoin.loc[combine_results_tjoin["mlx/3"] <= 0.5,
'vote_class'] = 0
combine_results_tjoin.loc[combine_results_tjoin["mlx/3"] > 0.5, 'vote_class']
= 1
combine_results_tjoin["bks_class"] =
pd.to_numeric(combine_results_tjoin['bks_class'],errors='coerce')
combine_results_tjoin = combine_results_tjoin.dropna()

combine_results_tjoin_g1 = combine_results_tjoin.dropna()
combine_results_tjoin_g1 = combine_results_tjoin_g1[['y_Actual', 'bks_class',
'vote_class']]
combine_results_tjoin_g1 = combine_results_tjoin_g1.rename(columns =
{'bks_class':'bks_class_g1', 'vote_class':'vote_class_g1'})
```

To properly illustrate the BKS mechanism, a numerical example is provided. A basic binary classification problem is illustrated in Table 3.2. There are two agents (classifiers), six input samples, and anticipated and actual classes.

Table 3.2 Prediction outputs of Agents 1 and 2

Data	**Actual class**	**Predicted Class**	
		Agent 1	**Agent 2**
Sample 1	1	1	1
Sample 2	1	2	1
Sample 3	2	2	2
Sample 4	1	1	1
Sample 5	2	2	2
Sample 6	2	1	2

As shown in Table 3.3, a BKS can be constructed. For example, for input samples 1 and 4 (Table 3.2), both agents 1 and 2 predict class 1, and the actual class is 1. This information is captured in the highlighted (grey) BKS unit in Table 3.3. As indicated in equation, the predictions from all agents are utilised to activate a BKS unit with a fresh test sample, and the combined predicted class (final output) is obtained by taking the most samples from the majority class (3.5). The combined (final) forecast is Class 1 when the specified (grey) BKS unit is triggered during the test phase.

Table 3.3 Creation of BKS for the classification scenario in Table 3.2

Agent 1 / **Agent 2**	Predicted Class=1	Predicted Class=2
Predicted Class=1	No. of Actual Class 1 samples=2	No. of Actual Class 1 samples=1
	No. of Actual Class 2 samples=0	No. of Actual Class 2 samples=0
Predicted Class=2	No. of Actual Class 1 samples=0	No. of Actual Class 1 samples=0
	No. of Actual Class 2 samples=1	No. of Actual Class 2 samples=2

3.3 *Cross Validation*

Estimating a model's performance on unseen data is a common application of Cross Validation (CV). This is done so that the model's expected performance when used to generate predictions on data that was not utilised during training can be estimated using a small sample. It is commonly used as it is a simple to use method and provides less biased results as compared to other methods, such as the simple train to test split.

A typical value of *k* is either 5 or 10, though any number which is less than the dataset length can be used. The structure of the CV is shown in Fig. 3.4, where the light grey shade represents the training fold and the orange represents the test fold.

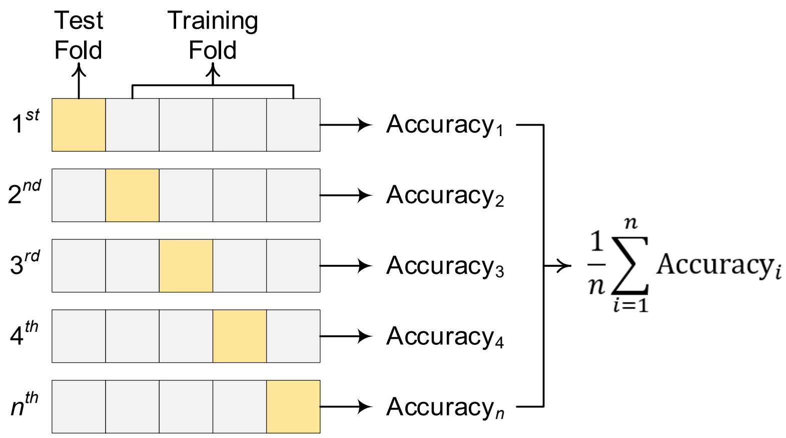

Fig. 3.4 A *n*-fold cross validation structure

Based on Fig. 3.4, the general procedure of the CV is given as follows.

(1) Split the dataset into *k* equal parts
(2) Choose *k* – 1 fold as the training set, with remaining as test set. The remaining fold will be the test set
(3) Train the model on the training set. On every iteration of CV, a new model is independently trained.
(4) Validate using the test set
(5) Save the result of the validation
(6) Repeat steps (3) to (6) *k* times.
(7) Get the final score average based on all runs.

As with the previous procedure, an example of a CV with k = 5 is given in Algorithm 3.7. The sklearn model in Python is used. The value of k can be easily changed under n_splits, signifying the simplicity of the algorithm.

Algorithm 3.7 A 5-fold cross validation

```
from sklearn.model_selection import KFold

# create the 5-fold cross-validation object
kf = KFold(n_splits=5, shuffle=True, random_state=42)

# iterate through each fold and train/test the model
for train_index, test_index in kf.split(X):
    # Split the data into training and testing sets
    X_train, X_test = X[train_index], X[test_index]
    y_train, y_test = y[train_index], y[test_index]

    # train and test the model on this fold
    model = ... # create your model here
    model.fit(X_train, y_train)
    accuracy = model.score(X_test, y_test)

    print(f"Accuracy on this fold: {accuracy}")
```

3.4 Summary

Three different algorithms were described in this chapter. For classification and regression, RF employs the ensemble learning method. It is a bagging technique in which the trees are run parallel to each other. While creating the trees, there is no interaction between them. GLM employs models that generalise the standard linear regression model to models that do not always include normally distributed response variables and in which a function of the mean of the response is linear in the explanatory variables. GBM employs a boosting method. It is predicated on the idea that the best potential next model, when paired with past models, minimises overall prediction error.

H2O.ai was used to create the MCS model using the aforementioned models, with all models written in Python. In the first model, the majority voting model is used, in which each individual classifier votes for a class and the majority wins. The ensemble's anticipated target label is the mode of the distribution of individually predicted labels. The second model used, is developed in this work, BKS. BKS is a $\boldsymbol{K}$-dimensional space, where every dimension indicates the decision from one classifier.

Cross-validation is utilised while determining the quality of the models. The use of CV helps to prevent overfitting, which can occur when one fails to generalise a pattern. The experiments make use of a wide variety of CVs, the specifics of which are described in the following chapters.

4 Benchmark Data and Discussions

Publicly available data sets from the UCI Machine Learning Repository (2020), the KEEL Repository (2020), and Kaggle (2020) are used in this empirical study. For evaluation, a real-world data set is also used.

4.1 Setup

The configuration of the majority voting framework is shown in Fig. 4.1, while the hierarchical agent-based framework employed in the experiments is shown in Fig. 4.2. It is divided into three categories, each with three agents. Random Forest (RF), Generalized Linear Model (GLM), and Gradient Boosting Machine (GBM) are the three agents chosen after comprehensive testing of individual and group performance.

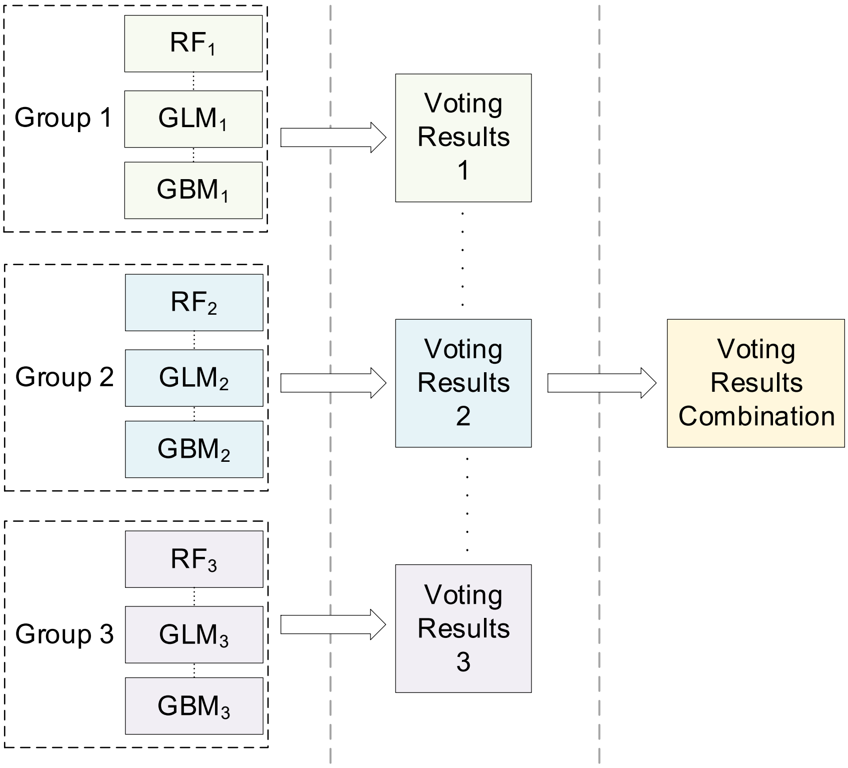

Fig. 4.1 Configuration of the majority voting framework used in the experiments

Three agent managers, each with a BKS module, are established for Fig. 4.2. These three agent managers' predictions are passed to the decision combination module, which uses another BKS to construct the final projected class.

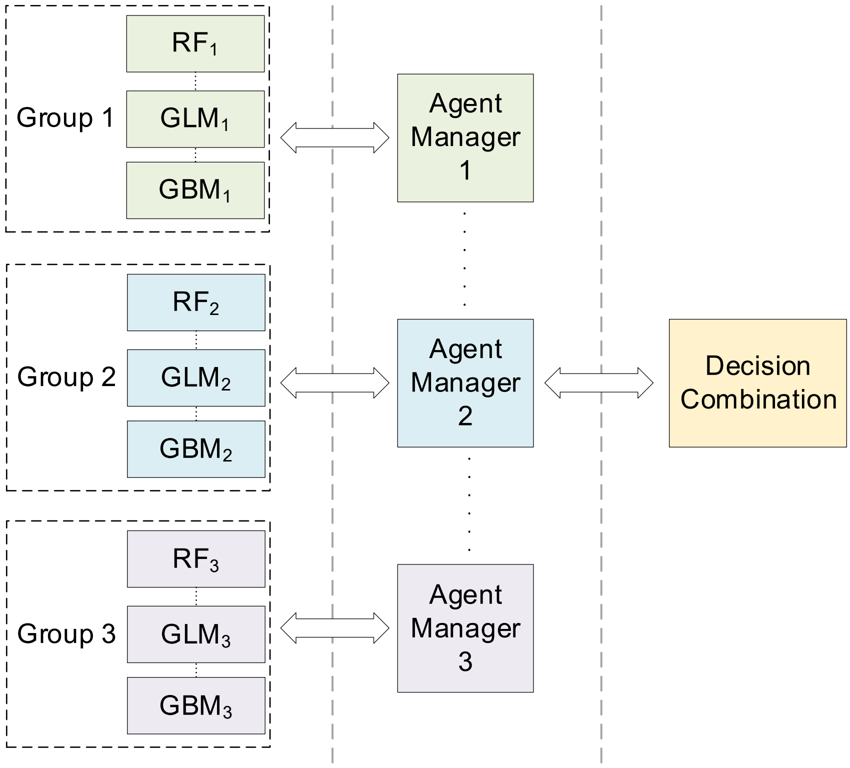

Fig. 4.2 Configuration of the hierarchical agent-based framework used in the experiments

The training approach begins with the randomization of data samples and ends with a validation phase. As a result, three group-based BKS modules are produced (one for each group). The outputs of BKS modules 1 through 3 are then combined using training data and another randomised sequence to produce another overall (final) BKS module that incorporates the outcomes of the previous three group-based BKS modules. Outputs of the group-based BKS modules are coupled with the outputs of the overall BKS module given a test sample to provide a final projected class for the computation of performance metrics such as classification accuracy and F1-score.

For performance comparison between majority voting and BKS statistically, the sign test (Sheskin, 2020) is adopted. The number of wins in the sign test is distributed according to a binomial distribution. Provided a huge variety of cases, the number of wins under the null hypothesis is distributed according to $n\left(\frac{n}{2}, \frac{\sqrt{n}}{2}\right)$, allowing the use of the z-test, i.e., should the number of wins be at least $(\frac{n}{2} + 1.96\frac{\sqrt{n}}{2})$, then the result is statistically significant with $p < 0.05$.

Derrac *et al*. (2011) calculated the number of wins required for a comparison of k = 25 experimental findings as follows: 18 wins for (the significance level) α = 0.05 (i.e., 95% confidence level) and 17 wins for a less stringent α = 0.1 (i.e., 90% confidence level). Furthermore, a total of 19 victories are necessary for a more stringent setting of α = 0.01 (i.e., 99% confidence interval).

Both accuracy rates and F1-scores are typically represented on a scale that spans from 0 to 1. In this context, values that are closer to 1 signify superior performance and reliability of the model, making them more desirable in evaluations and comparisons.

4.2 Benchmark Experiments Set 1

Publicly available data sets from the UCI Machine Learning Repository (UCI, 2020), the KEEL Repository (KEEL, 2020), and Kaggle (Kaggle, 2020) are used in this empirical study. For evaluation, a real-world data set is also used.

The experiments make use of a total of ten data sets. Table 4.1 shows the details of each data set, i.e., B1 through B10, including the number of instances and characteristics as well as the unbalanced ratio (IR) information. The datasets of B1 to B3, B5, B7 to B9 were sourced from KEEL (2020), B4 and B6 from UCI (2020) and B10 from Kaggle (2020).

Table 4.1 List and descriptions of benchmark set 1

Data set	Problem	Instances	Features	IR
B1	abalone-17_vs_7-8-9-10	2,338	8	39.3
B2	abalone-20_vs_8-9-10	1,916	8	72.7
B3	flare-F	1,066	11	23.8
B4	pima	768	8	1.9
B5	ring	7,400	20	1.0
B6	spambase	4,597	57	1.5
B7	twonorm	7,400	20	1.0
B8	winequality-red-4	1,599	11	29.2
B9	winequality-white-3-9_vs_5	1,482	11	58.3
B10	Credit card transactions by European cardholders	284,807	30	577.9

Algorithm 4.1 provides a demonstration of the loading of the data set in Python. The data is obtained over a Dropbox link and is afterwards input into the codes in directly.

Algorithm 4.1 Loading of data set

```
!wget -O spambase.csv
https://www.dropbox.com/s/2ehfrkxmzvpaeos/spambase.csvdl=0
df = pd.read_csv("spambase.csv")
```

The accuracy rates and F1-scores are listed in Table 4.2. Average accuracy of BKS and majority voting was at 0.9440 and 0.9385, respectively. The accuracy rates were all above 0.9, with the exception of B4, which only acquired 0.7204, mainly due to outliers in the dataset. The other datasets, though having multiple features and different number of instances managed to acquire good accuracy rates.

Table 4.2 Accuracy rates of benchmark set 1

Data set	**BKS**	**Voting**
B1	0.9734	0.9717
B2	0.9864	0.9864
B3	0.9453	0.9439
B4	0.7204	0.7203
B5	0.9541	0.9466
B6	0.9514	0.9498
B7	0.9782	0.9352
B8	0.9501	0.9521
B9	0.9823	0.9809
B10	0.9981	0.9980

A comparison between the accuracy rates of BKS and voting is given in Fig. 4.3. It can be seen that out of the ten data sets, only one (B8) where voting acquired slightly better rates as compared to BKS.

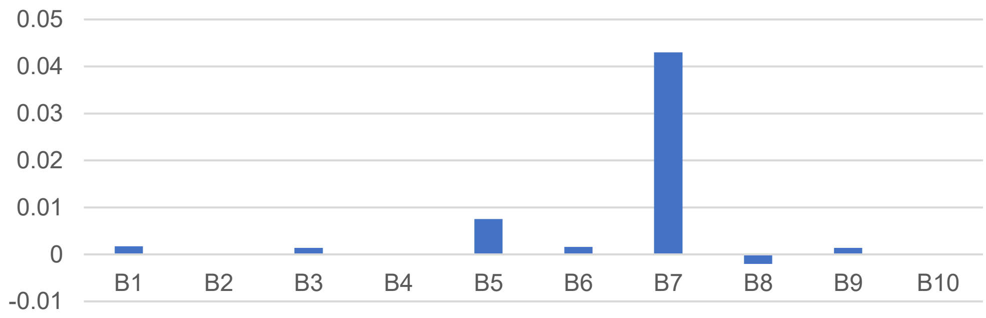

Fig. 4.3 Differences in accuracy rates of benchmark set 1

In general, for both performance indicators, BKS results are marginally higher than those obtained through majority vote. Similar results are seen in the F1-scores, as listed in Table 4.3. The averages were 0.9575 and 0.9523 for BKS and majority voting, respectively.

Table 4.3 F1-scores of benchmark set 1

Data set	**BKS**	**Voting**
B1	0.9863	0.9855
B2	0.9931	0.9931
B3	0.9715	0.9704
B4	0.7675	0.7700
B5	0.9539	0.9451
B6	0.9601	0.9588
B7	0.9782	0.9353
B8	0.9742	0.9753
B9	0.9911	0.9904
B10	0.9991	0.9990

In Fig. 4.4, a comparison between the F1-scores of BKS and voting is shown. Only one (B4) data set had voting acquiring slightly better results as compared to BKS, while for the remaining, BKS outperformed voting.

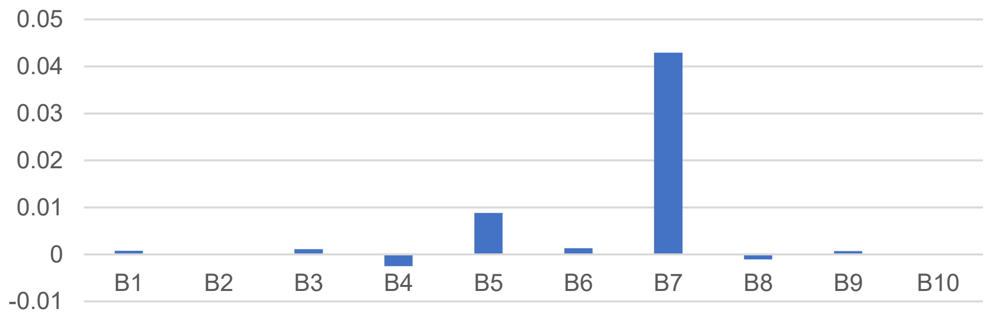

Fig. 4.4 Differences in F1-scores of benchmark set 1

The data samples are messed up with 10% and 20% more noise than normal so that the resilience of BKS may be evaluated. A total of 25 iterations are carried out for each data set, and Table 4.4 contains an average of the outcomes of these iterations. The numbers of victories attributable to the BKS in comparison to majority voting are listed. In general, it can be seen that the accuracy rate drops with the increase of noise. The drop for BKS however is slightly lower as compared to a bigger drop in voting, as with the increase of noise levels.

Table 4.4 Accuracy rates with and without noise of benchmark set 1

Data set	Noise	BKS	Voting
B1	0%	0.9734	0.9717
	10%	0.9402	0.9263
	20%	0.912	0.8923
B2	0%	0.9864	0.9864
	10%	0.9716	0.9625
	20%	0.9438	0.9296
B3	0%	0.9453	0.9439
	10%	0.9405	0.9221
	20%	0.9334	0.9027
B4	0%	0.7204	0.7205
	10%	0.7391	0.6859
	20%	0.7235	0.6822
B5	0%	0.9541	0.9466
	10%	0.9534	0.9483
	20%	0.9528	0.9460
B6	0%	0.9514	0.9498
	10%	0.8697	0.7874
	20%	0.8627	0.8150
B7	0%	0.9782	0.9352
	10%	0.9468	0.9226
	20%	0.9365	0.9004
B8	0%	0.9501	0.9521
	10%	0.9155	0.8967
	20%	0.8650	0.8236
B9	0%	0.9823	0.9809
	10%	0.9469	0.9341
	20%	0.8910	0.8709
B10	0%	0.9981	0.9980
	10%	0.9629	0.9535
	20%	0.9571	0.9230

A two-tailed sign test is performed to determine whether BKS performs statistically better than majority voting. Fig. 4.5 depicts the number of BKS victories over majority voting based on experimental data (plotted at 16 wins and above). The purple, blue, and orange lines show the number of wins required for a significance level of α = 0.1, 0.05, and 0.01, respectively in Fig. 4.5.

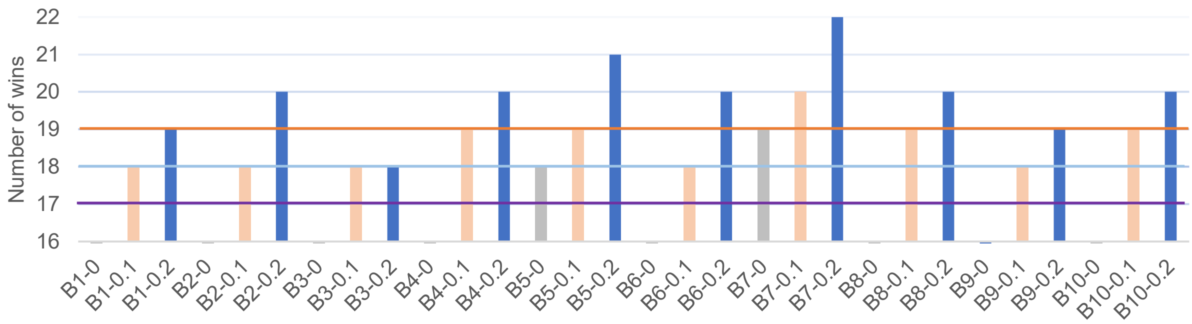

Fig. 4.5 Number of BKS wins in data sets with and without noise

In each of the ten noisy data sets (10% and 20% noise levels), BKS earns at least 18 victories out of 25 experimental runs, implying that it beats majority voting when dealing with noisy data samples for α = 0.05 (95% confidence level). When a stricter statistical significance criterion of α = 0.01 (i.e., 99% confidence level) is used for evaluation, BKS outperforms majority voting in 9 out of 10 data sets with a noise level of 20%. This finding shows that BKS is more effective than majority voting in reducing the negative impact of noise on performance.

Table 4.5 presents a comparison of the F1-score to the published results of GEP (Jedrzejowicz and Jedrzejowicz, 2020) and CUSBoost (Zhao *et al*., 2020). The purpose of this comparison is to determine whether or not BKS is more effective than other methods that have been described in literature. As compared to the results obtained by BKS and majority voting, CUSBoost (Zhao *et al*., 2020) achieves the worst performance, whereas GEP [26] achieves results that are comparable to those obtained by BKS. Overall, BKS receives the highest F1 ratings in four out of the six data sets, while the scores for the remaining two data sets are somewhat lower by 0.01 when compared with those that were acquired through majority voting.

Table 4.5 Comparison of F1-scores with literature (best in bold) of benchmark set 1

Dataset	BKS	Voting	GEP	CUSBoost
B1	**0.9863**	0.9855	0.9048	0.3231
B2	**0.9931**	0.9931	-	0.3363
B3	**0.9715**	0.9704	0.927	0.1809
B4	0.7675	**0.7700**	-	0.5543
B8	0.9742	**0.9753**	0.9005	0.0939
B9	**0.9911**	0.9904	0.8964	0.1674

4.3 Benchmark Experiments Set 2

The experiments make use of a total of 20 data sets. Table 4.6 shows the specifics of each data set, from C1 to C20, including the number of instances and features. The datasets of C1, C2, C4 to C16, C18 and C19 were sourced from UCI (2020), C3 from Kaggle (2020), and C17 and C20 and OpenML (2020).

Table 4.6 List and descriptions of benchmark set 2

Data set	Problem	Instances	Features
C1	adult	48,842	15
C2	Australian	690	14
C3	bank	4,521	17
C4	Breast_cancer	699	9
C5	Cervical_cancer	858	32
C6	chess-krvkp	3,196	37
C7	congressional-voting	435	16
C8	connect-4	67,557	43
C9	Cylinder-bands	540	39
C10	Diabetes	768	8
C11	german-credit	1,000	25
C12	Indian-liver	583	10
C13	magic	19,020	11
C14	Mammographic	961	5
C15	monks-3	3,190	6
C16	musk-2	6,598	167
C17	ringnorm	7,400	21
C18	spambase	4,601	58
C19	Transfusion	748	4
C20	twonorm	7,400	21

A wide range of instances, from 435 to 67,557, with an average of 8,980 instances. On the number of features, from 4 all the way to 167, with an average of 28 features. This shows that the experiments cover a wide range of examples, from simple to complex.

Experiments were conducted using CV with k = 10. The accuracy rates are shown in Table 4.7. The average scores for the 20 datasets were 0.8626 and 0.8511 for BKS and majority voting, respectively. Generally, BKS results are slightly higher than majority voting outcomes for both performance indicators.

Table 4.7 Accuracy rates of benchmark set 2

Data set	BKS	Voting
C1	0.8551	0.8536
C2	0.8823	0.8823
C3	0.9931	0.9207
C4	0.7447	0.7447
C5	0.9189	0.9117
C6	0.9767	0.9751
C7	0.6512	0.6226
C8	0.8882	0.8851
C9	0.8389	0.8207
C10	0.7204	0.7203
C11	0.7891	0.7877
C12	0.7671	0.7412
C13	0.8745	0.8732
C14	0.8366	0.8365
C15	0.8292	0.8222
C16	0.9985	0.9855
C17	0.9541	0.9466
C18	0.9514	0.9498
C19	0.7993	0.7982
C20	0.9821	0.9452

The number of instances of features do not have any effect on the accuracy rates, where some small size dataset such as C12 has an accuracy rate of 0.7671 as compared to C8 at 0.8882, which is over 100 times larger.

The accuracy rates of BKS and voting are compared in Figure 4.6. BKS outperforms voting in the majority of data sets, with some data sets yielding identical results. None of the other data sets had voting with greater accuracy than BKS.

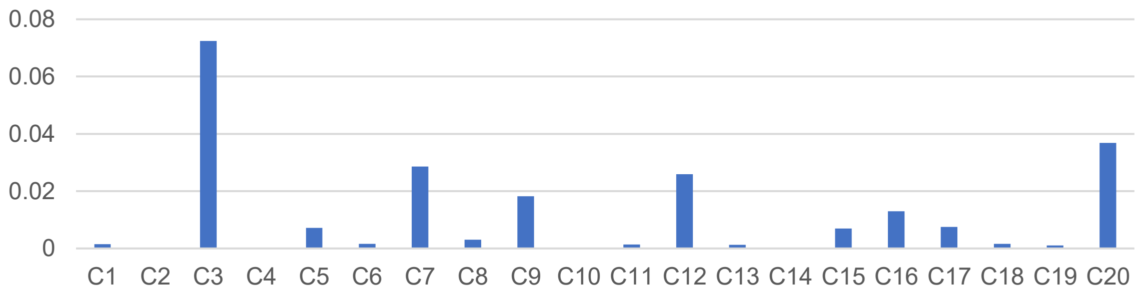

Fig. 4.6 Differences in accuracy rates of benchmark set 2

As with accuracy rates, the F1-scores for the 20 data sets were provided as well in Table 4.8. The scores showed a similar trend to the accuracy rates. BKS again had the higher scores, with 0.8483 as compared to 0.8412 with majority voting.

A comparison of the differences in the F1 scores is presented in Fig. 4.7. In every other data set, BKS either outperformed voting or produced the same outcomes as voting did. The only exception to this was C10, which had a greater accuracy rate than voting.

Table 4.8 F1-scores of benchmark set 2

Data set	BKS	Voting
C1	0.8439	0.8417
C2	0.8763	0.8763
C3	0.8875	0.8859
C4	0.6990	0.6969
C5	0.9191	0.9131
C6	0.9678	0.9659
C7	0.6512	0.6225
C8	0.8654	0.8640
C9	0.8246	0.8129
C10	0.7675	0.7700
C11	0.7788	0.7767
C12	0.7563	0.7363
C13	0.8509	0.8494
C14	0.7853	0.7829
C15	0.8289	0.8235
C16	0.9783	0.9763
C17	0.9539	0.9451
C18	0.9601	0.9588
C19	0.7922	0.7905
C20	0.9782	0.9353

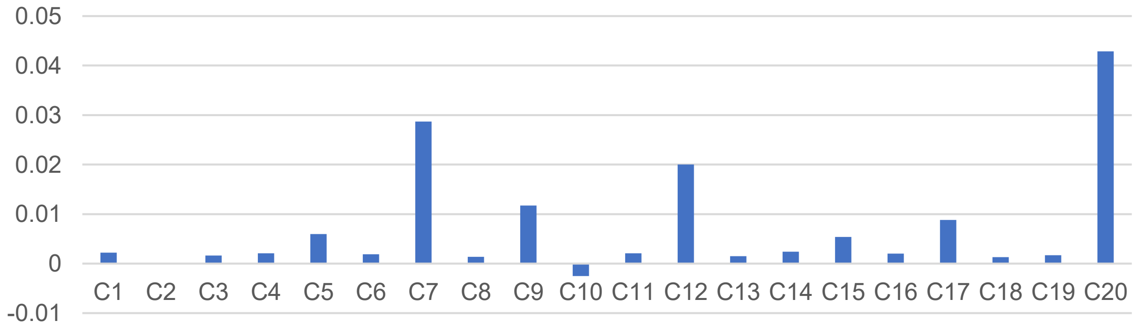

Fig. 4.7 Differences in F1-scores rates of benchmark set 2

As with the previous experiment, for 20 different experimental runs, a classifier needs to have at least 15 wins (Derrac *et al*., 2011) to indicate its superior performance over another classifier for $\alpha = 0.05$ (95% confidence level). As such, a number of literatures were used for comparison of the results.

The first comparison was from Hu *et al*. (2022), which used a total of 9 different algorithms. Algorithms used and compared with include Layer-wise Randomized Feedforward Networks (ALR), Ensemble Deep Randomized Vector Functional Link (edR), Residual networks (ResNet), and Self normalizing networks with SELUs (SNN).

The second comparison was done with the work in Shi *et al*. (2022). A total of 15 algorithms were used, which includes Pruning based edRVFL with re-normalization (PedR), Weighting based edRVFL with re-normalization (WedR), and Combination of PedRVFL and WedRVFL with renormalization (WPedR).

Third and final comparison was made with the work in Shiue *et al*. (2021). A 10-fold CV was used in the experiments. The algorithms used included AdaBoost (AB), Diversity in ensembles using artificial data (Deco), ensemble learning approach based on balanced accuracy and diversity (ELBAD), extremely randomized trees (ET), and rotation forest (RoF).

Comparisons of the results are shown in Table 4.9. For C1 to C4, C7 to C9, C11 to C16, and C18 to C20, BKS was the winner, which was 16 of the 20 datasets. In C5, RF (Shiue *et al*., 2021) acquired accuracy rates of 0.07 higher as compared to BKS. In C6, WPedR (Shi *et al*., 2022) was superior to BKS, by a very close margin. In C10, RoF (Shiue *et al*., 2021), acquired accuracy rates of 0.04 higher from BKS. In C17, ALR (Hu *et al*., 2022) was superior to BKS by 0.03.

It can be seen while BKS lost for some datasets, its accuracy rates were very close to the winner. In total, BKS had 16 wins from the 20 data sets, claiming its superiority in terms of $\alpha = 0.05$ significance.

Table 4.9 Comparison of accuracy rates with literature (best in bold) of benchmark set 2

				Hu *et al.* (2022)				Shi *et al.* (2022)				Shiue *et al.* (2021)			
Dataset	**BKS**	**Voting**	**ALR**	**edR**	**Resnet**	**SNN**	**PedR**	**WedR**	**WPedR**	**AB**	**Deco**	**ELBAD**	**ET**	**RF**	**RoF**
C1	**0.8551**	0.8536	0.8525	0.8525	0.8484	0.8476	0.8542	0.8530	0.8546						
C2	**0.8823**	**0.8823**								0.8696	0.8519	0.8710	0.8616	0.8657	0.8657
C3	**0.9931**	0.9207	0.9260	0.8960	0.8796	0.8930	0.9190	0.9920	0.9114						
C4	**0.7447**	0.7447								0.7411	0.7229	0.7431	0.7310	0.7136	0.7225
C5	0.9189	0.9117								0.9138	0.9340	0.9152	0.9110	**0.9880**	0.9610
C6	0.9767	0.9751	0.9912	0.9912	0.9912	0.9912	0.9933	0.9936	**0.9947**						
C7	**0.6512**	0.6226	0.6330	0.6110			0.6193	0.6191	0.6216						
C8	**0.8882**	0.8851	0.8251	0.8477	0.8716	0.8870									
C9	**0.8389**	0.8207								0.7648	0.6993	0.8267	0.8300	0.8360	0.7970
C10	0.7204	0.7203								0.7370	0.7590	0.7532	0.7492	0.7573	**0.7644**
C11	**0.7891**	0.7877	0.7880	0.7770	0.7720	0.7560									
C12	**0.7671**	0.7412								0.7167	0.6950	0.7254	0.7219	0.7660	0.7113
C13	**0.8745**	0.8732	0.8683	0.8681	0.8723	0.8692									
C14	**0.8366**	0.8365								0.8262	0.8218	0.8234	0.7780	0.7869	0.8340
C15	**0.8292**	0.8222					0.8198	0.8290	0.8235						
C16	**0.9975**	0.9855	0.9939	0.9857	0.9964	0.9891	0.9928	0.9954	0.9974						
C17	0.9541	0.9466	**0.9838**	0.9797	0.9811	0.9751									
C18	**0.9514**	0.9498	0.9426	0.9387	0.9461	0.9490	0.9418	0.9411	0.9472						
C19	**0.7993**	0.7982								0.7861	0.7810	0.7736	0.7297	0.7388	0.7948
C20	**0.9821**	0.9452	0.9811	0.9781	0.9735	0.9850									

4.4 Summary

Two frameworks were used, which consisted of the majority voting and hierarchical agent-based. Both frameworks used three groups, where each contained three agents of RF, GLM and GBM. For the hierarchical framework, each agent manager had its own BKS module. The final predicted class is generated by sending the predictions from the three agent managers to the decision combination module, which employs yet another BKS.

Details of each hierarchical framework including the train and test details were described. In the first set of experiments, 10 datasets from KEEL (2020), UCI (2020) and Kaggle (2020). The datasets had different imbalance ratios. The average accuracy of BKS was at 0.9440, which was slightly higher than majority voting. Comparison with literature was made as well. BKS received the greatest F1-scores in four of six data sets, while the values in the remaining two are 0.01 lower than those of majority voting.

In evaluating the robustness of BKS, the data was corrupted with up to 20% noise. 25 runs were conducted for each dataset. In all ten noisy data sets (10% and 20% noise levels), BKS won at least 18 of the 25 experimental runs, proving its superior performance over majority voting in noisy data sets for $\alpha = 0.05$ (95% confidence level).

In addition to the first set, a second set of experiments with 20 datasets were used. The datasets were sourced from UCI (2020), Kaggle (2020), and OpenML (2020). The average accuracy rates of BKS were 0.8626, which was higher than majority voting of 0.8511. Similar results are seen in the F1-scores as well. Comparison of the results was made with the literature. Out of the 20, BKS was the winner for 16 of the datasets. In the other 4, BKS acquired results that were close to the winner. Overall, BKS claims its superiority in terms of $\alpha = 0.05$ significance.

5 Real-world Data and Discussions

In order to conduct this analysis, real financial transaction records were obtained from a financial company in Southeast Asia. The records were collected from September to November 2017. Jiang and Yu (2018) emphasise that Southeast Asia has been experiencing strong economic growth, with a GDP growth rate that is over 6%, which makes it a region of interest for the purpose of financial research because it has been growing at such a quick rate.

The dataset for the research consists of a total of 60,595 transaction records gathered from 9,685 different customers. The business takes place across more than 23 countries and involves a diverse array of activities, including everything from making purchases on websites to shopping for groceries.

The company has found evidence of fraudulent activity in 28 of the 60,595 transactions it has processed. The remaining transactions are divided into two categories: those that involve fraud and those that do not involve fraud. It is most likely that the purpose of this study is to locate patterns or abnormalities within the information that will assist the company in spotting fraudulent acts in the future and preventing them from occurring.

5.1 Data Pre-processing

The precision and reliability of any analysis are affected by the quality of the data. Pre-processing data is the first step in assuring high-quality data. Real-world data is frequently incomplete and contains inaccuracies, making analysis difficult. As a result, prior to analysis, the data must be cleaned and transformed.

Data cleansing is an important step in the pre-processing of data. It entails discovering and dealing with missing values, deleting duplicates, and dealing with outliers. Missing data can pose substantial issues, resulting in biased or erroneous conclusions, and should thus be handled carefully. Outliers are data points that go significantly beyond the expected range of values and might skew conclusions if not handled properly.

Variable selection is frequently required in larger datasets to reduce computing effort and avoid overfitting. This entails finding and removing irrelevant or redundant factors while identifying and selecting the most relevant variables that contribute to the analysis. Overall, data cleaning, handling missing values, dealing with outliers, and variable selection serve to assure the quality of the data for accurate and trustworthy analysis.

5.1.1 Data Cleaning

Missing values are a prevalent problem in real-world data, and their existence can have a big impact on the accuracy and validity of data analysis outcomes. The empty values can occur for a variety of reasons, including data entry problems, storage device malfunction, or incomplete surveys. Missing values can render a record incomplete, rendering the results invalid.

One typical strategy for dealing with missing values is to use imputation algorithms to replace them. These methods rely on statistical techniques such as mean, median, or mode imputation or regression-based imputation to approximate missing data. However, the technique of imputation used is determined by the nature and distribution of the missing values.

The total number of missing values in the dataset in the cited thesis was less than 0.1% of the overall dataset, which is relatively low. As a result, the impacted rows were excluded from the experiment to ensure that only complete and original records were used. This method can help to reduce data bias and avoid introducing errors or distortions into the analysis results.

Finally, missing values are a regular problem in real-world data, and how they are handled is critical to ensuring the validity and integrity of data analysis outcomes. Various imputation approaches can be used to replace missing values, depending on the nature and extent of the missing data. However, if the number of missing values is very low, it may be desirable to exclude the affected rows in order to maintain data integrity.

5.1.2 Feature Selection

Each transaction record contains the following information: account number, transaction amount, date, time, device type used, merchant category code (MCC), country, and transaction type. Customers' account numbers are anonymised to protect their privacy. Together with the nine initial features, feature aggregation is used to create eight new features. Throughout a three-month period, these aggregated features use the transaction amount, acquiring nation, MCC, and device type. Table 5.1 provides a summary of the features.

Table 5.1 List of features and description

No	Features	Description
1	Account Number	Account number anonymized
2	Transaction Amount	The amount of money spent on the transaction
3	Transaction Date	The transaction's date
4	Transaction Time	The date and time of the transaction
5	Device Type	The type of transaction device
6	MCC	Code for the merchant category
7	Acquiring Country	Countries in which the transaction took place
8	For Country	The country where the card was issued
9	Transaction Type	Purchase or cancellation
10	Transaction Amount Count	Transaction totals by cardholder
11	Transaction Amount Sum	Total number of transactions by cardholder
12	Acquiring Country Count	Count of distinct acquiring countries
13	Acquiring Country Sum	Amount paid by the acquiring country for the transaction
14	MCC Count	Total number of MCC
15	MCC Sum	MCC total for a specific transaction
16	Device Type Count	The number of distinct gadget types used
17	Device Type Sum	The total of the specific device types used for the transaction

Feature significance scores can provide useful data set information. The scores might emphasise the significance of each feature for classification. A feature significance research is conducted on the 17 features using the DT, RF, and XGB classifiers.

The findings are depicted in Fig. 5.1. It can be seen that all of the variables have varying degrees of value, with feature 12 (the number of unique acquiring countries) appearing to be the most important factor in all three classifiers. When compared to the original features, the remaining aggregated features (features 10–17) have slightly better relevance scores.

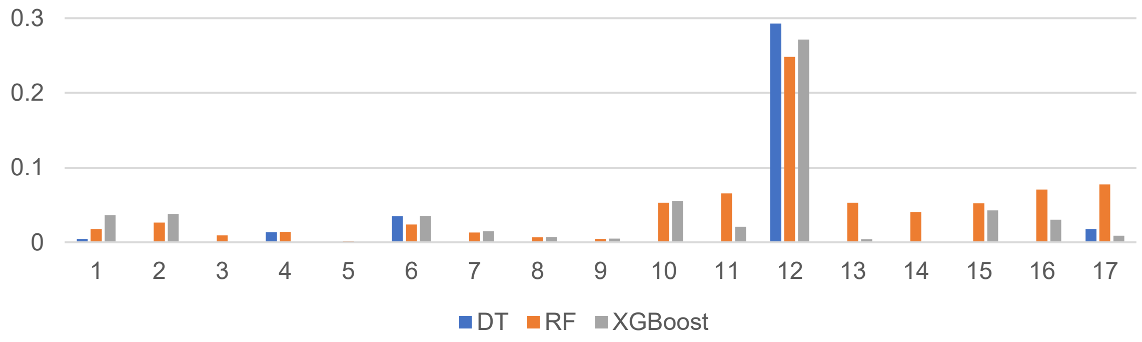

Fig. 5.1 Feature importance using DT, RF, and XGB

5.1.3 Data Imbalance

In practice, the number of financial fraud cases are very small as compared with genuine cases. As noted at the start of the chapter, 28 of the 60,595 transaction records were fraudulent, hence there is a great imbalance in the data. There are a number of sampling methods for imbalanced data, which include under and over-sampling. Over-sampling is used when a class of data is the under-represented minority class. The Synthetic Minority Oversampling Technique (SMOTE) is a popular oversampling method that generates synthetic samples by randomly sampling attributes from minority class occurrences.

If a class of data is over-represented majority, under-sampling can be used in balancing it with the minority class. This can be used when the amount of collected data is enough. In this thesis, under-sampling using two ratios were evaluated, as discussed in the following sub-chapter. An example of how undersampling is done using Python is given in Algorithm 5.1. The RandomUnderSampler package is utilized, which does the sampling using the set ratio.

Algorithm 5.1 Undersampling of data

```
from imblearn.under_sampling import RandomUnderSampler
from collections import Counter

# define undersample strategy
undersample = RandomUnderSampler(sampling_strategy=0.01)

# fit and apply the transform
X_over, y_over = undersample.fit_resample(X, y)

# summarize class distribution
print("New class:",Counter(y_over))
```

5.2 Experimental Setup

The hierarchical agent-based framework utilised in the studies was explained in Chapter 4.1 and is shown in Fig. 4.2. It is divided into three categories, each with three agents. Random Forest (RF), Generalized Linear Model (GLM), and Gradient Boosting Machine (GBM) are the three agents chosen after comprehensive testing of individual and group performance. Three agent managers are set up, each with their own BKS module. These three agent managers' predictions are passed to the decision combination module, which uses another BKS to construct the final projected class.

Assessing the robustness of a machine learning model is a crucial step in evaluating performances. Conducting multiple experiments across a variety of environments can help identify weaknesses and limitations, and inform strategies for improving the model's accuracy and reliability.

5.3 Real-world Experiments Set 1

Feature aggregation can be a useful strategy in credit card fraud detection as it allows to combine several raw features into a smaller collection of more meaningful features that can capture the underlying patterns and relationships in the data. As an example, the original transaction data only contained a total of 9 raw features, as listed in Table 5.1. An additional 8 features were aggregated, such as the total spending of the cardholder. The features which were selected in the comparisons are given in Table 5.2.

Table 5.2 Features selection

Set	Type	Features
2-1	Original	1-9
2-2	Original + Aggregated	1-17

For example, the additional features you collect may aid in the detection of unexpected spending patterns or transactions that depart from a cardholder's customary spending habits, indicating potential fraud. Furthermore, by aggregating transaction frequency, time, and location information, you may be able to spot patterns associated with fraudulent conduct, such as several transactions made in a short period of time or transactions made from multiple places in a short period.

The experimental outcomes are presented in Table 5.6. Clearly, the use of aggregated features significantly aids the classifier in making a more accurate determination.

Table 5.3 Accuracy rates with different features used

Set	BKS	Voting
2-1	0.9352	0.9263
2-2	0.9765	0.9667

Especially for the detection of fraudulent transactions, a 3 to 4% increase in accuracy rate is crucial. Feature aggregation can help improve the accuracy and efficacy of machine learning algorithms in credit card fraud detection by reducing data dimensionality, identifying fraudulent behaviour patterns, and mitigating the effects of noise and redundancy. As a result, all subsequent experiments will incorporate the inclusion of aggregated features.

5.4 Real-world Experiments Set 2

Similar to the benchmark data experiment, noise is applied to this real-world data collection in increments of 10% to 40%. Table 5.4 summarises the findings. When the level of noise grows, BKS outperforms majority voting, proving its robustness to noisy data.

Table 5.4 Accuracy rates and BKS wins with noise added

Noise	**BKS**	**Voting**	**BKS Wins**
0%	0.9765	0.9667	0
10%	0.9705	0.9555	9
20%	0.9560	0.9367	**18**
30%	0.9317	0.9107	**18**
40%	0.9238	0.9008	**19**

When the noise level is 20% or higher, BKS wins by 18 times (20% and 30% noise) and 19 times (40% noise) to majority voting. This result indicates that BKS outperformed majority voting at 95% confidence level (α = 0.05) when dealing with noisy data (20% noise and above) in this real-world experiment.

The F1-scores obtained from the experiments are detailed in Table 5.5. When there is no additional noise, the F1-scores for BKS and voting are identical to one another. Again, when dealing with noisy data sets, BKS almost always ends up with superior F1-scores as compared to those that result from majority voting.

Table 5.5 F1-scores with noise added

Noise	**BKS**	**Voting**
0%	0.9807	0.9734
10%	0.9752	0.9679
20%	0.9660	0.9587
30%	0.9495	0.9424
40%	0.9435	0.9364

Two experiments with under-sampling approaches are carried out in addition to the studies involving additive noise. There are two alternative ratios of minority (fraud transactions) to majority (genuine transactions) that are analysed, which are 1:100 and 1:500 respectively, and Table 5.6 displays the overall findings of each of these evaluations.

Table 5.6 Accuracy rates with different ratios of minority to majority samples

Sampling	**BKS**	**Voting**
Original	0.9765	0.9667
1:100	0.9767	0.9763
1:500	0.9775	0.9762

The performance of the BKS classifier sees very slight gains when the under-sampling ratio is set to 1:100. Based on this ratio, it appears that there are 100 instances of the majority class that are preserved for every one occurrence of the minority class. This results in a dataset that is more evenly distributed, which enables the BKS classifier to more successfully capture the patterns that are present in both classes.

Under-sampling is a beneficial method for handling imbalanced datasets, but the extent to which it is effective is contingent on a number of aspects, such as the classifier that was used, the structure of the data, and the ratio of under-sampling. Even though there is a possibility that under-sampling will not always increase the performance of the voting classifier, it has the potential to marginally improve the performance of the BKS classifier by balancing the dataset and lowering the bias towards the majority class.

5.5 Real-world Experiments Set 3

Some adjustments are made to the classifier design in this series of real-world studies. Previously, three groups were employed in set 1, as illustrated in Figs. 4.1 and 4.2 for majority voting and BKS, respectively. The same grouping, but with three RF, GLM, and GBM agents in each location. Instead of the three groups, it was decided to utilise five, with two additional groups made up of the same agents as the first two.

The number of classifiers utilised in a machine learning algorithm can have a substantial impact on the accuracy and dependability of the model. In general, increasing the number of classifiers can enhance model accuracy up to a point.

Using five classifiers instead of three can enhance accuracy since it provides a more diverse range of opinions regarding each data point's classification. There is a larger likelihood of a tie between two classifiers with three classifiers, resulting in a less conclusive decision. With five classifiers, the majority vote is more likely to be decisive, lowering the possibility of a tie and improving the model's accuracy.

Fig. 5.2 depicts the changed setup for the majority voting framework. It depicts five groups that feed into a decision combination. Fig. 5.3 also depicts the hierarchical agent-based structure for BKS. The predictions of the five agent managers are forwarded to the decision combination module, which applies another BKS to construct the final projected class.

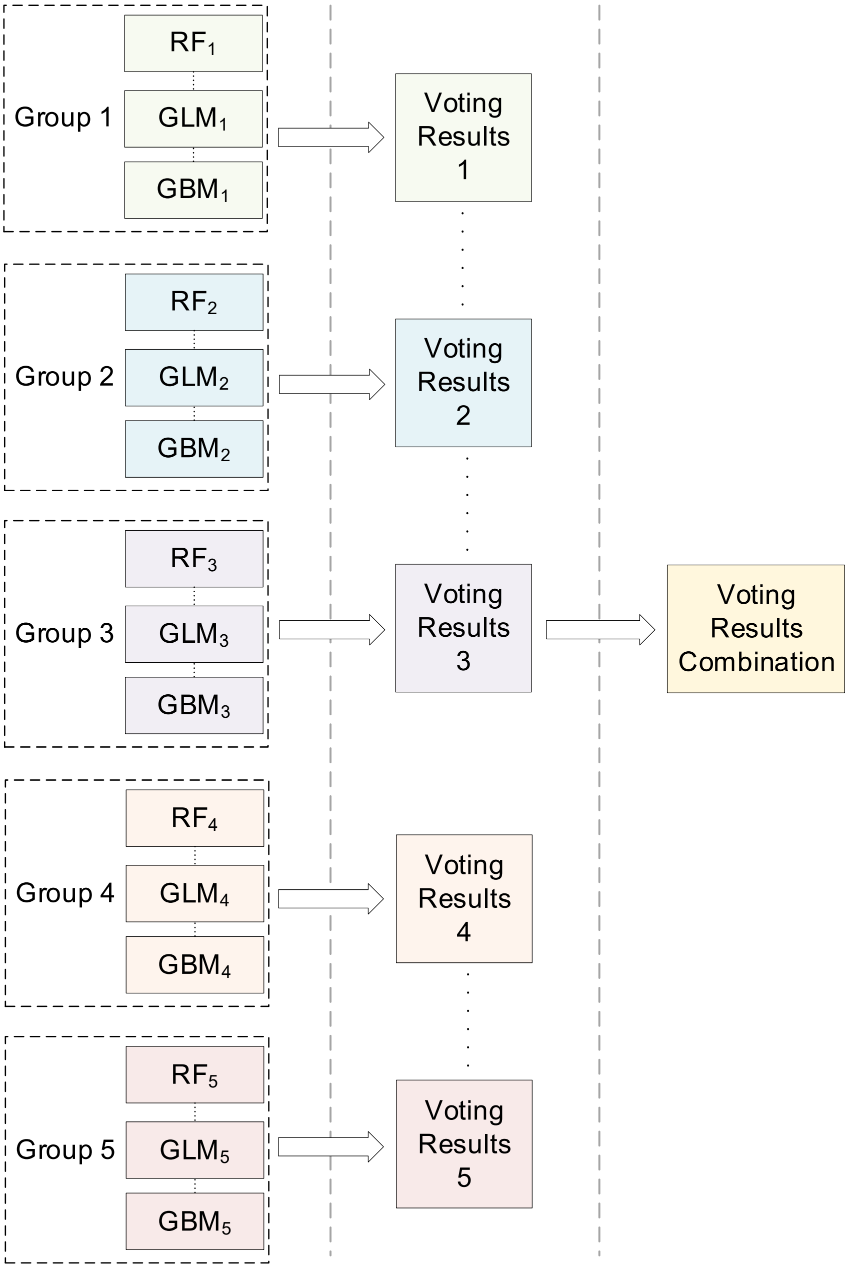

Fig. 5.2 Configuration of the updated majority voting framework used in the experiments

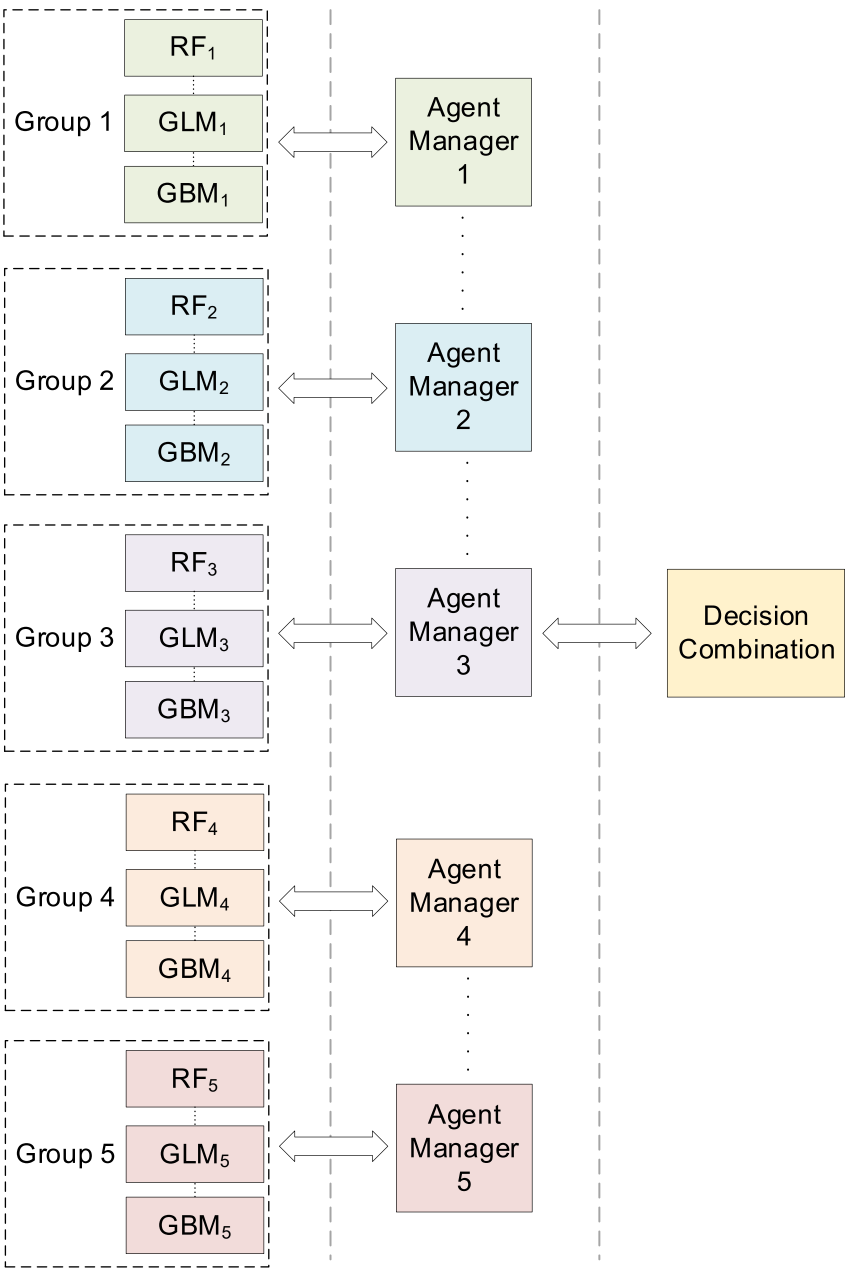

Fig. 5.3 Configuration of the updated hierarchical agent-based framework used in the experiments

As shown in Fig. 5.3, training begins with randomised ordering of data samples and is followed by a validation process. This results in five group-based BKS modules (one for each group). The outputs of BKS modules 1 to 5 are then coupled with another randomised sequence utilising training data, resulting in the construction of another overall (final) BKS module that combines the outputs of the previous three group-based BKS modules. The outputs of the group-based BKS modules are coupled with the outputs of the overall BKS module given a test sample to provide a final projected class for the computation of performance metrics such as classification accuracy and F1-score.

Results from the experiments are summarized in Table 5.7. Similar trend is seen from set 1, as in Table 5.2. When the noise levels are 10% and below, BKS does not achieve any significant wins. However, as the noise levels increase from 20% to 40%, BKS outperforms majority voting by up to 19 times. This demonstrates that BKS outperforms majority voting statistically at 95% confidence level (α = 0.05) while dealing with noisy data (20% noise and above) in this real-world scenario.

Table 5.7 Accuracy rates and BKS wins with noise added for updated model

Noise	BKS	Voting	BKS Wins
0%	0.9766	0.9668	0
10%	0.9706	0.9557	9
20%	0.9567	0.9374	**18**
30%	0.9320	0.9109	**18**
40%	0.9239	0.9009	**19**

Similar trend is seen in the F1-scores of the experiments, which is listed in Table 5.8. As noise is gradually increased from 10% to 40%, BKS consistently achieves higher F1-scores, as compared with those from majority voting. The scores are similar to those listed in Table 5.3.

Table 5.8 F1-scores with noise added for updated model

Noise	BKS	Voting
0%	0.9808	0.9730
10%	0.9754	0.9669
20%	0.9666	0.9537
30%	0.9498	0.9359
40%	0.9436	0.9303

Based on the results in Tables 5.7 and 5.8, the scores are identical to those in Tables 5.2 and 5.3, respectively. The addition of two additional groups does not provide any overall advantage in terms of the accuracy rates and F1-scores, instead it increases the computational complexity and the time taken for a decision.

As the trials shown that the outcomes of the BKS may be the same regardless of the number of groups used, the next revision of the BKS will use only three groups rather than five. This is done to lighten the load on the computer and cut down on the total amount of time needed.

5.6 Modified BKS

Based on the BKS detailed in Chapter 3.2.2, modifications to the BKS are made to further improve its efficacy. An example comprising 2 agents from Chapter 3.2.2 is presented to explain its mechanism, as presented in Table 5.9.

Table 5.9 Prediction outputs of Agents 1 and 2

Predicted Class		Actual class	Class count	Total count	Vote class
Agent 1	Agent 2				
0	0	0	10	10	0
0	1	0	10	10	1
1	0	1	10	10	1
1	1	1	10	10	1

The combined outcome is based on a simple majority voting scheme. Consider a given input sample that belongs to class 0, when agent 1 predicts class 0 while agent 2 predicts class 1, based on Table 5.9, the voting outcome is class 1.

In Chapter 3.2.2, the BKS leverages the predicted class of 0 or 1 for decision combination. In the first modification, the confidence level associated with each predicted class is exploited. An example is shown in Table 5.10. As such, when the predicted outcome of class 0 with a confidence level of 0.80 is selected, then the information of “class 0” and “0.80” is forwarded for the next level of decision combination, instead of just the predicted outcome of class 0.

Table 5.10 Values received by Agent Manager 1

Predicted Class (Confidence Level)	
Agent 1	Agent 2
0 (0.80)	0 (0.60)
0 (0.76)	1 (0.55)
1 (0.58)	0 (0.65)

It can be seen that the confidence level differs for each prediction, indicating an agent's certainty in making that particular prediction. Referring to the first row in Table 5.10, when both agents make the same prediction (class 0), the combined prediction is straightforward, i.e., class 0. When conflicting predictions are made by both agents, as in the second and third rows in Table 5.10, the confidence level plays a role. In the second row, the predicted outcome is class 0, since it has a higher weight as compared with that of class 1.

The use of confidence levels contributes significantly to the BKS method since it allows for a more sophisticated approach to decision combination. The BKS can better assess the certainty of each prediction and make more informed judgements by evaluating the confidence level associated with each anticipated class. Furthermore, the different levels of confidence for each forecast underscore the significance of this approach. Overall, confidence levels are a useful tool for increasing the accuracy and dependability of the BKS approach.

By manipulating the confidence information, several experiments by varying a threshold pertaining to the confidence scores are conducted. In the second modification, the threshold of confidence level for accepting a predicted outcome is varied from 0.60 to 0.90, with an increment of 0.05. The setting of threshold is listed in Algorithm 5.2.

Algorithm 5.2 Setting of threshold

```
threshold = [0.60, 0.65, 0.70, 0.75, 0.80, 0.85, 0.90]
```

After setting of the threshold, predictions are combined into a single data frame, as listed in Algorithm 5.3. Should there be no result, an X is marked as the prediction, as detailed in the following paragraphs.

Algorithm 5.3 Prediction combination in BKS

```
# BKS table #1: combining prediction into single dataframe

combine_results = pd.concat([results_ml1v1, conf_ml1v1, results_ml4v1,
conf_ml4v1, results_ml5v1, conf_ml5v1, validate_actual1], axis=1, sort=False)

combine_results.columns = ['out_1', 'cf_1', 'out_2', 'cf_2', 'out_3', 'cf_3',
'actual']

combine_results['out_1a'] = np.where(combine_results['cf_1'] >= threshold,
combine_results['out_1'], X)

combine_results['out_2a'] = np.where(combine_results['cf_2'] >= threshold,
combine_results['out_2'], X)

combine_results['out_3a'] = np.where(combine_results['cf_3'] >= threshold,
combine_results['out_3'], X)
```

Considering a total of 40 samples and 2 agents, the results based on two thresholds of confidence level of 0.70 and 0.90 are shown in Tables 5.11 and 5.12, respectively.

Table 5.11 Prediction outputs at threshold level of 0.70

Predicted Class		**Actual class**	**Class count**	**Total count**	**Vote class**
Agent 1	**Agent 2**				
0	0	0	10	10	0
0	1	0	10	10	1
1	0	1	6	7	1
1	1	1	6	6	1
X	0	1	3	3	0
X	1	1	3	4	1

Table 5.12 Prediction outputs at threshold level of 0.90

Predicted Class		Actual class	Class count	Total count	Vote class
Agent 1	Agent 2				
0	0	0	7	7	0
0	1	0	6	6	1
1	0	1	7	7	1
1	1	1	6	6	1
X	0	0	2	3	0
0	X	1	2	3	0
X	1	0	4	4	1
1	X	1	1	3	1
X	X	1	1	1	X

The threshold of confidence level at 0.70 means that any predicted outcome with a confidence level lower than 0.70 is rejected, hence the values in row 3 of Table 5.10 is marked as "X" (i.e., non-prediction) for both agents 1 and 2. As the threshold of confidence level increases, the number of non-predictions increases. The number of incorrect predictions or non-predictions for the simple majority voting scheme increases as well.

As the threshold of confidence level is set higher, the confidence of an agent is higher, hence the predictions result in a higher accuracy rate, as detailed in the next sub-chapter. In the third modification, as the number of non-predictions increases, the coverage of a classifier decreases.

The use of threshold levels contributes significantly to the decision-making process since it provides for more exact control over the acceptance or rejection of projected outcomes. Setting a confidence level threshold of 0.70, for example, means that any projected outcome with a confidence level less than 0.70 is rejected, leading in a higher accuracy rate, as shown in the next sub-chapter. However, as the threshold is raised, the frequency of non-predictions rises, potentially affecting a classifier's coverage. It is vital to note that the accuracy rate of the simple majority voting system improves with increasing confidence level, but so does the frequency of wrong or non-predictions. As a result, determining the best threshold level is critical for balancing accuracy and coverage. Overall, the usage of threshold levels is a significant technique for increasing decision-making precision and control.

In general, the total number of input samples that receive a class label, is defined by

$$\boldsymbol{s} = \sum_{i=1}^{c}\sum_{j=1}^{c}\boldsymbol{m_{i,j}} \tag{5.1}$$

Then, the number of non-predicted samples is determined using

$$\boldsymbol{u} = \sum_{i=c}^{c+3}\sum_{j=1}^{c}\boldsymbol{m_{i,j}} \tag{5.2}$$

The coverage of a classifier can be calculated using

$$\boldsymbol{\Omega} = \frac{\boldsymbol{s}}{\boldsymbol{s}+\boldsymbol{u}} \tag{5.3}$$

The coverage of samples is an important factor in machine learning since it defines the proportion of the overall dataset that is accurately represented by a given model or classifier. While using confidence levels and threshold levels can improve accuracy, they can potentially restrict the classifier's coverage. Setting a higher confidence level criterion, for example, may result in reduced coverage since more samples are rejected due to low confidence levels. This trade-off between coverage and accuracy must be carefully evaluated, since decreasing coverage may result in the loss of crucial information as well as potential model biases. As a result, striking a balance between coverage and accuracy is critical to ensuring that the classifier is both reliable and representative of the dataset. Overall, while confidence levels and threshold levels are useful tools in machine learning, it is critical to examine the influence on coverage and strive for a balance of coverage and accuracy.

5.7 Real-world Experiments Set 4

Based on the modifications proposed in the previous sub-chapter, experiments are conducted using three groups containing three agents. The three agents utilized are the same as previous, which consists of RF, GLM and GBM. Three agent managers are established, each with a BKS module. The prediction is passed to the decision combination module, which has another BKS to construct the final projected class based on the adjustments from these three agent managers.

A total of seven threshold levels are set. It starts from 0.60 all the way to 0.90, in increments of 0.05. Results are summarized in the following tables. The accuracy rates are first given in Table 5.13.

Table 5.13 Accuracy rates by threshold for modified model

Threshold	**BKS**	**Voting**
0.60	0.9848	0.9676
0.65	0.9862	0.9581
0.70	0.9863	0.9487
0.75	0.9862	0.9489
0.80	0.9931	0.9322
0.85	0.9936	0.9231
0.90	0.9983	0.9153

In general, as the threshold level increases, the accuracy rates for BKS increases. However, the opposite is seen for majority voting. As an example, at from threshold level of 0.6 to 0.9, the accuracy rate of BKS increases from 0.9848 to 0.9983, while voting reduced from 0.9676 to 0.9153. The jump in accuracy rates is not on a linear scale, instead it hovers around 0.9862 level for threshold levels of 0.65 to 0.75. Same goes with majority voting. At certain threshold levels, the accuracy rates remain almost the same. Only with a big jump of threshold levels, a significant jump can be seen.

The F1-scores of the same experiments are given in Table 5.14. A similar trend can be seen from Table 5.11. As with the other experiments, the F1-scores follow the same trend as the accuracy rates.

Table 5.14 F1-score by threshold for modified model

Threshold	BKS	Voting
0.60	0.9891	0.9715
0.65	0.9901	0.9629
0.70	0.9883	0.9525
0.75	0.9881	0.9536
0.80	0.9941	0.9364
0.85	0.9946	0.9272
0.90	0.9993	0.9180

Based on the Tables 5.13 and 5.14, the best threshold rate is at 0.90, with the highest accuracy rate and F1-score. Using threshold level of 0.90, an additional experiment with noise added, with levels of 10% to 40%, in increments of 10% was done. Results are given in Tables 5.13 and 5.14. The -t in the tables indicate results from the new experiment, using the threshold.

In Table 5.15, the accuracy rates using threshold of 0.90 can be seen to be at 0.02 higher as compared to without the threshold. At 40% noise level, the results are still at 0.9444, as compared to 0.9238. However, the reverse is seen for voting. As the threshold level increases, the number of samples available for voting reduces, hence the number of available samples drop, which translates to a lower accuracy level.

Table 5.15 Accuracy rates with noise added for modified model

Noise	BKS-t	Voting-t	BKS	Voting
0%	0.9983	0.9153	0.9765	0.9667
10%	0.9922	0.9048	0.9705	0.9555
20%	0.9779	0.8874	0.9560	0.9367
30%	0.9527	0.8624	0.9317	0.9107
40%	0.9444	0.8529	0.9238	0.9008

The same trend can be seen for the F1-scores, as in Table 5.16. F1-scores at threshold level of 0.90 are higher than those without threshold, while for voting, it is at the reverse. It is evident that the use of threshold level, and the higher the threshold is, it does help to increase the accuracy rates and F1-scores, making it a better classifier overall.

Table 5.16 F1-scores with noise added for modified model

Noise	BKS-t	Voting-t	BKS	Voting
0%	0.9993	0.9180	0.9807	0.9734
10%	0.9938	0.9123	0.9752	0.9679
20%	0.9848	0.8998	0.9660	0.9587
30%	0.9677	0.8830	0.9495	0.9424
40%	0.9613	0.8778	0.9435	0.9364

As of the large volume of transactions performed daily, computational speed is critical for fraud detection in credit card transactions utilising machine learning. Traditional fraud detection methods require manual review, which is a time-consuming process and costly.

By analysing massive volumes of data and discovering patterns and abnormalities that may suggest fraudulent conduct, machine learning algorithms can aid in the detection of fraud. However, in order to be effective, these algorithms must process transactions in real time. Customers and institutions might suffer financial losses and reputational damage if fraudulent transactions are not detected quickly. As a result, quick and efficient transaction processing is important for accurate fraud identification and prevention.

The computational time for detecting credit card fraud using machine learning versus human approaches can vary based on a number of factors, including the size and complexity of the dataset, the algorithms used, and the hardware and software infrastructure. Manual approaches for detecting credit card fraud often include human analysts evaluating transaction data, which can be time-consuming and error-prone.

Machine learning-based fraud detection systems, on the other hand, can analyse vast volumes of transaction data in real-time, offering faster and more accurate results. However, depending on the complexity of the methods and the quantity of the dataset, the computing time for training and optimising machine learning models might be significant. Furthermore, machine learning-based fraud detection systems necessitate continual monitoring and updating to assure accuracy and efficacy, which can necessitate additional computational time and resources.

Computational speed for two different thresholds is computed, with the results given in Table 5.17. The computational time is for 2,500 real-world test data set only, excluding all training time. Only test time is taken as the model would have been trained beforehand.

Table 5.17 Computational time (in seconds) for models

Threshold	BKS	Voting
0.60	1.92	1.68
0.90	1.76	1.53

It can be seen that voting takes slightly less computing time than BKS. This is primarily due to the ease with which predictions can be made in voting as opposed to reading up tables in BKS. However, the somewhat longer calculating time can be mitigated by the greater accuracy rates obtained. One method of reducing the time is to use parallel processing, which does multiple computations simultaneously. The distribution of computation will greatly reduce the time by almost half, which means that accurate predictions can be made quickly in the real-world environment.

5.8 Discussions

Machine learning fraud detection for credit cards can considerably improve the banking sector by enhancing the accuracy and speed of fraud detection when compared to conventional systems. Traditional credit card fraud detection solutions use rule-based algorithms that highlight transactions that depart from pre-defined rules, such as unusual purchasing locations or high-value transactions. These approaches, however, have limitations in their capacity to detect complicated fraud patterns or react to emerging fraud strategies.

Machine learning algorithms, on the other hand, can analyse massive amounts of data from many sources and uncover patterns that human analysts may not see right away. Machine learning algorithms, for example, may analyse a wide range of transaction-related parameters, such as purchase location, transaction amount, user behaviour patterns, and device information, to discover aberrant behaviour that may be indicative of fraud.

Furthermore, machine learning models may learn from previous fraud patterns and adapt to new fraud strategies in real-time, allowing for a more proactive approach to fraud detection. This can assist financial organisations in reducing losses and safeguarding their clients against fraudulent transactions.

Machine learning-based fraud detection systems can also drastically minimise false positives, which are normal transactions that are misclassified as fraudulent. False positives can cause customer annoyance and revenue loss for financial firms. Machine learning algorithms can increase fraud detection accuracy by lowering false positives and false negatives, reducing fraud risk and giving a better customer experience.

Overall, machine learning fraud detection for credit cards can assist financial institutions enhance the accuracy, speed, and efficiency of their fraud detection systems, resulting in a more secure and trustworthy financial ecosystem.

5.9 Summary

A real-world dataset of financial records from a financial firm was used in the experiments. The dataset contained over 60,000 records from 23 countries. As real-world data typically is not clean, as compared to simulated data, some pre-processing steps such as data cleaning and feature selection were required. The original dataset came with nine features, and feature aggregation was used in adding it to 17 features. A feature importance study using DT, RF and XGB was then done.

Three different experiment sets were conducted. Using the hierarchical BKS classifier, noise up to 40% was added, in increments of 10%. BKS transcends majority voting by 18 times (20% and 30% noise) and by 19 times (40% noise) when noise levels are 20% or higher. This result indicates that BKS outperforms majority voting statistically at 95% confidence level (α = 0.05) when dealing with noisy data.

In the following set of experiments, changes were made in the design of the classifier. An addition of two groups were added, making it a total of five groups, each with three agents of RF, GLM and GBM. Predictions from the five agent managers are sent to the decision combination module, which employs another BKS to generate the final projected class. The same is updated for majority voting.

Similar results were seen from the first experiment. The addition of two groups does not provide much assistance in increasing the accuracy rates and F1-scores. It does however increase the computational complexity; hence it is not recommended to be further utilized.

In the third set of experiments, modifications to the BKS is made. Instead of using accuracy rates, confidence levels were utilized. In increments of 0.05, the threshold levels from 0.60 to 0.90 were used. Any results that are below the set threshold level is rejected.

As the threshold of confidence level is set higher, the confidence of an agent is higher, hence the predictions result in a higher accuracy rate. Based on the experiments, as the threshold level increases, the accuracy rates for BKS increases. However, the opposite is seen for majority voting. As an example, at from threshold level of 0.6 to 0.9, the accuracy rate of BKS increases from 0.9848 to 0.9983, while voting reduced from 0.9676 to 0.9153. At certain threshold levels, the accuracy rates remain almost the same. Only with a big jump of threshold levels, a significant jump can be seen.

6 Conclusions

6.1 Summary of Research

For the construction of a prediction model, a number of steps need to be executed sequentially. The needs for developing the model should be reviewed first. Then, data pre-processing or data cleaning (removing inconsistent data, data transformation, feature selection, or sample selection) should be performed. Finally, to maximise pattern extraction, a suitable prediction model that takes feature qualities into account should be used.

It has been experimentally demonstrated that building a flawless classifier using a single paradigm is frequently difficult. As a result, designers have attempted to integrate many classifiers, resulting in the development of MCS, which strive to combine the advantages of different individual classifiers, frequently generated using distinct training paradigms, to get better outcomes. Classifier combination is a potential alternative to utilising a single classifier and has evolved into a well-established research subject based primarily on heuristic answers.

Intuitively, a collection of classifiers should produce better outcomes than a single decision maker. Such systems mitigate the consequences of errors generated by its components, which is especially important for tough samples. MCSs should be designed to ensure a balance of diversity and compatibility among individual members in order to minimise redundancy and achieve robustness.

Three different algorithms were presented in the work. For classification and regression, RF employs the ensemble learning method. It is a bagging technique in which the trees are run parallel to each other. While creating the trees, there is no interaction between them. GLM employs models that generalise the standard linear regression model to models that do not always include normally distributed response variables and in which a function of the mean of the response is linear in the explanatory variables. GBM employs a boosting method. It is predicated on the idea that the best potential next model, when paired with past models, minimises overall prediction error.

Using the abovementioned models, H2O.ai were employed to establish the MCS model, with all models done in Python. In the first model, the majority voting model is used, in which each individual classifier votes for a class, and the majority wins. The ensemble's anticipated target label is the mode of the distribution of individually predicted labels. The second model used, is developed in this work, BKS. BKS is a *K*-dimensional space, where every dimension indicates the decision from one classifier.

A series of experiments using publicly available data sets and real financial records were carried out to evaluate the proposed multi-classifier system. Two frameworks were used, which consisted of the majority voting and hierarchical agent-based. Both frameworks used three groups, where each contained three agents of RF, GLM and GBM. For the hierarchical framework, each agent manager had its own BKS module. Predictions from the three agent managers are sent to the decision combination module, which employs another BKS to generate the final projected class.

Details of each hierarchical framework including the train and test details were described. In the first set of experiments, 10 datasets from KEEL (2020), UCI (2020) and Kaggle (2020). The datasets had different imbalance ratios. The average accuracy of BKS was at 0.9440, which was slightly higher than majority voting. Comparison with literature was made as well. BKS received the greatest F1-scores in four of six data sets, while the values in the remaining two are 0.01 lower than those of majority voting.

The data used to assess the robustness of BKS was tainted with up to 20% noise. Each dataset underwent 25 runs. BKS achieved at least 18 victories out of 25 trial runs in all ten noisy data sets (10% and 20% noise levels), indicating its superior performance over majority voting in undertaking noisy data samples for $\alpha = 0.05$ (95% confidence level).

In addition to the first set, a second set of experiments with 20 datasets were used. The datasets were sourced from UCI (2020), Kaggle (2020), and OpenML (2020). BKS average accuracy rates were 0.8626, which was higher than majority voting accuracy rates of 0.8511. Similar results can be seen in F1-scores. The findings were compared to the literature. BKS was the clear victor in 16 of the 20 datasets. BKS obtained outcomes that were close to the winner in the other four. Overall, BKS claims to be the best in terms of α = 0.05 significance.

A real-world dataset of financial records from a financial firm was used in the experiments. Over 60,000 records from 23 nations were included in the dataset. As real-world data is often not as clean as simulated data, some pre-processing techniques such as data cleaning and feature selection were necessary. The original dataset had nine features, which were expanded to 17 via feature aggregation. Then, employing DT, RF, and XGB, a feature significance investigation was performed.

Four distinct experimentation sets were conducted. In the first set, original features are compared to aggregated feature combinations. Clearly, the use of additional aggregated features contributes significantly to the improvement of accuracy rates. As a result, all subsequent experiments incorporate the inclusion of aggregated features.

In the second set, noise of up to 40% was added in 10% increments using the hierarchical BKS classifier. BKS outperforms majority voting 18 times (20% and 30% noise) and 19 times (40% noise) when noise levels are 20% or higher. This result indicates that BKS outperforms majority voting statistically at 95% confidence level (α = 0.05) when dealing with noisy data.

The classifier's design was changed in the next series of experiments. Two other groups were added, bringing the total to five, each with three agents of RF, GLM, and GBM. Predictions from the five agent managers are sent to the decision combination module, which employs another BKS to generate the final projected class. The same applies to majority voting.

The addition of two groups has little effect on raising accuracy rates and F1-scores. It does, however, increase computing complexity, hence it is not advised for continued use. The second set of experiments involves changing the BKS. Confidence levels were used rather than accuracy rates. Then, threshold levels from 0.60 to 0.90 were applied in steps of 0.05. Any result below the predetermined threshold level are rejected.

As an agent has more confidence when the threshold of confidence level is set higher, the predictions have a better accuracy rate. Based on the experimental results in Fig. 6.1, accuracy rates for BKS grow as the threshold level rises.

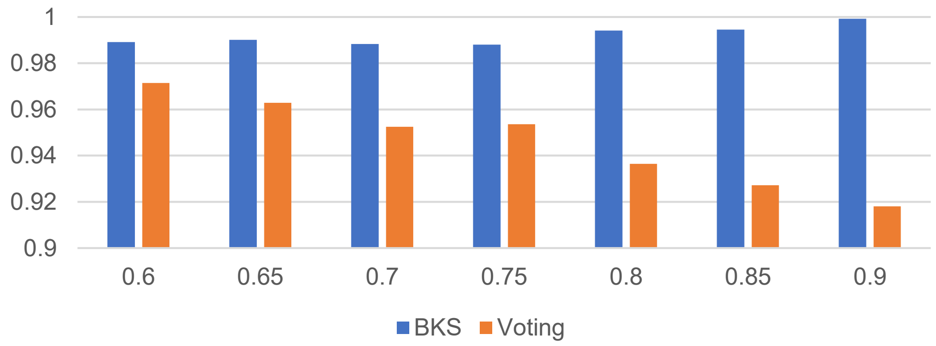

Fig. 6.1 Comparison of accuracy by different threshold

As shown in Fig. 6.1, the accuracy rate of the BKS increases from 0.9891 to 0.9983 at the threshold level of 0.6 to 0.9, while voting decreases from 0.9715 to 0.9180. The accuracy rates are nearly constant at some threshold values. A noticeable jump can only be seen with a large jump in the threshold levels.

In order to provide a clear and concise alignment between the research objectives set out at the beginning of this study and the findings derived from the conducted experiments, a summary of how each objective was addressed is presented in Table 6.1.

Table 6.1 Alignment Map: Research Objectives *vs*. Findings

Research Objectives	Findings
1) To develop a multi-classifier system model using machine learning algorithms for data classification	○ Developed a hierarchical agent-based framework using the BKS as a decision-making mechanism. ○ Utilized RF, GLM, and GBM for classification and regression.
2) To apply the Behaviour Knowledge Space (BKS) method for combination of predictions from multiple classifiers and analyse the performance with benchmark data sets	○ BKS was applied in two frameworks: majority voting and hierarchical agent-based. ○ BKS achieved higher average accuracy rates compared to majority voting in multiple datasets.
3) To enhance the BKS method with confidence measures for effective combination of predictions from multiple classifiers	○ Introduced confidence levels in BKS, leading to improved accuracy rates as the threshold level rises. ○ As shown in Fig. 6.1, the accuracy rate of the BKS increases with varying threshold levels.
4) To ascertain the efficacy of the devised multi-classifier system with enhanced BKS for payment card fraud detection with noise-induced transactions records	○ BKS outperforms majority voting statistically at 95% confidence level (α = 0.05) when dealing with noisy data. ○ Real-world dataset of financial records was used to validate the robustness of the devised system.

By mapping the objectives to the corresponding findings, it becomes evident how each research goal was systematically approached and fulfilled.

6.2 Contributions of Research

The key contribution of this thesis is the development of a multi-classifier system to address the classification problem associated with credit card fraud. In particular, a hierarchical agent-based framework that makes use of the BKS as a decision-making mechanism has been developed in order to categorise credit card transaction data as fraudulent or non-fraudulent. This was done in order to combat credit card fraud. The successful completion of this goal was made possible by combining the two strategies. This combination paves the way for the accumulation of knowledge, which ultimately results in improved performance over the course of time.

Traditionally, financial institutions have relied on rule-based systems to manage their operations. Machine learning-based fraud detection models have the potential to improve the overall performance of fraud detection systems by identifying intricate patterns and relationships hidden within the data. This is accomplished through the process of "pattern mining". When using conventional methods, it can be challenging to identify patterns and connections of this nature and nature's own relationships. If a fraud detection system is able to identify potentially fraudulent behaviour at an early stage, it can play an important role in the prevention of illegal transactions and the protection of cardholders' financial information. By first identifying fraudulent transactions and then taking active steps to halt them, a fraud detection system can assist financial institutions and merchants in lowering the amount of money that is fraudulently taken from them.

Not only on financial losses, but also on improved customer experience, which is important because a fraud detection system can help improve customer experience by reducing the number of cases in which legitimate transactions are blocked by minimising the number of false positives (i.e., legitimate transactions that are mistakenly flagged as fraudulent). This is important because a fraud detection system can help improve customer experience by reducing the number of cases in which legitimate transactions are blocked by minimising the number of false positives. The term "false positive" refers to the situation in which perfectly legitimate transactions are incorrectly identified as fraudulent. In order to provide adequate protection for the customer, it is essential to have a fraud detection system that operates as intended. This can be accomplished by quickly identifying any illegal transactions and putting a stop to them. This protects not only the customer but also the financial institution from potential liability.

According to Fiore *et al*. (2019), it is essential for any business that deals with online transactions or issues credit cards to implement reliable fraud detection solutions in order to reduce the risk of financial loss while simultaneously boosting customer confidence. When illegal transactions are mixed in with legitimate ones, modern fraud detection systems use sophisticated analytics and data mining techniques to uncover suspicious usage patterns in transaction logs. These patterns can be found when fraudulent transactions are mixed in with valid ones. In order to accomplish this, it is necessary to gain insights from very large datasets and perform binary classification in order to differentiate between legitimate and fraudulent transactions.

6.3 Suggestions for Future Work

The evolution of the BKS models is an ongoing process, and as with any scientific endeavour, there are always areas that can be improved, refined, and expanded upon. The journey towards a more robust and efficient model is marked by several challenges and opportunities, each of which offers a unique perspective on enhancing the system's capabilities.

One of the primary concerns that have been identified in the current iteration of the BKS models is the potential presence of blank cells within the BKS table. These gaps, although seemingly trivial, can have significant implications. They indicate an absence of prediction for specific data samples, which can lead to incomplete or skewed results. This issue becomes even more pronounced with the addition of more classifiers, as the knowledge space becomes increasingly vast. Addressing this challenge is not just about filling in the gaps; it's about understanding why they exist in the first place and developing strategies to ensure they don't occur in future iterations.

Noisy datasets, especially those tainted with noisy class labels, present another formidable challenge. Such datasets can introduce a myriad of inaccuracies within the BKS cells, leading to predictions that may not truly represent the underlying data patterns. To combat this, there's a growing interest in incorporating probabilistic methodologies into the BKS framework. Bayesian inference, for instance, offers a structured approach to interpret the BKS predictions in the presence of noise. By understanding the probability distributions and leveraging them to make informed predictions, the model can achieve a higher degree of accuracy and resilience against noisy data.

The issue of unbalanced data is another area that warrants significant attention. In the real world, data rarely presents itself in neatly balanced packages. Some classes may be overrepresented, while others might be underrepresented. This imbalance can skew the model's predictions, leading to biases that can affect the overall efficacy of the system. To address this, there's a plan to employ a combination of over-sampling and under-sampling techniques. These strategies aim to balance out the representation from each category, ensuring that the model remains unbiased. Additionally, the use of statistical hypothesis testing will provide a structured framework to evaluate the impact of different classification techniques on the model's performance. This rigorous evaluation will shed light on the strengths and weaknesses of each approach, guiding future development efforts.

Looking ahead, there's also an intent to explore an online adaptation of the BKS model. In today's dynamic digital landscape, data is continuously evolving. An online variant of the model would allow it to learn in real-time, adapting to new data samples as they become available. This continuous learning approach ensures that the model remains current, adaptive, and reflective of the latest data trends.

On the regulatory front, the importance of compliance cannot be overstated. Financial institutions operate within a complex web of regulations, with the Anti-Money Laundering legislation being a prime example. Ensuring adherence to these regulations is a multifaceted challenge. By integrating advanced fraud detection systems with compliance checks, financial institutions can navigate this regulatory maze more efficiently. The integration of these technologies can streamline the compliance process, making it more transparent, efficient, and cost-effective.

In conclusion, the journey towards a more refined BKS model is marked by challenges and opportunities. Each of these areas offers a unique perspective on enhancing the system's capabilities. By addressing these challenges head-on and leveraging the opportunities they present, the BKS model can continue to remain at the forefront of fraud detection, offering unparalleled accuracy and efficiency.

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